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Chapter - 1

Advancements in NGS Technologies (DNA-seq and RNA-seq) and Infectious Fungal Diseases of Rice

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Chapter - 1

Advancements in NGS Technologies (DNA-seq and RNA-seq) and Infectious Fungal Diseases of Rice

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Abstract

The Next Generation Sequencing technologies are a great invention of the scientific community that is constantly working towards the discovery of genetic variations in the genome that lead to enhanced tolerance to biotic and abiotic stresses. After phenotyping of the strong culm genotypes, it is necessary to focus on the genotype's genetic makeup to demystify the genetic nexus. In the present scenario, few types of research have been carried out that determine the causal SNPs for climate resilience and strong culm. The Next generation sequencing-mediated discoveries are picking up and causal SNPs may be discovered with a wide array of technologies has been described in this chapter. Also, the twenty-first century has been flooded by the incidence of new fungal diseases that impacts not only the yield but also the survival nature of the rice plant. This review also emphasizes fungal diseases and the resistance factors present at this present time.

Keywords: DNA-seq, RNA-seq, KASP assay, Infectious fungal diseases

Introduction

Next Generation Sequencing applications have been successful and surpassed Sanger sequencing approaches in terms of the data produced from complex genomes. The whole genome sequencing approach has hastened the process to identify the causal SNPs compared to the reference genome in various population types ^[1]. Similarly, RNA-seq procedures have largely replaced the need for microarray analysis using tissue samples. In addition to genic analysis, RNA-seq is also useful for studying alternatively spliced transcripts of identical genes. Also, the detection of causal SNPs in transcriptomes associated with climate-resilient traits through alignment with the reference genome is conducive to highly accurate SNP detection. The blessings of the NGS technologies have extended themselves to De Novo

Genome Assembly, Interaction analysis between Protein-DNA (CHIP-seq), Epigenome studies, Methylation patterns on DNA, study through Bisulfite sequencing, and Metagenomic studies. The types of sequencing facility present are depicted. The development of resistant cultivars has been an effective and economic strategy for control of diseases in rice. Rice (*Oryza sativa* L.) is an important source for the understanding of complex processes that control high giveaway and value-appended traits. The world population is expected to boom beyond 9 billion by 2050 [2]. Rice is the primary food crop world-wide, providing one-fifth of the calories in-taken by humans.

Crop protection and food security are inter-dependent, and breeding for the development of resistance towards biotic and abiotic stresses is essential to ensure food availability to population. Owing to the labour-intense character of breeding, integrated disease control is often reduced to chemical treatment, leaving every purpose of this environment-friendly approach in void. Advancements in breeding for rice, however, make breeding an increasingly easy effort in staying forward for pathogen adapting. An added advantage to plant breeding is the discovery of the new age genotyping arrays that have made identification and confirmation of SNPs an easier task across diverse crop-based systems and disease detection. The ultimate aim for a plant breeder must be to identify, and phenotypes for climate-resilient lines and unravelling of SNPs associated with the trait of interest. The figure representing a graphical abstract of the properties of climate resilient rice that is the need of the hour and the various facets of NGS (Fig. 1).

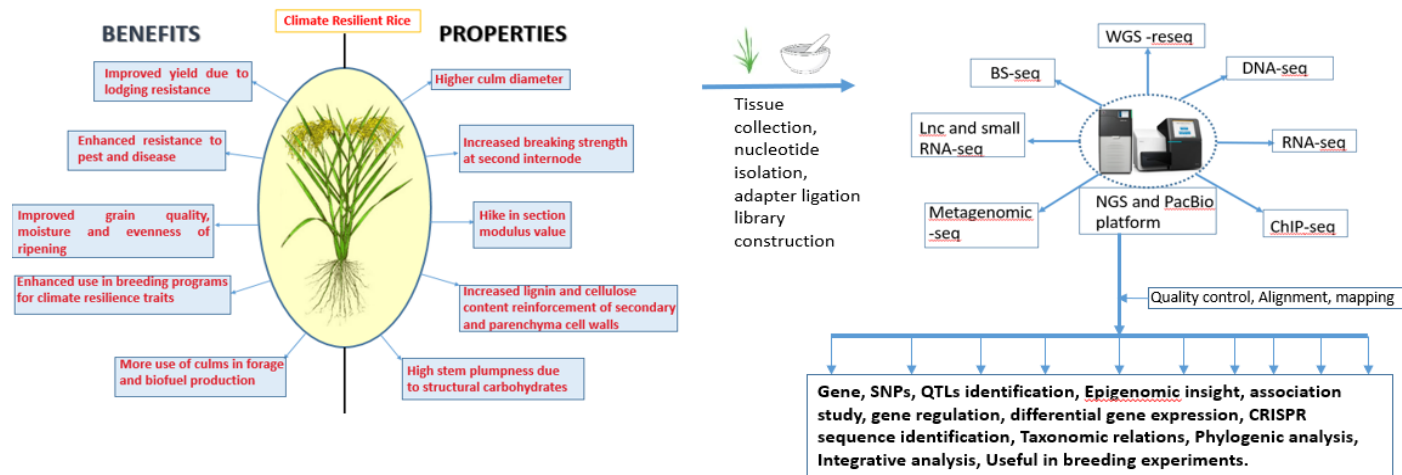


Fig 1: NGS and the climate-resilient properties of rice

NGS approaches basics as a gateway for golden standard climate resilience in rice

The advancements made in rice research through the 3K rice genome sequence project have made it possible for research fraternity to go deep into the rice genome nooks and corners. The basic workflow is dependent on data of NGS quality control and pre-processing, mapping of reads involving the use of specific algorithms and reference sequences followed by quality control checks, visualization of data, and functional annotations. Usually, quality control procedures are followed after every step so as to specifically pinpoint the correct genomic regions associated with the trait of interest.

Even though the factor of the cost involved in the extraction of NGS sequencing data is enormously greater than traditional sequencing systems, the enormous amount of data produced from whole genome sequencing is easily worth the cost in comparison to Sanger sequencing producing same amount of data. Moreover, the recent drops in the cost of NGS only warrant picking up the data generated for the genotyping of climate-resilient traits as well as climate-resilient rice and other GM crops ^[3,4]. A General overview of the premature stage of NGS data analysis includes sequencing target (DNA/RNA), shearing into smaller fragments that include nebulization, sonication, enzyme-mediated cleavage, and acoustic shearing. This fragmentation step followed by size selection step in set target range. For RNA fragment, it must be converted into complementary DNA using specialized reverse transcriptase. Following that adapters should be ligated to the DNA/RNA fragments and immobilized to a solid support. These adapters are NGS specific to platform sequences requiring recognition of sequencing reads by NGS instrument.

The adapters are made of 4 components i) universal sequencing primer sequence for reaction initiation in each fragment, ii) PCR-specific primer sequences amplification for sequencing template enrichment; iii) anchoring sequences for attaching DNA fragments to solid support and indexing barcodes for differentiating multiple samples when sequenced together. This is followed by the generation of sequencing reads or contigs and the alignment of sequencing read. The core Illumina sequencing technology employs fluorescently labelled nucleotides with a reversible terminator. The terminator moiety temporarily prevents new DNA strands from extending. After optical detection of the incorporated nucleotide based on a specific fluorescence label, terminator moiety is cleaved and thereby fresh strand synthesis resumes for subsequent cycle of nucleotide incorporation. For simultaneous detection of nucleotide inclusion in millions of sequencing reactions, deoxy ATP, deoxy

CTP, deoxy GTP, and deoxy TTP are labelled with the different fluorescent signals are emitted. Each nucleotide can be detected by the different fluorescent signal emitted ^[5, 6].

There are advanced indexing approaches have been created by IDT in collaboration with Illumina. These adapters may have single indexing, unique dual indexing (UDI), and unique molecular identifiers (UMI) present in the IDT xGen UDI-UMI indexes specifically designed for multiplexing solution for obtaining faster results (<https://www.lubio.ch/applications/ngs/adapters>) ^[7].

The adapter design is as follows:

- i) 8 to 10 nucleotides index lengths (CDI or UDI).
- ii) Unique/Combinatorial dual indexes (UDI/CDI).
- iii) Multiplex capacity up to 1.5 kb unique indexes.
- iv) Indexing primers designed to perform enzymatic library normalizations.

Sample indexing allows manifold samples for sequencing in single attempt (i.e. multiplexed) in a flow cell also termed a chip. Each sample index, usually 8-10 bases, is specific to a given sample library and is used for de-multiplexing for data analysis for assignment of individual sequence reads to the true sample. Primer indexing involves no GC bias and also maintains colour balance for 2-and 4-color Illumina instruments. The usage of unique dual indexes (UDIs) mitigates errors introduced by index-hopping (associated with primer multiplexing). Molecular barcoding is by using unique molecular identifiers (UMI) enables reliable error correction and increases throughput. UMIs are short sequences that incorporate a unique barcode to each molecule in a given sample library and will share the same insert and UMI tag. Ideally, UMIs reduce the rate of false-positive variant call and increase variant detection. Adapters may also contain UMIs, such as the xGen UDI-UMI Adapters, available with a UDI design that identifies low-frequency variants.

Pacific biosciences single molecule real-time sequencing

The Pacific Biosciences Single Molecule Real-Time Sequencing is the most advanced third generation sequencing platform. Pac Bio SMRT is a 3rd generation technology created by Pacific Biosystems, is developed based on sequencing-by-synthesis principle. It is widely used in epigenetics to mark essential markers for identification of new species in plants ^[8]. During library preparation, template DNA fragments are provided with a hairpin-loop-adapters affixed on either side making circular template that also carries

universal primer binding sites and initiation sequence. Sequencing is carried out employing special nanophotonic visualization chambers known as zero-mode waveguide (ZMW) which permits minimum volume for light detection, capable of detecting single nucleotide incorporated by DNA polymerase.

SMRT technology utilizes fluorophores (four unique fluorophores for each dNTP) that are linked to terminal phosphate rather than that fluoresces on cleavage during replication and the fluorescence is detected. A mutated DNA polymerase $\phi 29$ that has increased processivity, attached at the bottom of the chamber as the DNA passes through in the form of a loop. Library is generated by creating DNA fragments followed by end-repair and hairpin-adaptor ligations in order to obtain circular SMRT DNA template fragments. Closed-ended circle DNA template allows the polymerase for continuing around to the second adapter sequence and then onto the antisense strand, and so on so as to generate long reads with over 70,000 bases [9].

Whereas NGS platform provides read length of 150 to 400 base pairs, the PacBio generates longer reads in average length of 8 to 15 kilo bases. This high contiguousness leads to de novo assemblies using PacBio sequencing enables gap closure in various reference assemblies and characterizes structural variation. As this technology doesn't use a separate fluorescent label step, sequence detection signal is continuously recorded at 75 frames per second movie instead of scanning image. However, this technology has several constraints in terms of error rate and several correction steps need to be implemented, and also possess high run cost [10].

Other than PacBio sequencing, many hybrid sequencing strategies have been developed to make use of additionally accurate short reads in conjunction with PacBio long reads. Generally, these hybrid sequencing strategies are more budget-friendly rather than sequencing solely by PacBio. In recent times two *Oryza sativa indica* genomes have been sequenced in this fashion. These indica genomes may well be gold standard references for comparison against indica cultivars as done in many studies [11, 12]. The SMRT sequencing also enables causative variant detection of a disease, sex determination of garden asparagus plant, and determination of essential microbes for cheese making as has been reported [13, 14].

DNA-seq and RNA-seq

The second-generation or the next-generation sequence data analysis involve various primary and secondary analysis. It starts with base calling from fluorescence images of Illumina flow cells with platform-specific patented algorithms. The algorithms utilize a sliding window mechanism for sequence reads. The base calling results are reported via FASTQ format. Data

call quality is determined by Q scores. A successful run only completes with an average Q score of 20. The most frequently used NGS data QC software includes FastQC (Falco) ^[15], FASTX-Toolkit (FastproNGS), and NGS QC Toolkit ^[16]. Low-quality read trimmed using tools like FastP trimming tool.

The QC data and pre-processing are followed by read mapping to the reference genome (Nipponbare, IRGSP 1.0) or de novo genome assembly. Alignment is followed by Burrows-Wheeler Aligner (BWA) and indexed by SAMtools ^[17]. Duplicate reads are de-duplicated and read categories are created using the Picard MarkDuplicates tool and for multiple samples are grouped using AddOrReplaceReadGroups tools (Broad Institute) and again QC check is proceeded with using GATK tools. The reads mapped to the reference genome can be used for transcriptomic analysis (for RNA-Seq) ^[18], epigenomics and DNA methylation analysis (Enzymatic methyl-Seq) ^[19, 20], Protein-DNA interaction analysis (CHIP-Seq) ^[21], mutation and SNP variation discovery (whole genome resequencing) and metagenomics analysis ^[22]. Finally, tools or genome browsers are used for analysis, visualization and interpretation in standard genome browsers.

The whole genome resequencing involves data processing and quality control which is then followed by read mapping and local realignment conducted by algorithms like BWA, Novoalign. Following local realignment base quality is recalibrated by GATK RealignerTargetCreator. Following this step involves variant calling, variant annotation, and trait association with genotype. The whole genome sequencing approach has been put to use for marker-assisted selection of elite rice cultivars ^[23], an association of various agronomic traits for Basmati rice ^[24] and contrasting salinity stress tolerance evaluation between rice species to call for SNPs and indels associated with salt tolerance trait ^[25] have been analysed. The whole genome sequencing methodology is enabling the identification of SNP variants as a causal marker to differentiate between stress responses or between genotypes in recent times. Recent publication also demonstrated evolution in patterns of structural variations in Asian rice that played crucial roles in domestication as revealed through a combined approach of Illumina and PacBio sequencing platforms ^[26]. A recent study involving QTL-seq identified QTLs associated with soluble starch-synthase, starch debranching enzymes in rice ^[27]. DNA-seq related study of OsPL encoding MYB-family transcription factor imparts anthocyanin synthesis and response to heat stress and triggers hormone signalling only because it possesses a cytosine insertion at the third exon of OsPL gene ^[28] has been deciphered by use of this technology.

Specific algorithms *viz.* SOAPsnp (SOAP abbrv. Short Oligonucleotide Alignment Program) ^[29] for SNP alignment, ELAND (Efficient Large-Scale Alignment of Nucleotide Databases) ^[30], Novoalign are used for alignment of both genomic and epigenomic datasets ^[31]. For alignment of PacBio algorithms SMRT ^[32], BWA-MEM platform is used in the case of studying de novo genome assembly of yeast *Kazachstania bulderi*. For Oxford Nanopore sequencing-related algorithms L-INS-I is used ^[33]. SAM and Binary Alignment File formats (BAM) are mapping file formats. Graphical visualization for mapping results provided by overlaying indexed read sequences (SAM & BAM file formats) against reference genome. Commonly used visualization tools are Ensembl Plants ^[34], and Integrative Genomics Viewer (IGV) ^[35]. ‘Galaxy’ is a “bridging” system that provides user affectionate interface until command line tools. Galaxy allows the creation for data analysis work-flow from consistency achievement and reproducibility, sharing of analytical procedures for researchers ^[31].

For designing an RNA-seq experiment, it is mandatory to have two contrasting groups of tissue from phenotypic differences in terms of climate resilient or non-resilient or time-dependent expression variation in an experimental design. In order to determine the RNA expression and related real-time expression differences stringent RNA isolation procedures must be followed for obtaining the highest RNA integrity (RIN) for sequencing purposes. An important factor behind extracting high-quality RNA samples with zero or low levels of degradation includes snap freezing of tissue samples in liquid nitrogen to avoid potential RNA degradation. In absence of liquid nitrogen, RNA stabilizing agents (eg: RNA later) may be used. Young tissues from either leaf or developing and elongating tissues will provide much better and high-quality RNA integrity rather than using mature tissues. The reason is extracting nucleic acids from older tissues is cumbersome and reduces RNA integrity as high amounts of polyphenols are existing in mature tissues that interferes with the integrity of RNA used for sequencing purposes. TRIZOL method of RNA isolation followed by DNase treatment is applied for extracting total RNA including small RNAs like miRNAs or specialized miRNA isolation kits ^[36, 37]. Also, it may be noted that using a greater number of replicates yields more accuracy in terms of the detection of differential gene expression.

This procedure is followed by reverse transcription to cDNA and ligation of sequencing adapters. The use of the poly-A enrichment procedure is followed by transcription to cDNA. The poly-A enrichment captures mRNAs and no other rRNA forms that don’t possess the tail structure. The RNA-seq

is carried out in Illumina NGS platform (HiSeq X10 or other platforms) [38]. After the sequencing is completed the data QC is carried out in RSeQC and RNA-SeQC [39]. Tools such as Trimmomatic provides options for read trimming [40]. The de-novo transcriptome assembly is best carried out in Trinity, SPAades, Trans-ABYSS, Bridger and SOAPdenovo-Trans [41]. Tools such as Scotty are used to determine sequencing depth (meaning ratio of total number of bases obtained by sequencing to the size of the genome or an average number of times each base is measured in the genome) and sample size. RNA-seq reads are searched against reference transcript databases using standard non-gapped annotation-guided mappers like PASTA, SAMMate, SpliceSeq, X-Mate. BWA-MEM is a long-read mapping tool that has also surfaced in the mapping arena and offers fast speed and split alignment options [42]. Overlaying of mapped reads in reference genome provides more insight into mapping pattern visualization. Databases may next be developed from the sequenced data of RNA-seq e.g.: IC4R-2.0 [43].

In the de novo Genome assembly one has to assemble contigs by TIGR assembler [44], followed by construction of scaffolds by SOAPdenovo, Velvet, SSAKE, followed by genome assembly and quality evaluation [45]. Various microRNA genes have been studied in various newly developed landraces of rice using this procedure, as has been developed in case of other crops [46, 47]. Degradome sequencing of Magnaporthe oryzae-elicited rice novel miRNAs have been carried out through preparation of small RNA libraries and sequencing validation with qRT-PCR [48]. RNA-seq data is normalized by dividing of raw read count by the effective library size. Errors in data normalization may hamper downstream analysis and in terms of puffed-up false positives in differential expression analysis. Also, housekeeping genes that do not alter in expression under study condition are also identified [49]. The importance of RNA-seq studies have picked up recently and involves the integration of RNA-seq other next-generation sequencing procedures with DNA-seq or ChIP-seq or BS-seq. Various Next Generation RNA-seq studies revealed the key candidate transcription factors belonging to the NAC family for drought tolerance and grain filling in rice [50, 21].

DEseq is a tool that performs data normalization through statistical tests prior to expression analysis and has been used in differential gene expression studies for coleoptile elongation in rice under submerged conditions. The gene ontology to seek the functional annotation can be done in terms of GO enrichment analysis on DE genes was carried out using the web-based toolkit “AgriGO” followed by statistical tests like Fisher’s exact test and Benjamini-Hochberg Procedure that helps in the correct identification of truly expressed

genes ^[18]. Gene ontology analysis may also be performed utilizing the KEGG pathway annotation database via ribosomal protein acetylation in the rice roots system ^[51]. In a most recent study identification of alternative splicing events have been identified in rice through dRNA-seq and degradome sequencing ^[52]. Standard Splice junctions between exons may be detected by using specialized splice junction detection tools. The visualization of protein annotation of the RNA may be found in ProtAnnot together with Integrated Genome Browser may be used for visualization and assessing the effects of differential splicing on gene function ^[53].

Identification and calling of indels can be done in GATK, SNPSeeker, Dindel and various other tools as were been used in Rice SNP-seek database update ^[54, 55]. Variant calling performed with minimum Phred-scaled confidence threshold of 30 using the GATK-HaplotypeCaller. However, the detection of structural variant in homologous regions may be cumbersome because of genome reference bias or uneven read coverage that results in false calls. For accurate variant calling customized scaffold option may be applied for identification and phasing of complex variants by means of NGS by means of algorithms like SOPRA, SCARPA or SSPACE ^[56].

Variant calling in GVCF file and GATK-GenotypeGVCF for genotyping of the variant files in multiple sampling occasions. Raw variants generated from the variant calling procedures would be again filtered for identification of good-quality variants with tools like GATK-Variant Filtration ^[57, 17]. The identified variants may be further validated by designing of core SNPs for SNP genotyping assays such as the KASP assay (Kompetitive allele specific PCR) ^[58]. Even though the RNA-seq through NGS are robust enough to not validate it by qPCR ^[59], yet performing the real-time PCR involving the expression of upregulated or downregulated genes provides further value to the results ^[60]. The fact of added value has also been proven in various recent research and metagenomics approaches in rice and other crops ^[61, 62, 63].

Core KASP SNP assays for genotyping

Kompetitive Allele-Specific PCR is a method that successfully finds out SNPs and InDels at sites of interest. SNPs are very promising because of their widespread distribution in the genome and are suitable for elevated-throughput automation in genotyping. KASP is based on fluorescence on terminus similar to TaqMan or rhAmp detection of probe ^[64]. A two-color probe detects individual samples from different genotypes at a single site and a mixed fluorescence at middle eigenvalues; different fluorescent signals are reflective of DNA templates carrying the mutation. For the KASP test, an indel

or SNP-specific KASP assay mix buffer (specific assay buffer consisting of two forward primers consisting of either of SNPs; each primer and a common reverse primer), KASP master mix (Consisting of the FAM/HEX/ROX dyes associated fluorophores and quencher probes, and other PCR components). FAM dye is associated with one of the alleles. HEX dye is associated with another allele and the ROX dye acts as a passive reference. The DNA sample is mixed with the assay mix at recommended concentration and volume along with the KASP master mix for a thermal cycling reaction and fluorescence is read by endpoint scanning using a FRET-capable plate reader. This strategy is quite effective in cases of polyploid crops and the determination of haplotypes [65].

KASP assays are highly accurate, less expensive than TaqMan, simple and rapid rate of allele identification, and wide range of applications in healthcare and agriculture. Several functional SNPs, Indels are being converted into functional markers via KASP assays, thereby improving efficiency, speed of selection in various crop breeding endeavours. Notably, these markers can be put to use in GWAS mapping experiments wherein the identified SNPs may be designed and validated by KASP assays. Transition mutations in form of A<->G and C<->T. Transversion mutation whereby A or G may be substituted by C or T. Brown plant hopper resistance has been validated for a mutation in gene BPH25(t). Another mutation in the Pita locus that contains the resistance gene for Blast. At the Pita locus at the 918 positions, a mutation leads to change in amino acid alteration from alanine into serine. One SNP caused base pair differences, wherein G/G means resistance and T/T means susceptibility in rice. This difference leads to amino acid change. This was visually detected by means of KASP assay and further validated by Sanger sequencing [66]. Similarly, these markers may be useful for phenotyping of various yield related traits.

Table 1: Protocols for SNP identification PCRs

Platform	PCR protocol						
	Pre-read	Hold 1 st Stage	PCR 1 st stage		PCR 2 nd stage		Post read
	Temperature/time	Temperature/time	Temperature/time	Number of Cycles	Temperature/time	Cycles	Temp/time
TaqMan	60 °C/30s	95 °C/10m	95 °C/15s	40	-	-	60 °C/30s
			60 °C/1m				
KASP	30 °C/1m	94 °C/15m	94 °C/20s	10	94 °C/20s	26	37 °C/1m
			61 °C/1m (decrement of 0.6 °C per cycle to a final 55 °C)		55 °C/1m		
rhAmp	60 °C/30s	95 °C/10m	95 °C/10s	36	-	-	60 °C/30s
			60 °C/30s				
			68 °C/20s				

PCR protocols for the TaqMan, KASP and rhAmp

The protocols for the genotyping wherein 4 individual steps are involved and the involvement of a plate-read and cooling step is represented in a table (Table 1). In an experiment performed, results were documented as the success rate of KASP marker design was 39/50 for Taqman, 49/50 for KASP, and 49/50 for rhAmp. After genotyping is performed, Taqman demonstrated 93% and KASP demonstrated 93.5% of amplifications whereas rhAmp demonstrated 97% amplification. Allele call quality yielded 97% for rhAmp and 98% for both KASP and TaqMan. rhAmp and KASP demonstrated extremely significant allelic discrimination in comparison to Taqman. Taqman showed the most compact cluster. In conclusion, rhAmp SNP genotyping assays were less expensive and more reliable, KASP little more expensive and the most expensive was Taqman [64].

For example, in the below pictorial representation of a KASP assay, Homozygous allele C:C leads to wild type cluster are having (high FAM and low HEX signal); Homozygous allele T:T that is parallel to the Y-axis is having (high FAM and low HEX signal); Heterozygotes cluster diagonally (nearly equal FAM and HEX signals).

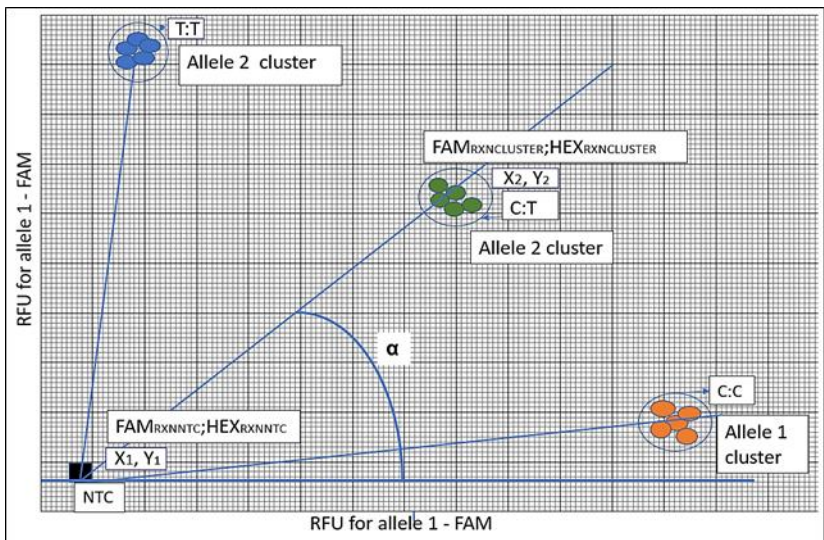


Fig 2: The KASP assay cluster analysis

Statistics of the graph

Relative fluorescence units are measured relative to the Non-Template Control (NTC). Using the co-ordinate geometry distance between the

FAM_{RXNNTC} and HEX_{RXNNTC}, and the FAM_{RXNCLUSTER}; HEX_{RXNCLUSTER} is given as:

$$D = \sqrt{\Delta x^2 + \Delta y^2} = \sqrt{[(FAM_{RXNCLUSTER} - FAM_{RXNNTC})^2 + (HEX_{RXNCLUSTER} - HEX_{RXNNTC})^2]}$$

RXNCLUSTER here denotes the coordinates of the heterozygote allele calling cluster.

The slope of the line that makes angle α is $\tan \alpha = (y_2 - y_1) / (x_2 - x_1) = \Delta y / \Delta x$. Hence the angle of separation given as $(\alpha) = \tan^{-1} \Delta y / \Delta x$, where Δx is FAM_{RXNCLUSTER} - FAM_{RXNNTC} and Δy is VIC_{RXNCLUSTER} - VIC_{RXNNTC}.

For the data points involving a medoid type of cluster may be computed by algorithms that are designed to handle K-type clustering of the data. However, for very compact graphs as generated by rhAmp, as stated by [64], average coordinate value of cluster is $S = \sqrt{\sum (x_i - \bar{x}) / n - 1}$, wherein x_i is distance of data point from the average cluster coordinate, $\bar{x}$ is the mean of all data points in cluster whereas 'n' being number of data points in cluster (Fig. 2).

Infectious diseases of rice

Sheath blight infection

Rice sheath blight fungal pathogen *Rhizoctonia solani* belongs to the basidiomycete having septate hyphae and spores borne on a basidium derive their energy from dead or dying cells of the rice sheath. Symptoms include oval-shaped or elongated grey spots on leaf sheaths. They lead to blighting/discolouration features on leaves.

Major resistance QTLs identified

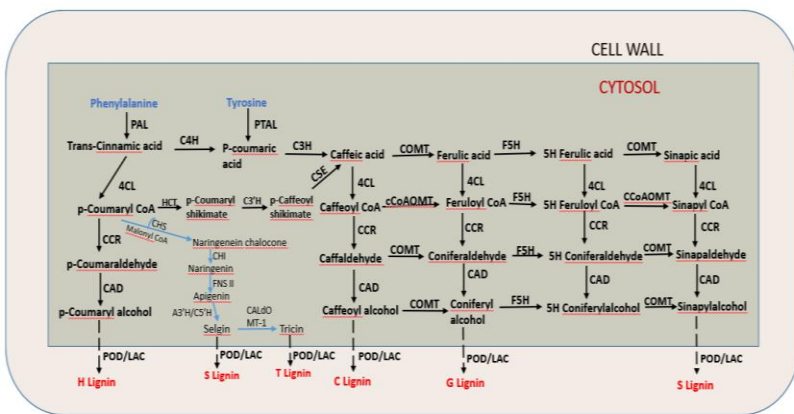
Genes targeted by the sheath blight pathogens include pectins, pectin methylesterases, polygalacturonases are expressed and mediate the fungal pathogenesis [67]. After infection by the pathogen, several key microRNAs are expressed in the rice plants that dictate their resistance or susceptibility these include Osa-miR1320-5p, Osa-miR156d, Osa-miR159b, Osa-miR820c, and Osa-miR1876 are differentially regulated in susceptible and resistant genotypes [68]. Major resistance related QTLs qShB9-2, that encodes for β -1,3-Glucanases that drives in trafficking of cell wall materials through plasmodesmata. Another major resistance includes qShB9-2 out of which a C/T SNP in OsWAK91 sequence found to be linked to resistant phenotype [69]. Another essential chitinase gene expressed in ShB susceptible cultivar Taipei 309 confers varying level of tolerance mediated by transgenic approaches [70].

Rice blast

Rice blast infection is caused by *Magnaporthe oryzae* that is hemibiotrophic pathogen belongs to Magnaporthaceae family. Rice plants in various stages of growth is infected at essential parts of the plant involving roots, stems, nodes, leaves, panicles. The conidium germinates to produce germ tube. That differentiates into appressorium. The appressorium has differentiated cell wall, separate melanin layer lying in between cell wall and plasma membrane. Inside cell hyphae rapidly multiply and this arises the symptoms of blast. To overcome the plant immunity are Avr effectors. In addition, several small non-coding RNA genes like MGG_02329 (Isotrichodemin C-15 hydroxylase), MGG_02378 (glutamate decarboxylase), MGG_10932 codes a C2H2 zinc finger transcription factor associated with development of the appressoria; MGG coding for a FAD oxidoreductase; MGG_02065 coding for kinesin light chain; and MGG_10027 coding for calcium transporting ATPase 1 ^[71].

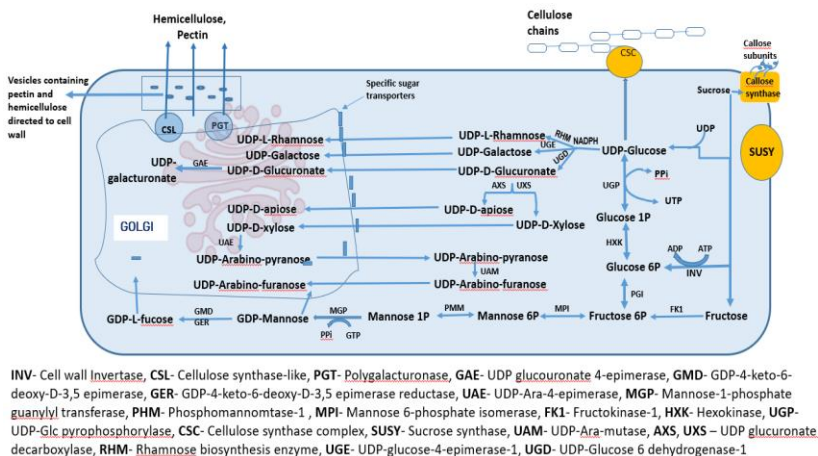
Resistance genes

There are innumerable number of resistance genes termed as R genes involved among which Pi9, Pi54, pi21, Pi50, Pi7, Pi57, Pigm, and Ptr reported to provide wide spectrum blast resistance. Genes such as Pi5-1, Pi63 and Pb1 are induced by pathogen are expressed in an organized manner. After a series of signalling steps MAPK6 (belonging to the Map kinase pathway) triggers transcription factors like WRKY, bZip, MADs box, NAC6, Dof zinc finger, triggers immunity to rice plants and drives transcription of essential genes like biosynthesis of flavonoids, aminopropanols, lignin synthesis, amino propanol, phaseic acid, terpenoids (PAMPs) and phytocassane genes. Infection in resistance genotypes also leads to upregulation of genes related to biosynthesis of cytokinins, jasmonic acid, auxins and suppression of ethylene mediated senescence. MiroRNAs like miR1849, miR164c, miR164e, miR1854-3p, miR2925, and miR812f are also upregulated as found in study. In addition, several breeding programs have been carried out at Indian Institute of Rice Research to integrate resistance genes into rice through the process of marker assisted selection ^[71]. The lignin biosynthesis pathway that provides the rice plant resistance to broad area of biotic and abiotic stress tolerance and climate resilience is attached in the picture (Fig. 3). The lignin biosynthesis pathway is an important pathway that is explained in review by ^[72]. Further the synthesis of structural and non-structural carbohydrates as it operates in plants in is represented (Fig. 4).



PAL- L phenylalanine ammonia lyase, **4CL**- 4 Hydroxycinnamate:CoA ligase, **CCR**- Cinnamaldehyde CoA reductase, **CAD**- Cinnamyl Alcohol Dehydrogenase, **POD/LAC** Peroxidase/ Laccase, **C4H** Cinnamate 4 hydroxylase, **PTAL**- Bifunctional L-phenylalanine/L-Tyrosine ammonia lyase, **C3'H** 4 coumaroyl shikimate/quinic 3' hydroxylase, **CSE**-Caffeoyl shikimate esterase, **COMT**- Caffeate/5-hydroxyferulate 3-O-methyltransferase, **FSH**- Ferulate 5-hydroxylase/coniferaldehyde 5 hydroxylase, **CCoAOMT**- CaffeoylCoA 3-O methyl transferase, **CHI**- Chalcone synthase, **CHS**- Chalcone isomerase, **FNSII**- Flavone synthase II, **A3'H/CS'H** Apigenin 3' hydroxylase/chrysoeriol 5' hydroxylase, **HCT**- 4 hydroxycinnamoyl CoA:shikimate, **CalDOMT**-1.5 hydroxylconiferaldehyde O-methyl transferase; **T lignin** - Tricin lignin, **S lignin** - Selagin lignin

Fig 3: The lignin biosynthetic pathway



INV- Cell wall Invertase, **CSL**- Cellulose synthase-like, **PGT**- Polygalacturonase, **GAE**- UDP glucuronate 4-epimerase, **GMD**- GDP-4-keto-6-deoxy-D-3,5 epimerase, **GER**- GDP-4-keto-6-deoxy-D-3,5 epimerase reductase, **UAE**- UDP-Ara-4-epimerase, **MGP**- Mannose-1-phosphate guanylyl transferase, **PHM**- Phosphomannomutase-1, **MPI**- Mannose 6-phosphate isomerase, **FK1**- Fructokinase-1, **HXK**- Hexokinase, **UGP**- UDP-Glc pyrophosphorylase, **CSC**- Cellulose synthase complex, **SUSY**- Sucrose synthase, **UAM**- UDP-Ara-mutase, **AXS**, **UXS** - UDP glucuronate decarboxylase, **RHM**- Rhamnose biosynthesis enzyme, **UGE**- UDP-glucose 4-epimerase-1, **UGD**- UDP-Glucose 6 dehydrogenase-1

Fig 4: Synthesis of structural and non-structural carbohydrates in plant cells

Bakanae

Bakanae disease is a disease caused by seed-borne *Fusarium fujikuroi*, a fungus that is belonging to the Ascomycota division and belong to the class Sordariomycetes. These infect the plants in the seedling stage wherein the hyphae directly penetrate the epidermal region of basal stems and roots and progress through vascular bundles. This disease usually infects the lower region of the rice plants especially stems and roots. qBK1_628091 present in

the short arm of Chr. 1 and qBK4_31750955 present in the long arm of Chr. 4 provide resistance to the rice plant ^[73].

False smut

False smut in rice caused by the pathogen of fungus *Ustilaginoidea virens*. Pathogen when infects rice grain converts it into RFS ball established from mycelia clad with powdery chlamydospores and may also give rise to sclerotia. Toxins produced by the rice balls include ustiloxin and ustilaginoidin which are toxic and harmful to animals ^[74]. qFsr8-1, is a major quantitative trait locus that explains the resistance of the fungus with great phenotypic variance when mapping was performed by the composite interval mapping method ^[75]. In a study, it was tested 43 *U. virens*, specific simple sequence repeat (SSR) markers to study the genetic variation in sixty isolates of *U. virens* collected from different rice growing areas of India. Out of 34 SSR primers, primers viz. FS_S522 and FS-S211 produced 7 different alleles and hence can be used to study the genetic variability in *U. virens* isolates. However, no grouping was observed based on geographical regions. Disease cause yield loss from 2.8% to 81% in rice-producing areas depending on variety and disease intensity. False smut disease development is favoured due to high doses of N2 fertilizers, extensive cultivation of rice hybrids, close plant spacing, and weather conditions such as higher rainfall occurring during the booting and heading stage. Treatment shows false smut infected hills/m² of 12%, false smut infected panicles of 14.9% false smut infected spikelets/panicle of 6.4%. In another experiment, 88 SSR primers are found to be polymorphic. The polymorphic SSRs are used to characterize false smut resistance in IC334233 x BPT-5204 using F2 mapping population ^[76].

The table below denotes the various widespread diseases caused by the bacteria, fungi, insects Table nos. 2:

Disease	Symptom	Causal organism	Resistance gene	Reference
1.	Stem rot-Culm infection cylindrical in nature, causes yield compromise and lodging	<i>Sclerotium oryzae</i>	^[77]	
2.	Bacterial leaf blight-Necrosis observed in leaves, infected leaves get dry and wilt	<i>Xanthomonas oryzae pv. oryzae</i> (Xoo)	Xa1, Xa2 upto Xa33 (33 in number)	^[78]
3.	Bakanae-Produce adventitious roots at lower region of culm, affected tillers die before maturity	<i>Fusarium fujikoroii</i>	SHR5 receptor kinase, sulphate transporter	^[73]

4. Rice blast-elliptical, with grey-white centers and brown to red-brown margins, occurs in leaves, nodes, panicles, leaf collars *Pyricularia grisea*, *Magnaporthae oryzae* Pi54, Pi56, Pi9, Pizt, OsMPK15 ^[79, 80].
5. Sheath blight-Infects leaf sheath, the lesion appears 1-2 cm and 1-4 cm forming necrotic areas and narrow reddish-brown bands, weakens culm *Rhizoctonia solani* qSB3, qSB6 ^[81, 69].
6. Bacterial foot rot-Nodes of culm turn black *Erwinia chrysanthemi* Resistance gene not discovered ^[82].
7. Yellow stem borer-Entry point through nodes causes rotting of sheath and culm, causing white panicles *Scirpophaga insertulans* Cry IA, Cry 2A proteins ^[83].
8. Gall midge-After entry into the culm, the culm stops growth and produces non-productive tillers *Orseolia oryzae*-Gm6, Gm8 ^[84].
9. Sheath spot-Lesion oval with 0.5-2 cm long, 0.5-1 cm wide, the centre of lesion pale, cream with white reddish-brown margin, weakened culms lodge *Rhizotonia oryzae* Resistance gene not discovered ^[85].
10. Root rot-Cause black rot on roots *Lasiodiplodia theobromae* Resistance gene not discovered ^[86].

Conclusion

The above discussion on the sequencing strategies may help unravel various key features of the microorganisms and the diseases it causes to various crops. Also, this review may also help plant breeders and geneticists and botanists to discover key genes related to the resistance to diseases in rice. Overall, the new technologies discovered in relation to the next generation sequencing are key to further discoveries in relation to the genetic makeup of the cultivars or new varieties depicting new phenotypes for disease resistance in rice and also other crops as well.

References

1. Zhou X, Bai X, Xing Y. A Rice Genetic Improvement Boom by Next-generation Sequencing. *Curr Issues Mol Biol.* 2018;27(1):109-126.
2. Béné C, Barange M, Subasinghe R, Pinstруп-Andersen P, Merino G, Hemre GI, *et al.* Feeding 9 billion by 2050-Putting fish back on the menu. *Food Sec.* 2015;7:261-74.
3. Zhang H, Zhang Y, Xu W, Li R, Zhang D, Yang L. Development and performance evaluation of whole-genome sequencing with paired-end

- and mate-pair strategies in molecular characterization of GM crops: One GM rice 114-7-2 line as an example. *Food Chem Mol Sci.* 2022;4:100-061.
4. Patel N, Ferns BR, Nastouli E, Kozlakidis Z, Kellam P, Morris S, *et al.* Cost analysis of standard Sanger sequencing versus next generation sequencing in the ICONIC study. *Lancet*; 2016. p. 388.
 5. Head SR, Komori HK, LaMere SA, Whisenant T, Van Nieuwerburgh F, Salomon DR, *et al.* Library construction for next-generation sequencing: overviews and challenges. *Biotechniques.* 2014;56(2):61-77.
 6. Wheeler DA, Srinivasan M, Egholm M, Shen Y, Chen L, McGuire A, *et al.* The complete genome of an individual by massively parallel DNA sequencing. *Nature.* 2008;452(7189):872-6.
 7. MacConaill LE, Burns RT, Nag A, Coleman HA, Slevin MK, Giorda K, *et al.* Unique, dual-indexed sequencing adapters with UMIs effectively eliminate index cross-talk and significantly improve sensitivity of massively parallel sequencing. *BMC Genomics.* 2018;19:1-10.
 8. Kakoulidou I, Avramidou EV, Baránek M, Brunel-Muguet S, Farrona S, Johannes F, *et al.* Epigenetics for crop improvement in times of global change. *Biology.* 2021;10(8):766.
 9. Jain C, Rhie A, Hansen NF, Koren S, Phillippy AM. Long-read mapping to repetitive reference sequences using Winnowmap2. *Nat Methods.* 2022;19(6):705-10.
 10. Dohm JC, Peters P, Stralis-Pavese N, Himmelbauer H. Benchmarking of long-read correction methods. *NAR Genomics Bioinformatics*, 2020, 2(2).
 11. Zhang J, Chen LL, Sun S, Kudrna D, Copetti D, Li W, *et al.* Building two indica rice reference genomes with PacBio long-read and Illumina paired-end sequencing data. *Sci Data.* 2016;3(1):1-8.
 12. Rhoads A, Au KF. PacBio sequencing and its applications. *Genomics Proteomics Bioinformatics.* 2015;13(5):278-89.
 13. Jin H, Mo L, Pan L, Hou Q, Li C, Darima I, *et al.* Using PacBio sequencing to investigate the bacterial microbiota of traditional Buryatian cottage cheese and comparison with Italian and Kazakhstan artisanal cheeses. *J Dairy Sci.* 2018;101(8):6885-96.
 14. Harkess A, Huang K, Van Der Hulst R, Tissen B, Caplan JL, Koppula A, *et al.* Sex determination by two Y-linked genes in garden asparagus. *Plant Cell.* 2020;32(6):1790-6.

15. De Sena Brandine G, Smith AD. Falco: high-speed FastQC emulation for quality control of sequencing data. *F1000Res*; 2019. p. 8.
16. Patel RK, Jain M. NGS QC Toolkit: a toolkit for quality control of next generation sequencing data. *PLoS One*, 2012, 7(2).
17. Woldegiorgis ST, Wang S, He Y, Xu Z, Chen L, Tao H, *et al.* Rice stress-resistant SNP database. *Rice*. 2019;12(1):1-2.
18. Hsu SK, Tung CW. RNA-Seq analysis of diverse rice genotypes to identify the genes controlling coleoptile growth during submerged germination. *Front Plant Sci*. 2017;8:762.
19. Vaisvila R, Ponnaluri VC, Sun Z, Langhorst BW, Saleh L, Guan S, *et al.* Enzymatic methyl sequencing detects DNA methylation at single-base resolution from picograms of DNA. *Genome Res*. 2021;31(7):1280-9.
20. Hoppers A, Williams L, Ponnaluri VC, Sexton B, Saleh L, Campbell M, *et al.* Enzymatic methyl-seq: next generation methylomes. *J Biomol Tech.*; 2020, 31.
21. Chung PJ, Jung H, Do Choi Y, Kim JK. Genome-wide analyses of direct target genes of four rice NAC-domain transcription factors involved in drought tolerance. *BMC Genomics*. 2018;19(1):1-7.
22. Imchen M, Kumavath R, Vaz A, Góes-Neto A, Barh D, Ghosh P, *et al.* 16S rRNA gene amplicon based metagenomic signatures of rhizobiome community in rice field during various growth stages. *Front Microbiol*. 2019;10:2103.
23. Duitama J, Silva A, Sanabria Y, Cruz DF, Quintero C, Ballen C, *et al.* Whole genome sequencing of elite rice cultivars as a comprehensive information resource for marker assisted selection. *PLoS One*, 2015, 10(4).
24. Kishor DS, Seo J, Chin JH, Koh HJ. Evaluation of whole-genome sequence, genetic diversity, and agronomic traits of basmati rice (*Oryza sativa* L.). *Front Genet*. 2020;11:86.
25. Subudhi PK, Shankar R, Jain M. Whole genome sequence analysis of rice genotypes with contrasting response to salinity stress. *Sci Rep*. 2020;10(1):1-3.
26. Kou Y, Liao Y, Toivainen T, Lv Y, Tian X, Emerson JJ, *et al.* Evolutionary genomics of structural variation in Asian rice (*Oryza sativa*) domestication. *Mol. Biol. Evol*. 2020;37(12):3507-24.

27. Arikrit S, Wanchana S, Khanthong S, Saensuk C, Thianthavon T, Vanavichit A, *et al.* QTL-seq identifies cooked grain elongation QTLs near soluble starch synthase and starch branching enzymes in rice (*Oryza sativa* L.). *Sci Rep.* 2019;9(1):8328.
28. Akhter D, Qin R, Nath UK, Eshag J, Jin X, Shi C. A rice gene, OsPL, encoding a MYB family transcription factor confers anthocyanin synthesis, heat stress response and hormonal signalling. *Gene.* 2019;699:62-72.
29. Li X, Wu L, Wang J, Sun J, Xia X, Geng X, *et al.* Genome sequencing of rice subspecies and genetic analysis of recombinant lines reveals regional yield-and quality-associated loci. *BMC Biol.* 2018;16(1):1-2.
30. Arai-Kichise Y, Shiwa Y, Ebana K, Shibata-Hatta M, Yoshikawa H, Yano M, *et al.* Genome-wide DNA polymorphisms in seven rice cultivars of temperate and tropical japonica groups. *PLoS One.* 2014;9(1).
31. Wang X. Next-generation sequencing data analysis. CRC Press; 2016 Apr. Available from: <https://doi.org/10.1201/b19532>.
32. Balarezo-Cisneros LN, Timouma S, Hanak A, Currin A, Valle F, Delneri D, *et al.* High quality de novo genome assembly of non-conventional yeast *Kazachstania bulderi* a new potential low pH production host for biorefineries. *bioRxiv.* 2023:2023-2101.
33. Choi JY, Lye ZN, Groen SC, Dai X, Rughani P, Zaaier S, *et al.* Nanopore sequencing-based genome assembly and evolutionary genomics of circum-basmati rice. *Genome Biol.* 2020;21(1):1-27.
34. Bolser D, Staines DM, Pritchard E, Kersey P. Ensembl plants: integrating tools for visualizing, mining, and analyzing plant genomics data. In: *Plant Bioinformatics: Methods and Protocols.* 2016. p. 115-40.
35. Robinson JT, Thorvaldsdóttir H, Winckler W, Guttman M, Lander ES, Getz G, *et al.* Integrative genomics viewer. *Nat Biotechnol.* 2011;29(1):24-6.
36. Puchta M, Boczkowska M, Groszyk J. Low RIN Value for RNA-Seq Library Construction from Long-Term Stored Seeds: A Case Study of Barley Seeds. *Genes.* 2020;11(10):1190.
37. Phule AS, Barbadikar KM, Maganti SM, Segutuvell P, Subrahmanyam D, Babu MP, *et al.* RNA-seq reveals the involvement of key genes for aerobic adaptation in rice. *Sci Rep.* 2019;9(1):1-10.

38. Nair SS, Luu PL, Qu W, Maddugoda M, Huschtscha L, Reddel R, et al. Guidelines for whole genome bisulphite sequencing of intact and FFPE DNA on the Illumina HiSeq X Ten. *Epigenetics Chromatin*. 2018;11(1):1-20.
39. Kumar RS, Malathi S, Rao PJ. Screening of certain rice entries against Asian rice gall midge, *Orseolia oryzae* (Wood-Mason) in Warangal, Telangana. *J Entomol Zool Stud*. 2020;8(5):1888-93.
40. Bolger AM, Lohse M, Usadel B. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics*. 2014;30(15):2114-20.
41. Hölzer M, Marz M. De novo transcriptome assembly: A comprehensive cross-species comparison of short-read RNA-Seq assemblers. *Gigascience*. 2019;8(5).
42. Li H. Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM. *arXiv preprint*. 2013; arXiv:1303.3997.
43. Sang J, Zou D, Wang Z, Wang F, Zhang Y, Xia L, et al. IC4R-2.0: rice genome reannotation using massive RNA-seq data. *Genomics Proteomics Bioinformatics*. 2020;18(2):161-172.
44. Ouyang S, Zhu W, Hamilton J, Lin H, Campbell M, Childs K, et al. The TIGR rice genome annotation resource: improvements and new features. *Nucleic Acids Res*. 2007;35(1).
45. Barrero RA, Napier KR, Cunnington J, Liefing L, Keenan S, Frampton RA, et al. An internet-based bioinformatics toolkit for plant biosecurity diagnosis and surveillance of viruses and viroids. *BMC Bioinformatics*. 2017;18(1):1-2.
46. Krishnatreya DB, Baruah PM, Dowarah B, Chowrasia S, Mondal TK, Agarwala N. Genome-wide identification, evolutionary relationship and expression analysis of AGO, DCL and RDR family genes in tea. *Sci Rep*. 2021;11(1):1-8.
47. Yang H, Zhou C, Li G, Wang J, Gao P, Wang M, et al. Reference gene and small RNA data from multiple tissues of *Davidia involucrata* Baill. *Sci Data*. 2019;6(1):1-10.
48. Dong S, Zhang J, Sun D, Liu H, Yang Q, Wang H, et al. Identification of *Magnaporthe oryzae*-elicited rice novel miRNAs and their targets by miRNA and degradome sequencing. *Eur. J Plant Pathol*. 2018;151(3):629-647.

49. Evans C, Hardin J, Stoebe DM. Selecting between-sample RNA-Seq normalization methods from the perspective of their assumptions. *Brief Bioinform.* 2018;19(5):776-792.
50. Ren Y, Huang Z, Jiang H, Wang Z, Wu F, Xiong Y, *et al.* A heat stress responsive NAC transcription factor heterodimer plays key roles in rice grain filling. *J Exp Bot.* 2021;72(8):2947-2964.
51. Saha J, Giri K, Roy S. Identification and characterization of differentially expressed genes in the rice root following exogenous application of spermidine during salt stress. *Genomics.* 2020;112(6):4125-4136.
52. Li H, Li A, Shen W, Ye N, Wang G, Zhang J, *et al.* Global survey of alternative splicing in rice by direct RNA sequencing during reproductive development: landscape and genetic regulation. *Rice.* 2021;14(1):1-5.
53. Freese NH, Estrada AR, Blakley IC, Duan J, Loraine AE. Many rice genes are differentially spliced between roots and shoots but cytokinin has minimal effect on splicing. *Plant Direct.* 2019;3(5).
54. Mansueto L, Fuentes RR, Borja FN, Detras J, Abriol-Santos JM, Chebotarov D, *et al.* Rice SNP-seek database update: new SNPs, indels, and queries. *Nucleic Acids Res.* 2017;45(D1).
55. Kosugi S, Natsume S, Yoshida K, MacLean D, Cano L, Kamoun S, *et al.* Coval: improving alignment quality and variant calling accuracy for next-generation sequencing data. *PLoS One.* 2013;8(10).
56. Ghurye J, Pop M. Modern technologies and algorithms for scaffolding assembled genomes. *PLoS Comput Biol.* 2019;15(6).
57. Kishikawa T, Momozawa Y, Ozeki T, Mushiroda T, Inohara H, Kamatani Y, *et al.* Empirical evaluation of variant calling accuracy using ultra-deep whole-genome sequencing data. *Sci Rep.* 2019;9(1):1-10.
58. Yang G, Chen S, Chen L, Sun K, Huang C, Zhou D, *et al.* Development of a core SNP arrays based on the KASP method for molecular breeding of rice. *Rice.* 2019;12:1-8.
59. Coenye T. Do results obtained with RNA-sequencing require independent verification? *Biofilm.* 2021;100043.
60. Sanhueza D, Vizoso P, Balic I, Campos-Vargas R, Meneses C. Transcriptomic analysis of fruit stored under cold conditions using controlled atmosphere in *Prunus persica* cv. "Red Pearl". *Front Plant Sci.* 2015;6:788.

61. Pombo MA, Zheng Y, Fei Z, Martin GB, Rosli HG. Use of RNA-seq data to identify and validate RT-qPCR reference genes for studying the tomato-Pseudomonas pathosystem. *Sci Rep.* 2017;7(1):44905.
62. Tariq R, Ji Z, Wang C, Tang Y, Zou L, Sun H, *et al.* RNA-Seq analysis of gene expression changes triggered by *Xanthomonas oryzae* pv. *oryzae* in a susceptible rice genotype. *Rice.* 2019;12(1):1-4.
63. Pombo MA, Ramos RN, Zheng Y, Fei Z, Martin GB, Rosli HG, *et al.* Transcriptome-based identification and validation of reference genes for plant-bacteria interaction studies using *Nicotiana benthamiana*. *Sci Rep.* 2019;9(1):16-32.
64. Ayalew H, Tsang PW, Chu C, Wang J, Liu S, Chen C, *et al.* Comparison of TaqMan, KASP and rhAmp SNP genotyping platforms in hexaploid wheat. *PLoS One*; 2019, 14(5).
65. Makhoul M, Rambla C, Voss-Fels KP, Hickey LT, Snowdon RJ, Obermeier C, *et al.* Overcoming polyploidy pitfalls: a user guide for effective SNP conversion into KASP markers in wheat. *Theor Appl Genet.* 2020;133:2413-30.
66. Yang S, Yu W, Wei X, Wang Z, Zhao Y, Zhao X, *et al.* An extended KASP-SNP resource for molecular breeding in Chinese cabbage (*Brassica rapa* L. sp. *pekinensis*). *PLoS One*; 2020, 15(10).
67. Bhaskar Rao T, Chopperla R, Prathi NB, Balakrishnan M, Prakasam V, Laha GS, *et al.* A comprehensive gene expression profile of pectin degradation enzymes reveals the molecular events during cell wall degradation and pathogenesis of rice sheath blight pathogen *Rhizoctonia solani* AG1-IA. *J Fungi.* 2020;6(2):71.
68. Chopperla R, Mangrauthia SK, Bhaskar Rao T, Balakrishnan M, Balachandran SM, Prakasam V, *et al.* A comprehensive analysis of MicroRNAs expressed in susceptible and resistant rice cultivars during *Rhizoctonia solani* AG1-IA infection causing sheath blight disease. *Int J Mol Sci.* 2020;21(21):7974.
69. Molla KA, Karmakar S, Molla J, Bajaj P, Varshney RK, Datta SK, *et al.* Understanding sheath blight resistance in rice: the road behind and the road ahead. *Plant Biotechnol J.* 2020;18(4):895-915.
70. Richa K, Tiwari IM, Devanna BN, Botella JR, Sharma V, Sharma TR, *et al.* Novel chitinase gene LOC_Os11g47510 from indica rice Tetep provides enhanced resistance against sheath blight pathogen *Rhizoctonia solani* in rice. *Front Plant Sci.* 2017;8:596.

71. Devanna BN, Jain P, Solanke AU, Das A, Thakur S, Singh PK, *et al.* Understanding the dynamics of blast resistance in rice-*Magnaporthe oryzae* interactions. J Fungi. 2022;8(6):584.
72. Dixon RA, Barros J. Lignin biosynthesis: old roads revisited and new roads explored. Open Biol. 2019;9(12):190-215.
73. Volante A, Tondelli A, Aragona M, Valente MT, Biselli C, Desiderio F, *et al.* Identification of bakanae disease resistance loci in japonica rice through genome-wide association study. Rice. 2017;10:1-6.
74. Wang WM, Fan J, Jeyakumar JM. Rice false smut: an increasing threat to grain yield and quality. Protecting rice grains in the post-genomic era; 2019. p. 89-108.
75. Han Y, Li D, Yang J, Huang F, Sheng H, Sun W. Mapping quantitative trait loci for disease resistance to false smut of rice. Phytopathol Res. 2020;2:1-1.
76. Neelima T, Pazhanivelan S, Kodary AK, Vetriwar A, Jagdeeshwar R. International Conference on System of Crop Intensification for Climate-Smart Livelihood and Nutritional Security in: Extended Summaries; 2022.
77. Ghosh T, Pradhan C, Das AB. Control of stem-rot disease of rice caused by *Sclerotium oryzae* catt and its cellular defense mechanism-A review. Physiol. Mol. Plant Pathol. 2020;112:101-536.
78. Jiang N, Yan J, Liang Y, Shi Y, He Z, Wu Y, *et al.* Resistance genes and their interactions with bacterial blight/leaf streak pathogens (*Xanthomonas oryzae*) in rice (*Oryza sativa* L.)-an updated review. Rice. 2020;13(1):1-2.
79. Manojkumar HB, Deepak CA, Harinikumar KM, Rajanna MP, Chethana B. Molecular profiling of blast resistance genes and evaluation of leaf and neck blast disease reaction in rice. J Genet. 2020;99:1-10.
80. Kanyange L, Kamau J, Ombori O, Ndayiragije A, Muthini M. Genotyping for Blast (*Pyricularia oryzae*) Resistance Genes in F2 Population of Supa Aromatic Rice (*Oryza sativa* L.). Int. J Genomics; 2019.
81. Chen Z, Feng Z, Kang H, Zhao J, Chen T, Li Q, *et al.* Identification of new resistance loci against sheath blight disease in rice through genome-wide association study. Rice Sci. 2019;26(1):21-31.
82. Saha S, Garg R, Biswas A, Rai AB. Bacterial diseases of rice: an overview. J Pure Appl. Microbiol. 2015;9(1):725-36.

83. Chatterjee S, Mondal P. Management of rice yellow stem borer, *Scirpophaga incertulas* Walker using some biorational insecticides. J Biopestic. 2014;7:143.
84. Zhou H, Wang X, Mo Y, Li Y, Yan L, Li Z, *et al.* Genetic analysis and fine mapping of the gall midge resistance gene *Gm5* in rice (*Oryza sativa* L.). Theor Appl Genet. 2020;133:2021-33. doi: 10.1007/s00122-020-03575-3.
85. Lanoiselet VM, Cother EJ, Ash GJ. Aggregate sheath spot and sheath spot of rice. Crop Prot. 2007;26(6):799-808.
86. Claudius-Cole AO. *Lasiodiplodia theobromae* in the root rot disease complex of rice. J Rice Res. 2018;6(4):1-5.

Chapter - 2

Genetic Precision: CRISPR's Role in Plant Disease Management

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Chapter - 2

Genetic Precision: CRISPR's Role in Plant Disease Management

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Abstract

Crop biotic stresses have substantially led to threatening global security in many instances such as coffee rust in Ceylon, Bengal Famine, and many more which are emphasizing the importance of imparting resistant traits to the crops. The most efficient way to impart traits was through breeding. Concerns about classical breeding approaches have been raised as genetic gains using classical approaches are diminishing due to reduced genetic diversity in superior cultivars and issues surrounding linkage drag. Additionally, classical approaches are time- and resource-intensive. This presents a significant advantage for recombinant DNA technology in combating such issues. Transgenic approaches are highly regulated and contested by domestic and international consumers. CRISPR/Cas9 knockout mechanism has become increasingly popular in plant breeding and biological research due to its highly precise modifications, cost-effectiveness, and reduced regulations. Genomes of commercial crops such as rice, wheat, and tomato were precisely edited *via* RNA-guided endonuclease activity for disease resistance. Beyond genome editing, CRISPR/Cas can provide potential solutions to approaching plant disease management.

Keywords: CRISPR/Cas-9, genome editing, plant-pathogen interaction, biosafety

Introduction

Crop diseases, arising from pathogens such as fungi, bacteria, and viruses, pose a significant threat to global food security by reducing crop yields and quality. Currently, our primary approach to combat these diseases is through the use of chemical pesticides. While effective, this method raises environmental concerns due to pollution and harm to non-target organisms, and can lead to the development of pesticide-resistant pathogens. To achieve a more sustainable agriculture, there's a growing need for crops with innate

disease resistance, thereby reducing reliance on pesticides. Traditional breeding methods have been somewhat successful but are hindered by their time-consuming nature and imprecision, often resulting in the loss of desirable traits and introduction of unwanted ones over multiple generations of crossbreeding. To address these limitations, technologies such as genome editing offer a more efficient and precise means of enhancing plant disease resistance (Yin, K. and Qiu, J.L., 2019).

Genome editing, also known as genome editing with engineered nucleases (GEEN), involves the manipulation of DNA within a genome using engineered nucleases, often referred to as "molecular scissors". Unlike conventional genetic engineering, which typically involves the introduction of new genetic material, genome editing focuses on making precise alterations to existing genetic material. By directly targeting and modifying specific genes responsible for susceptibility to diseases, genome editing enables the creation of crop varieties that are inherently resistant to pathogens, with improved yields, without the need to introduce additional genes from donor varieties. Renowned for its precision and ability to target specific genes, Genome editing represents a powerful tool for genetic modification in agriculture. (Zafar *et al.*, 2020)

Background

The Clustered Regularly Interspaced Short Palindromic Repeat (CRISPR) and their associated protein (Cas-9) system is the most effective type of adaptive immunity in archaea and bacteria against viruses and other foreign genetic elements through targeted interference or cleavage of nucleic acids. CRISPR is a unique organization of short, partially repeated DNA sequences found in the genomes of prokaryotes. CRISPR was first discovered accidentally in 1987 by the Japanese scientist Ishino and his team. They found unusual repetitive palindromic DNA sequences interrupted by spacers in *Escherichia coli* while analysing a gene for alkaline phosphatase. In 1990 Francisco Mojica named CRISPR by identifying similar sequences in other prokaryotes. The CRISPR defence mechanism protects bacteria from repeated viral attacks *via* three basic stages: adaptation (Spacer acquisition), crRNA synthesis (expression), and target interference. CRISPR loci are an array of short repeated sequences found in chromosomal or plasmid DNA of prokaryotes. The Cas gene is usually found adjacent to CRISPR and codes for the nuclease protein (Cas protein) responsible to destroy or cleave viral nucleic acid.

It was in 2012 that Doudna, J, and Charpentier, E discovered CRISPR/Cas-9 could be used to edit any desired DNA by just providing the right template.(Gao, 2019) Since then, CRISPR/Cas-9 has become the most effective, efficient, and accurate method of genome editing tool in all living cells and utilised in many applied disciplines. Genome editing is a type of genetic engineering in which DNA is deliberately inserted, removed, or modified in living cells. Before the discovery of CRISPR/Cas-9, scientists relied on two gene-editing techniques using restriction enzymes, zinc finger nucleases (ZFN) and Transcription activator-like effector nucleases (TALENs) (Gaj, *et al.*, 2017). ZFN has a zinc finger DNA binding domain used to bind a specific target DNA sequence and a restriction endonuclease domain used to cleave the DNA at the target site. TALENs are also composed of a DNA binding domain and restriction domain like ZFN but their DNA binding domain has more potential target sequence than the ZFN gene-editing tool. In both cases, the difficulty of protein engineering, being expensive, and time-consuming were the major challenges for researchers and manufacturers. (Gaj, *et al.*, 2017)

The invention and application of CRISPR/Cas technology have revolutionised the field of genome editing and greatly contributed to the field of agriculture also. There are many examples to show that this technique has the potential to transform plant pathology research and disease management by providing a series of molecular tool kits to edit, modify, or detect plant pathogens DNAs and RNAs at an unprecedented pace and precision. (Puchta *et al.*, 2022)

Components of CRISPR/Cas-9

The CRISPR/Cas system can be divided into two classes based on the structure and functions of Cas proteins. The class I system consists of multi-subunit Cas- protein complexes and divided as type I, III, IV, while the class II systems utilise a single Cas-protein divided as type II, V and VI. The structure of the type II CRISPR/Cas-9 system is relatively simple and it is extensively used in genetic engineering. (Liu *et al.*, 2021). Guide RNA (gRNA) and CRISPR-associated (Cas-9) proteins are the two essential components in the CRISPR/Cas-9 system. The Cas-9 protein, the first Cas protein used in genome editing, was extracted from *Streptococcus pyogenes* (SpCas-9). It is a large (1368 amino acids) multi-domain DNA endonuclease responsible for cleaving the target DNA to form a double-stranded break and is called a genetic scissor. (Mei *et al.*, 2016). Cas-9 consists of two regions, called the recognition (REC) lobe and the nuclease (NUC) lobe. The REC lobe consists of REC1 and REC2 domains responsible for binding guide RNA,

whereas the NUC lobe is composed of RuvC, HNH, and Protospacer Adjacent Motif (PAM) interacting domains. The RuvC and HNH domains are used to cut each single-stranded DNA, while PAM interacting domain confers PAM specificity and is responsible for initiating binding to target DNA. (Yamano *et al.*, 2016). Guide RNA is made up of two parts, CRISPR RNA (crRNA) and trans-activating CRISPR RNA (tracrRNA). The crRNA is an 18-20 base pair in length that specifies the target DNA by pairing with the target sequence, whereas tracrRNA is a long stretch of loops that serve as a binding scaffold for Cas-9 nuclease. In prokaryotes, the guide RNA is used to target viral DNA, but in the gene-editing tool, it can be synthetically designed by combining crRNA and tracrRNA to form a single guide RNA (sgRNA) in order to target almost any gene sequence supposed to be edited (Mei *et al.*, 2016).

Mechanism of CRISPR/Cas-9

It involves three steps such as recognition, cleavage and repair.

Step 1: The designed sgRNA directs Cas-9 and recognizes the target sequence in the gene of interest through its 5' crRNA complementary base pair component. The Cas-9 protein remains inactive in the absence of sgRNA

Step 2: The Cas-9 nuclease makes double-stranded breaks (DSBs) at a site 3 base pair upstream to PAM. PAM sequence is a short (2-5 base-pair length) conserved DNA sequence downstream to the cut site and its size varies depending on the bacterial species. Cas-9 recognizes PAM at 5'-NGG-3' (N can be any nucleotide base), triggers local DNA melting followed by the formation of RNA-DNA hybrid, then activated for DNA cleavage. HNH domain cleaves the complementary strand while the RuvC domain cleaves the non-complementary strand of target DNA to produce predominantly blunt-ended DSB's.

Step 3: Double stranded break (DSB) is repaired by the host cellular machinery by using two mechanisms such as Non-homologous end joining (NHEJ) and Homology- directed repair (HDR).

There are many different vectors that can be used for genome editing in a variety of organisms. Some examples of these vectors include pRGE31, pRGE31, lentiCRISPR v2, eSPCas9, and others. These vectors contain all of the necessary sequences, such as promoters, selection markers, multiple cloning sites, restriction sites, and the coding sequence for the Cas9 protein (Xie and Yang 2013). These vectors have unique restriction sites that allow the guide RNA that has been generated for a particular gene to be cloned into them. This process is then followed by the transformation of plants *via Agrobacterium*-mediated transformation. Inside plant cells, the guide RNA

provides instructions to the Cas-9 enzyme, telling it to cleave the target sequence. This, in turn, initiates the DNA mending machinery, such as NHEJ. Due to the fact that these DNA mending systems are prone to errors, they cause frameshift mutations, which in turn lead to site-specific gene editing. These mutations are caused by insertions and deletions (Jiang *et al.*, 2013a).

Applications of CRISPR/Cas-9

Bacterial blight, caused by *Xanthomonas oryzae* pv. *oryzae* (Xoo), is a prevalent and destructive rice disease. Xoo pathogenicity depends on a specific class of virulence factors, called transcription activator-like (TAL) effectors, which mimic plant transcriptional activators. Rice Os8N3 (also known as OsSWEET11), which belongs to the Sugar Will Eventually be Exported Transporters (SWEET) family of sugar transporters, represents one of the susceptibility genes induced by TAL effectors (Yang *et al.*, 2006). The CRISPR/Cas9 system was highly efficient in generating Os8N3 gene editing in rice. It targets mutagenesis of Os8N3 in Kitaake, a Japonica rice cultivar. The mutant lines carrying edited Os8N3 displayed significantly enhanced resistance to Xoo with normal pollen development. This study provides a successful example of improving bacterial blast resistance using CRISPR/Cas9 technology. (Kim *et al.*, 2019).

The bacterium *Erwinia amylovora*, the causal agent of fire blight disease in apples, triggers its infection through the DspA/E effector which interacts with the apple susceptibility protein MdDIPM4. MdDIPM4 knockout has been produced in two *Malus × domestica* susceptible cultivars ‘Gala’ and ‘Golden Delicious’ using the CRISPR/Cas9 system delivered via *Agrobacterium tumefaciens*. This work demonstrates for the first time the development and application of a CRISPR/Cas9-FLP/FRT gene editing system for the production of edited apple plants carrying a minimal trace of exogenous DNA. (Pompili *et al.*, 2020).

A classical study was conducted to engineer the resistance against DNA viruses, beet severe curly top virus (BSCTV), and bean yellow dwarf virus (BeYDV), in *Nicotiana benthamiana* and *Arabidopsis thaliana* through CRISPR-Cas approach. BSCTV- and BeYDV-resistant plants were developed by the sgRNA and Cas9 constructs targeting the coding replication-associated protein (Rep protein) gene and the non-coding intergenic region (IR). Simultaneously, Ali *et al.* (2023) reported on the engineering of sgRNA molecules targeting the intergenic region (IR), coat protein (CP), and metal-binding site involved in protein conformation and DNA cleavage (RCRII) of the bipartite begomovirus, tomato yellow leaf curl virus (TYLCV), in the

model plant *N. benthamiana*. The mutant plants showed a reduction in the viral load along with enhanced viral resistance. The generated TYLCV-resistant plants were also reportedly resistant against a mixed infection of monopartite beet curly top virus (BCTV) and bipartite *Merremia mosaic virus* (MeMV).

Crop	Causal organism	Disease	Targeted gene	Reference
Wheat	<i>Blumeria graminis</i> f. sp. <i>tritici</i>	Powdery mildew disease	TaMLO-A1	Wang <i>et al.</i> (2014)
Rice	<i>Xanthomonas oryzae</i>	Bacterial Blight of Rice	OsSWEET11, OsSWEET14	Jiang <i>et al.</i> (2013b)
Arabidopsis	<i>Magnaporthe oryzae</i>	Rice Blast disease	OsERF922	Wang <i>et al.</i> (2016)
Mexican cotton	Begomovirus	Cotton leaf curl disease	CLCuD IR and Rep regions	Iqbal <i>et al.</i> (2016)
Cucumber	<i>Cucumber vein yellowing virus</i>	Ring spot disease, vein yellowing disease	eIF4E	Chandrasekaran <i>et al.</i> (2016)
Tobacco	<i>Bean yellow dwarf virus</i> (BeYDV)	Leaf thickening chlorosis, curling	BeYDV	Baltes <i>et al.</i> (2014)
Tobacco	<i>Tomato yellow leaf curl virus</i> , <i>Beet curly top virus</i>	Leaf curl disease	TYLCV-IR, RCA regions	Ali <i>et al.</i> (2015)

Challenges of CRISPR/Cas-9

Although CRISPR/Cas9 technology is a straightforward and reliable method for changing a plant's genome to increase its immunity, it is nevertheless accompanied with several difficulties. The ability to guide RNA to match sequences with places in the genome other than the target site gives CRISPR/Cas9 its intrinsic ability to create remote targets (Ran *et al.*, 2013). To reduce off-site targeting, the precision with which the Cas9 targets a desired sequence must be optimised (Majeed *et al.*, 2018). Additionally, modified Cas9 proteins are being created to identify other PAMs (Agudelo *et al.*, 2020). There have also been new Cas12a and Cas13a of CRISPR/Cas9 nucleases from different bacterial Type II systems that target DNA and single-stranded RNA respectively (Koonin *et al.*, 2017). Although they share some characteristics with Cas9, these two systems outperform Cas9 in specific situations, especially in plant disease by using different ways to cleave target nucleotides and process pre-crRNA (Langner *et al.*, 2018).

The occurrence of unintended changes (translocations, inversions, massive deletions, and insertions) as a result of the intricate endogenous

pathways that repair the double-stranded DNA by Cas nucleases is another drawback of the CRISPR/Cas9 system (Kosicki *et al.*, 2018). Furthermore, Cas9-induced DSBs can be harmful to cells, triggering cell-death pathways and lowering the efficiency of transformation and editing (Roy *et al.*, 2018). Nuclease-deficient Cas9 proteins have been created and joined to other proteins, such as deaminases and recombinases to accomplish base editing and site specific recombination, overcoming these DSB-related restrictions (Standage-Beier *et al.*, 2019). The necessity of the PAM sequence adjacent to the protospacer DNA, which is used by the Cas9 complex in conjunction with the complementary sgRNA region to recognize and cleave the target DNA sequence, is the main drawback of employing the original *Streptococcus pyogenes* CRISPR/Cas9 (SpCas9). The PAM sequence 5'-NGG-3', where N can be any of the four nucleotide bases, is recognized by the SpCas9 complex. Although most genomes have this three-base-pair region, its necessity restricts the genes that can be targeted, especially when trying to examine genes engaged in extremely specialised pathways of interest (Langner *et al.*, 2018). Furthermore, studies have demonstrated that CRISPR/Cas9 can identify different PAM sequences, which raises the possibility of off-target alterations (Zhang *et al.*, 2017).

Biosafety challenges

While CRISPR-based gene editing is a powerful tool for improving plants, it raises concerns and challenges related to biosafety (Mushtaq *et al.*, 2021). One major challenge is off-target effects, where unintended mutations occur in non-targeted genome regions. This issue can be reduced by using advanced CRISPR variants like Cas12b (C2c1), which offer higher specificity and accuracy. Another concern is the potential for unintended outcomes, including allergens or pollutants. Therefore, strict biosafety assessments are necessary to ensure the safety of genetically modified plants for human consumption and the environment. Several regulatory frameworks have been developed to address the biosafety issues associated with gene-edited plants. A comprehensive understanding of its molecular characteristics is essential before a genetically modified plant can be considered for market release. This involves analysing the specific genomic modifications introduced through CRISPR and genome editing techniques. Modern molecular biology tools such as next-generation sequencing and bioinformatic analysis detect and characterise these modifications, ensuring the accuracy and predictability of the genetic changes made. (Li *et al.*, 2017).

The CRISPR-Cas system has modified genetic engineering by providing a specific and efficient tool for manipulating gene sequences in multiple

organisms, representing a significant milestone in scientific advancement. However, its unparalleled potential for off-target outcomes engenders significant concern among biosafety experts, as inadvertent genomic loci may be cleaved and modified, leading to harmful consequences such as altered gene expression, disrupted cellular processes, and unintended genetic mutations that might cause pathological conditions. Mitigating the prospect of off-target effects is imperative for responsible research and development in this field. Researchers have been investigating advanced CRISPR variants, such as Cas12b, that offer superior specificity and efficacy in DNA targeting, minimising the likelihood of off-target cleavage and attenuating biosafety concerns. Cas12b is significantly smaller than Cas9 or Cas12a, making it easier to deliver into cells. A recent study suggested that Cas12b could be reprogrammed to target particular sequences with high specificity and efficiency in various plant cells. Cas12c, also known as C2c3, is a CRISPR variant that shares structural similarities with Cas12a. Cas12c has been used to edit the genomes of bacteria and mammalian cells with high precision and flexibility. The small size of Cas12c might also facilitate its delivery into cells. However, it is also susceptible to off-target effects, which pose significant biosafety concerns.

Future prospects in plant disease management

Advancements in CRISPR/Cas9 technology have revolutionised gene editing in plants, offering plant virologists, geneticists, and molecular biologists a powerful tool to enhance crop yields and bolster disease resistance. However, despite its transformative potential, there are still several imperfections that need addressing to optimise plant modification for the benefit of humanity. The expression levels of cas9 and gRNA play a pivotal role in the precision of CRISPR/Cas9 editing in plants, alongside factors like the sequence composition of the target site (such as GC content) and the secondary structure of target-gRNAs (Majeed *et al.*, 2018).

The emergence of CRISPR/Cas9 technology has significantly reshaped gene editing in plants, particularly in crop species. By delving into the fundamental biology of plant development and stress response, researchers can now more easily design elite and superior crops. CRISPR/Cas9 enables the precise removal of undesired genes from wild type species and targeted insertion of genes at specific sites, paving the way for the creation of customized plants (Arora and Narula, 2017). The development of the CRISPR/Cas9 RNP system has circumvented the reliance on the target cell's ability to translate Cas9 and its interaction with gRNA, thereby enhancing editing efficiency. Successful strategies employing CRISPR/Cas9 sequence-

specific nucleases have been deployed to combat rice blast disease (Wang *et al.*, 2016). Furthermore, the combination of cytidine deaminase enzyme and Cas9 has facilitated high-throughput correction of target codons in rice, demonstrating promising prospects for crop protection through genetic modification with negligible effects on human health or the environment, particularly amidst growing political and social pressures to curtail pesticide usage (Li *et al.*, 2017).

Literature Cited

1. Arora L, Narula A. Gene editing and crop improvement using CRISPR-Cas9 system. *Front Plant Sci.* 2017;8:296-675.
2. Ali A, Zafar MM, Farooq Z, Ahmed SR, Ijaz A, Anwar Z, Maozhi R. Breakthrough in CRISPR/Cas system: Current and future directions and challenges. *Biotechnol J.* 2023;18(8):220-0642.
3. Ali Z, Abul-Faraj A, Li L, Ghosh N, Piatek M, Mahjoub A, Mahfouz MM. Efficient virus-mediated genome editing in plants using the CRISPR/Cas9 system. *Mol Plant.* 2015;8(8):1288-1291.
4. Baltes NJ, Hummel AW, Konecna E, Cegan R, Bruns AN, Bisaro DM, *et al.* Conferring resistance to geminiviruses with the CRISPR-Cas prokaryotic immune system. *Nat Plants.* 2015;1(10):1-4.
5. Chandrasekaran J, Brumin M, Wolf D, Leibman D, Klap C, Pearlsman M, *et al.* Development of broad virus resistance in non-transgenic cucumber using CRISPR/Cas9 technology. *Mol. Plant Pathol.* 2016;17(7):1140-1153.
6. Iqbal Z, Sattar MN, Shafiq M. CRISPR/Cas9: a tool to circumscribe cotton leaf curl disease. *Front Plant Sci.* 2016;7:190-600.
7. Gaj T, Ojala DS, Ekman FK, Byrne LC, Limsirichai P, Schaffer DV. In vivo genome editing improves motor function and extends survival in a mouse model of ALS. *Sci Adv.* 2017;3(12):39-52.
8. Gao C. Precision plant breeding using genome editing technologies. *Transgenic Res.* 2019;28(2):53-55.
9. Jiang W, Bikard D, Cox D, Zhang F, Marraffini LA. CRISPR-assisted editing of bacterial genomes. *Nat Biotechnol.* 2013;31(3):233.
10. Kim D, Kim DE, Lee G, Cho SI, Kim JS. Genome-wide target specificity of CRISPR RNA-guided adenine base editors. *Nat Biotechnol.* 2019;37(4):430-435.

11. Kosicki M, Tomberg K, Bradley A. Repair of double-strand breaks induced by CRISPR-Cas9 leads to large deletions and complex rearrangements. *Nat Biotechnol.* 2018;36(8):765-771.
12. Koonin EV, Makarova KS. Evolutionary plasticity and functional versatility of CRISPR systems. *PLoS Biol.* 2022;20(1).
13. Langner T, Kamoun S, Belhaj K. CRISPR crops: plant genome editing toward disease resistance. *Annu Rev Phytopathol.* 2018;56:479-512.
14. Liu S, Huang M, Xu Y, Kang J, Ye S, Liu S, *et al.* CRISPR/Cas12a technology combined with RT-ERA for rapid and portable SARS-CoV-2 detection. *Virol Sin.* 2021;36(5):1083-1087.
15. Li J, Sun Y, Du J, Zhao Y, Xia L. Generation of targeted point mutations in rice by a modified CRISPR/Cas9 system. *Mol Plant.* 2017;10(3):526-529.
16. Majeed M, Soliman H, Kumar G, El-Matbouli M, Saleh M. Editing the genome of *Aphanomyces invadans* using CRISPR/Cas9. *Parasit Vectors.* 2018;11:1-10.
17. Mei Y, Wang Y, Chen H, Sun ZS, Ju XD. Recent progress in CRISPR/Cas9 technology. *J Genet Genomics.* 2016;43(2):63-75.
18. Mushtaq M, Dar AA, Basu U, Bhat BA, Mir RA, Vats S, *et al.* Integrating CRISPR-Cas and next generation sequencing in plant virology. *Front Genet.* 2021;12:735-489.
19. Puchta H, Jiang J, Wang K, Zhao Y. Updates on gene editing and its applications. *Plant Physiol.* 2022;188(4):1725-1730.
20. Pompili V, Dalla Costa L, Piazza S, Pindo M, Malnoy M. Reduced fire blight susceptibility in apple cultivars using a high-efficiency CRISPR/Cas9-FLP/FRT-based gene editing system. *Plant Biotechnol J.* 2020;18(3):845-858.
21. Roy B, Zhao J, Yang C, Luo W, Xiong T, Li Y, *et al.* CRISPR/Cascade 9-mediated genome editing-challenges and opportunities. *Front Genet.* 2018;9:240.
22. Ran FA, Hsu PD, Lin CY, Gootenberg JS, Konermann S, Trevino AE, *et al.* Double nicking by RNA-guided CRISPR Cas9 for enhanced genome editing specificity. *Cell.* 2013;154(6):1380-1389.
23. Standage-Beier K, Brookhouser N, Balachandran P, Zhang Q, Brafman DA, Wang X. RNA-guided recombinase-Cas9 fusion targets genomic DNA deletion and integration. *CRISPR J.* 2019;2(4):209-222.

24. Wang H, La Russa M, Qi LS. CRISPR/Cas9 in genome editing and beyond. *Annu Rev Biochem.* 2016;85:227-264.
25. Yin K, Qiu JL. Genome editing for plant disease resistance: applications and perspectives. *Philos Trans R Soc Lond B Biol Sci.* 2019;374(1767):2018-0322.
26. Yamano T, Zetsche B, Ishitani R, Zhang F, Nishimasu H, Nureki O. Structural basis for the canonical and non-canonical PAM recognition by CRISPR-Cpf1. *Mol Cell.* 2017;67(4):633-645.
27. Xie K, Yang Y. RNA-guided genome editing in plants using a CRISPR-Cas system. *Mol Plant.* 2013;6(6):1975-1983.
28. Zafar SA, Zaidi SS, Gaba Y, Singla-Pareek SL, Dhankher OP, Li X, *et al.* Engineering abiotic stress tolerance via CRISPR/Cas-mediated genome editing. *J Exp. Bot.* 2020;71(2):470-479.
29. Zhang JP, Li XL, Li GH, Chen W, Arakaki C, Botimer GD, *et al.* Efficient precise knockin with a double cut HDR donor after CRISPR/Cas9-mediated double-stranded DNA cleavage. *Genome Biol.* 2017;18:1-18.

Chapter - 3

Seed Borne Viral Diseases in India and its Seed Certification Programme

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Chapter - 3

Seed Borne Viral Diseases in India and its Seed Certification Programme

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Abstract

Seed plays a vital role in associating microorganisms, which prove hazardous for the seed itself, or new plant created from it. Among the microorganism the fungus, bacterium, nematode and virus are either pathogenic or weak parasite. The seed transmission of viral diseases plays a vital role in the spread and survival of many important plant viral diseases. Infected seed is probably the most important source of viruses and subviral pathogens in commercial crop production. Seed transmitted viruses are distributed into 24 groups. Among them 14 groups are in India and cause the diseases like SMV, CGMV, TLCLV, TSV, PMoV, LCV, IPCV, ZYMV, BCMV. Totally 231 plant viruses transmitted through seeds in all over India. The diseases are detected by ELISA, RT-PCR, Grow out test, Dry seed examination and rapid Immune assay method. Management of seed borne viruses is hard nut to crack. Among all PDM strategies Seed certification is the only way to produce virus free seeds. It's a legally sanctioned system for quality control of seed multiplication and production. Hitherto, Government of India had taken steps in framing Seed Act (1966), Seed rules (1968), Seed (control) Order (1983), National seed policy (2002), Plant quarantine order (2003), PPV Act (2001) not only protect the breeders, researchers but also a common farmer.

Keywords: Seed borne virus, molecular identification, plant legislation.

Introduction

“Seed carried the pathogen one place to another place to cause the disease incidence in plant”

-Paul Neergaard, 1977.

Seeds play a important role in associating microorganisms, which establish hazardous and cause economic impact. The associated

microorganism may be a fungus, bacterium, nematode or virus. It may be pathogenic or weak parasite. They may be associated internally or externally with the seed (or) as contamination as sclerotial bodies, fungal bodies, nematode galls, infected plant parts, soil particles etc. Mixed with seed. Seed borne diseases assumed a greater importance to seed industries. Seed borne pathogens results in seed rots, seedling decay, pre and post emergence mortalities, abnormalities, discoloration, reduced seed size and shrivelledness of seeds. The seed borne pathogens not only affect the market value but also affect the nutritive value of the products.

Seed borne pathogens transmit diseases between field, regions and countries through seed or other planting material. Seed borne pathogens can be spread from the seed and infect the new plant in several ways. Some of the seed borne pathogens like smut fungi in number of cereals and viruses like *pigeon pea sterility mosaic virus* cause heavy seed abortion resulting in 80-100% yield losses. Global losses in food production due to seed borne diseases are important negative factors in world agriculture. So seed borne diseases must be controlled.

Virus

Virus means poison. It's a very small micro-organism, non-cellular parasites of cells. According to Bawden (1953) virus is an obligate parasite which require living host for its multiplication. The size of the virus particles was less than 200nm. Virus is a submicroscopic entity, which is capable of being introduced into specific living cells and reproducing inside such cells only (Luria, 1953). Virus is a capsid encoding microorganism composed of protein and nucleic acid. The nucleic acid may be RNA or DNA, either single standard (ss) or double standard (ds).

Definition: Seed borne diseases

Any malfunctioning of host or plants those results from continuous irritation by a pathogenic agent which are carried by seed and transferred to the plants and leads to development of symptoms are called seed borne diseases.

Seed borne diseases may result

- Loss in germination.
- Discoloration and shriveling.
- Development of plant disease.
- Development of pathogen to new areas.

- Introduction of new strains (or) physiologic races of the pathogen along with new germplasm from other countries.
- Toxin produced in infected seeds etc.

Seed borne viral diseases

- Seed transmission plays a vital role in the spread and survival of the number of important plant viral diseases. Infected seed is probably the most important source of virus and subviral pathogens in commercial crop production. About 1/7th of the viruses transmitted through seed. Seed is effective mean of introducing virus into crop at early stages with number of infection and help to disperse across area and survival.
- The seedling raised from the randomly dispersed infected seeds serve as initial sources of virus inoculums or loci of infection from which secondary spread occurs within and outside the field by suitable vectors. LMV and BCMV have the least number of alternate host. Seed transmission is the one of the chief way of virus survival from one season to another season. Besides, being a source of inoculum, the seed also helps for its perpetuation over long periods. For e.g. BCMV seed persist for about 34 to 36 years.
- About 15-20% of plant viruses (~100) are seed transmitted can be carried on the seed coat (contaminant) or actually infect the seed (e.g., in the embryo)
- Frequency of transmission (% infected seeds) depends on virus-host interaction and time of plant is infected usually between 1-30%, but can reach 50-100%.

Smith (1951) reported 8 seed transmitted viral diseases, Crowley (1957) reported 20 seed transmitted virus in 40 sp, Fluton (1964) reported 36 seed transmitted virus, Bennett (1969) reported 47 seed transmitted viral diseases, Phatak (1974) reported 85 seed transmitted viruses, Agarwal and Sindair (1988) reported 156 seed transmitted viruses, Power and Flecker (2003) reported 131 seed transmitted viruses and Subramanya Sastry K (2013) reported 236 seed transmitted virus.

Seed transmitted viruses are distributed in to 24 virus groups

- They are, *Alfamovirus*, *Bromovirus*, *Capillovirus*, *Carlavirus*, *Carmovirus*, *Caulimovirus*, *Comovirus*, *Cryptovirus*, *Cucumovirus*, *Enamovirus*, *Fabavirus*, *Furovirus*, *Hordeivirus*, *Ilarvirus*, *Necrovirus*, *Nepovirus*, *Sobemovirus*, *Tobamovirus*, *Tobravirus*, *Tombusvirus*, *Potexvirus*, *Potyvirus*, *Tospovirus*, *Tymovirus* groups.

Among the 24 virus group 14 groups are repoted in India as they are underlined here.

Virus characteristics

Most of the virus particles are elongate (or) isometric (or) bacilliform (or) spherical.

Pecluvirus

Caused the disease Peanut clump virus, Rod shapde, 21nm dia, (+)-ss RNA, Genomic size is 5.9 kb; 4.5 kbs, Capsid protein and size is 23 kDa, Transmitted by fungus *Polymyxa graminis* and by seed. *Polymyxa graminis* also transmit the diseases like *barley yellow mosaic virus*, *soil borne wheat mosaic virus*.

Symptoms

- Stunted growth.
- Rosette appearance.
- Chlorotic and necrotic streaks on the leaf.

Ilarvirus

Caused the disease Tobacco streak virus. Quasi-isometric particles, size of the virus particles spherical to bacilliform, it’s a salt liable virus. The sub genomic RNA encoding it is required for virus replication. Transmitted by pollen and seeds.

Symptoms

- In early stage it produce a chlorotic streak symptoms on the leaves.
- In mid stage it produce a necrotic streak symptoms on the leaves.
- In pre mature stage it cause death of the plant.

Potyvirus

Caused the disease Potato virus Y, Bean yellow mosaic virus, Plum pox virus, Bean common mosaic virus, Papaya ring spot virus.

Particle size range from 680-900nm long and 11-13nm in diameter. Genomic size is 9.7kb with multiple copies of single protein capsid of 30-40 K Da. Transmitted by aphids and seeds.

Table 1: List of the seed borne viral diseases in India

Seed borne virus	Crop	References
<i>Soybean mosaic virus</i>	Soybean	Mandhare and Gawade, (2010)

<i>Cucumber green mottle mosaic virus</i>	Cucurbits	Nagendran <i>et al.</i> , (2017)
<i>Tomato yellow leaf curl virus</i>	Tomato	Eui-joon kil <i>et al.</i> , (2016)
<i>Tobacco mosaic virus</i> <i>Tomato mosaic virus</i>	Pepper	Sunilkumar and Prakash (2012)
<i>Tobacco streak virus</i>	Leguminous host	Vemana <i>et al.</i> , (2011)
<i>Tobacco streak virus</i>	Sunhemp, sunflower, leguminous	Lavakumar <i>et al.</i> , (2008)
<i>Peanut mottle potyvirus</i>	Peanut	Puttaraj <i>et al.</i> , (2001)
<i>Leaf crinkle virus</i>	Urdbean	Sharma <i>et al.</i> , (2015)
<i>Indian peanut clump virus</i>	Peanut, millet.	Reddy <i>et al.</i> , (1998)
<i>Zucchini yellow mosaic virus</i>	Gherkin	Anthony johnson <i>et al.</i> , (2013)
<i>Garlic virus X</i>	Garlic	Baranwal <i>et al.</i> , (2011)
<i>Bean common mosaic virus</i>	Cowpea	Arakere <i>et al.</i> , (2009)
<i>Mung bean yellow mosaic</i>	Blackgram	Vijayalakshmi <i>et al.</i> , (2015)
<i>Tobacco ringspot virus</i>	Gladiolus	Meenu Katoch <i>et al.</i> , 2003

Mode of entry of viruses

- Contamination of seeds.
- Outside of embryo.
- Inside embryo.

Contamination of seeds

The seeds from systemically infected plants might well carry virus as a surface contaminant. This would be true seeds extracted from fleshy or pulpy fruits. However, the virus transmitted to the next generation, it would be necessary for the virus to remain active on the surface of the seeds until they germinate and then to find a method of entering in the seedling in early stages of its development. Only few viruses that could qualify for transmission of this type of viral diseases.

For e.g.: *Tobacco mosaic virus* in Tomato.

Outside of embryo

Not all the virus particles are transmitted through seeds. For seed development carbohydrates are moved in to the seed as food reserve. At the time of carbohydrate movement the virus particles move along with them and reach the phloem in considerable quantities and connect with mother plant vascular system. Because movement of virus through the phloem is most effective introduction of seed borne viruses and invasion in different parenchymatus tissue.

For e.g.:

- *Sugarbeet curly top virus* transmitted through periplasm reported by Bennett (1936).
- BCMV in testa and endosperm reported by Crowley (1959).

Inside the embryo

- Direct infection.
- Indirect infection.

Indirect infection

Introduction of virus particle in to the embryosac by two way

1. Introduction of virus into the embryo sac by the male gametophyte.
2. Ovule invasion by virus from the mother plant.

1. Introduction of virus into the embryo sac by the male gametophyte

After the germination of the infected pollen grain, pollen tubes grows in the embryo sac after it matures it releases two infected male gametes, one male gamete unite with polar nuclei and infect the endosperm and another gamete is unite with egg cell and infect the embryo.

For e.g.: *Tobacco rattle virus* (TRV)

Alfa-Alfa mosaic virus (AMV)

2. Ovule invasion by virus from the mother plant

Ovule may be infected through virus invasion from adjacent cells of the mother plant in the early stages of ovule development or later ovule infection is to be found in the correlation between seed transmission and time of infection of the mother plant.

For e.g.: *Soybean mosaic virus* in soybean- embryo infected through ovules.

direct infection

The virus directly enter to the embryo sac and infect the embryo in early stage.

For e.g.: BCMV

Detection of seed borne viruses

Virus is an obligate parasites and they differ in their genetic makeup and required special techniques for detection. Seed borne nature of the virus is found by either raising seedlings from infected seeds and observing

symptoms or by using assay of seed extract.

- Dry seed examination.
- Growing on test.
- Indicator test.
- Serological methods.

Dry seed examination

Visual observation of seeds may reveal abnormalities like discolouration, shriveling, reduced seed size, staining and seed coat necrosis.

Growing on test

Growing on test is used to determine virus seed transmission. Seeds are planted in blotter paper, sand, soil, vermiculite or in other growth medium. Seedlings are examined at regular intervals for the virus symptoms. The test should be conducted under insect proof condition. Light and temperature conditions should be provided for optimum plant growth and symptom expression. This method is used to detect the BCMV, LMV, SMV. The reliability of this test depends upon symptom expression. The test is not reliable when district symptoms are not produced.

Indicator test

Viruses can be detected in seeds by assaying the extracts of different parts of seeds and seedlings raised from infected seeds on suitable indicator plants. Susceptible host which produce local lesions or systemic symptoms are used as indicator plants. This test is used to detect the BCMV in bean.

Serological method

The method is used for characterization and determining relationship between viruses. The test is based on the reaction between an antiserum containing specific antibody produced by infecting plants with a pure virus preparation and an antigen virus protein. The tests are specific since an antibody combines only with the antigen which contains similar grouping of amino acid sequence.

Elisa

It is also known as double antibody sandwich. It is a serological test in which the sensitivity of the antibody antigen reaction is increased by attaching an enzyme to one of the two reactant. It is useful for the identification of large number of plant samples. This method is used to detect the SMV, LMV. Its is relatively complex and expensive but it is the

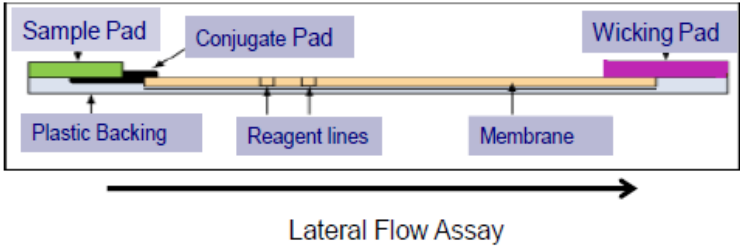
most sensitive and reliable test of all the plant virus serological test.

PCR

This technique is used to amplify a specific nucleic acid fragment that lies between two regions of known nucleotide sequence. PCR is not dependent on the presence of expressed proteins (antigens), but only on the presence of the target nucleic acid sequence.

Rapid diagnostic test; immunostrips

- First used widely for monitoring GMO.
- First for detecting plant viruses-TSWV.
- Supplied with buffer specific for each virus, or make you buffer.
- Place the strip along edge of bag, only to recommended depth, remove in 5-10minutes.
- Now available for TMV, CMV, ZYMV, PVY and several others for seed borne virus detection.



http://www.agdia.com/campusuite/modules/photo-gallery.cfm?box_id=268&grp_id=397&seq=1&kind=video&preview=yes

Yield losses due to seed borne viruses

The assessment of yield loss due to seed borne viral diseases difficult to estimate both quantitatively and qualitatively.

List of diseases	Yield loss	References
LMV	0.1-4%	Broadbent <i>et al.</i> ,
BCMV	50-60%	Lockhart <i>et al.</i> ,
SMV	35%	Ross
PMV	31-47%	Kuhn <i>et al.</i> ,
PSV	80%	Culp and Troutman
CMV	13-84%	Kaiser and mossahebi
CpBMV	43.5%	Sharma and Varma

To avoid these seed borne viral diseases seed certification is the one of the crucial way to produce a virus free seeds.

Seed certification programme

Introduction

In general, seed certification is a process designed to maintain and make available to the general public continuous supply of high quality seeds and propagating materials of notified kinds and varieties of crops, so grown and distributed to ensure the physical identity and genetic purity. Seed certification is a legally sanctioned system for quality control of seed multiplication and production.

In India the field evaluation of the seed crop and its certification started with the establishment of National Seeds Corporation in 1963. A legal status was given to seed certification with the enactment of first Indian Seed Act in the year 1966 and formulation of Seed Rules in 1968.

Objective of seed certification

The main objective of the Seed Certification is to ensure the acceptable standards of seed viability, vigour, purity and seed health.

Eligibility requirements for certification

Any variety to become eligible for seed certification should meet the following requirement:

- Field standards.
- Specific requirements.
- Seed Standards.

Field standards: It include the selection of site, isolation requirements, spacing, planting ratio, border rows etc.

Specific requirements: Presence of off types in any seed crop should be within the maximum permissible levels for certification.

Seed standards: The seed must be.

- Pure (minimum).
- Inert mater (maximum).
- Other crop seeds (maximum).
- Total weed seeds (maximum).
- Objectionable weed seeds (maximum).

- Germination (minimum).
- Moisture (maximum).

Seed certification agencies

Under the seed Act, 1966 the seed certification agencies was developed.

The main functions are

- It is an autonomous body.
- It should be involve itself in the production and marketing of seeds.
- The seed certification standards and procedures adapted by seed certification agency should be uniform, throughout the country.
- It is an long term objective there is no profit and no loss basis.
- Staff should be trained by seed certification agency.

Organizational set up of a seed certification agency

For certification various crop the seed certification agency go for planning, to check out acreage, area of operation, farm size etc..., planning should be done by directors and other staff. Each certification agency had own seed testing laboratory (or) get the seed sample from the seed testing laboratories.

Seed certification control measures

1. An Administrative check on the origin of the propagating material
2. Field Inspection.
3. Sample inspection.
4. Bulk Inspection.
5. Grow out test.

Phases of seed certification

- Verification of seed source.
- Field inspection.
- Seed standards.
- Tagging and sealing.

Seed export/import

Import the seeds under the regulation of PQO 2003. The seeds imported to ICAR for trial evaluation for one crop season. The import license is

granted by Directorate General of Foreign Trade (DGFT) only on the recommendation of Development Assistant Committee. If the seed is infected it is rejected. The report is submitted within 30 days, this effort is to be made to prevent the entry of exotic pest and diseases in to India.

Seed legislation programme in India

- Seed act (1966).
- Seed rules (1968).
- The seed (amendment) rules (1972).
- Seed (control) order (1983).
- Protection of plant varieties, (2003).
- Seed bill (2004).

Seed Act (1966): It is established in the year 1966, Dec 25. It is enacted by the parliament for the whole of India to regulate seeds. Under the seed Act central and state certification agencies are established. Karnataka is the first state to establish a seed certification agency under autonomous body and then all the state seed certification agency.

The seed (amendment) rules (1972): The rules should meet a prescribed standard provided that the standard is not lower than the minimum limits of germination and purity specified for the seed.

Seed (control) order (1983): If any person carrying on the business of selling, exporting and importing of seeds needs to obtain a license. The license is valid up to 3 years only. If any impurities are found the license is cancelled.

Production of plant varieties rules (2003): The rules provided detailed procedures while applying for protection, way of administering the national gene fund, procedure on application for compensation, procedure to alter the denomination of a registered variety, procedure for cancellation of certificate and all other procedures to be implemented as per the provision given in the PPV&FR Act, 2001.

Seed bill

- Regulate the quality seeds for sale, import and export.
- Facilitating production and supply the quality of seeds.
- Checking the poor quality seed.
- Increasing the private participation in the seed trade.

- Liberalizing import of seed and other planting material to the country.

The rules have been framed to implement various legislations given under seed act, 1966 and contain 11 sections

- Central seed committee.
- Central seed laboratory.
- Seed certification agency.
- Marketing or labeling.
- Requirements for certification.
- Certification of seeds.
- Seed analyst and seed inspectors.
- Sealing, dispatch, and analysis of samples.
- Miscellaneous.

Certified seed consist of following details

- Classes of seed.
- Certification no.
- Date and issue of certificate.
- Lot no.
- Variety.
- Name and address of the person etc.

Approved sources for quality seeds

- ISTA
- NBPGR
- NSC
- SSC

ISTA

- Established in the year 1918 in Netherland. Main aim is use of quality seed for maximum productivity.
- Seed certification laboratory accredited by ISTA.

Plant quarantine

“A legal restriction on movement of agricultural commodities for the purpose of exclusion, prevention or delay in the establishment of plants, pests and diseases in the area where they are not present”

Types of quarantine

- Domestic quarantine.
- Foreign quarantine.
- Embargo.

Effective production of seed against spread and establishment of seed-borne diseases in India

- Seed certification schemes.
- Equipment for seed-processing.
- Establishment of seed health testing laboratories.
- Establishment of research units and training centres.
- Establishment of an efficient quarantine control system.

All these efforts should be directed towards stepping up and prevention in defence of Indian agriculture against hazards from powerful forces of pathogens.

Grower's responsibility with respect to seed borne disease

- Knowledge of which diseases are seed borne.
- Careful scouting of crops to spot symptoms.
- Confirmation of diagnosis with extension or other diagnostic service if symptoms appear.
- Consultation with Seed Company to determine course of action.

Conclusion

Agriculture production purely based on the basic input, seed. If the seed is infected with virus it causes systemic infection and causes heavy devastating damage to the crop. To avoid this problem we have to go for quality seed production. Therefore, government of India framing several acts to prevent the seed from external pathogens. Quality of seed production only not to protect the researchers, breeders and also protect the farmers. Therefore, it is necessary that the information regarding seed borne diseases and seed legislation must reach the farmers and also make aware of their right.

References

1. Subramanya Sastry K. Seed-borne virus plant diseases; 2013. p. 1-73.
2. Sharma PN, Sharma A, Singh M. Effect of leaf crinkle disease on yield and quality of urdbean (*Vigna mungo* L. Hepper) in Himachal Pradesh. Himachal J Agric Res. 2015;41(1):80-82.
3. Puttaraju HR, Prakash HS, Shetty HS. Detection of peanut mottle potyvirus in leaf and seed of peanut and its effect on yield. Indian Phytopathol. 2001;54(4):479-80.
4. Bennet CW. Seed transmission of plant viruses; 1956. p. 221-58.
5. Kodhandaraman SV, Devadason A, Ganesan MV. Seed borne nature of Begomovirus, Mung borne yellow mosaic virus in Blackgram. Appl. Microbiol Biotechnol.; 2015.
6. Arakere C, Shankar U, Chandra C, Nayaka B, Bhuvanendra H, Kumar S, *et al.* Detection and identification of the blackeye cowpea mosaic strain of cowpea mosaic virus in seeds of cowpea from southern India. Phytoparasitica. 2009;37:283-93.
7. Johnson AM, Vidya S, Srinivasulu M, Mandal B, Saigopal DV. First report of Zucchini yellow mosaic virus infecting gherkin (*Cucumis anguira*) in India. Indian J Virol. 2013;24(2):289-90.
8. Sharma PN, Sharma V, Sharma A, Rajput K, Sharma SK. Identification and molecular characterization of Bean yellow mosaic virus infecting French bean in Himachal Pradesh. Virus Dis. 2015;26(4):315-8.
9. Vemana K, Jain RK. New experimental hosts of Tobacco streak virus and absence of true seed transmission in leguminous hosts. Indian J Virol. 2010;21(2):117-27.
10. Kumar L, Madhavi J, Waliyar F. Emergence and spread of Tobacco streak virus menace in India and control strategies. Indian J Plant Prot. 2008;36(1):1-8.
11. Mandhare VK, Gawade SB. Effect of seed borne soybean mosaic virus infection on quality and yield parameters in soybean. Legume Res. 2010;33(1):43-9.

Chapter - 4

Global and Regional Perspectives on Mango Cultivation and Disease Dynamics: A Comprehensive Analysis

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Chapter - 4

Global and Regional Perspectives on Mango Cultivation and Disease Dynamics: A Comprehensive Analysis

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Abstract

Mangoes (*Mangifera indica* L.), renowned as the "King of Fruits", have been cultivated since 4000 BC, particularly in the Himalayan region of South Asia. The fruit's exceptional nutritional content, including significant amounts of vitamins A and C, carbohydrates, and minerals, makes it a staple in various culinary applications. However, mango cultivation faces substantial challenges due to diseases like anthracnose, caused by fungal pathogen *Colletotrichum gloeosporioides*. This disease affects mangoes globally, leading to significant agricultural losses. It is responsible for economic losses, with approximately 25% to 30% of mango production being affected by anthracnose-related diseases. The disease thrives in hot and humid climates, making it a common occurrence both before and after harvest. This comprehensive analysis explores the global and regional perspectives on mango cultivation, identifying major growing belts and varieties in India, hybrid species developed to enhance productivity, and the causes of poor productivity. Additionally, it delves into the epidemiology and transmission of anthracnose, examining the disease's impact on mango production worldwide. The study underscores the necessity for effective disease management strategies to sustain and enhance mango cultivation globally. This chapter offers current insights into the mechanisms of disease spread, the biological aspects of infection, and different host resistance varieties of mango.

Keywords: *Colletotrichum gloeosporioides*, epidemiology, disease cycle, anthracnose

Introduction

The fundamental components required for the formation of human civilization are food, fibre, and raw materials. These essential elements are

entirely met through agricultural practices. Since 4000 BC, the cultivation of mangoes has been observed in the Himalayan region of South Asia, particularly along the Bay of Bengal, Eastern India, Burma, and the Andaman Islands. (Yadav & Singh, 2017). The Anacardiaceae family, particularly mangoes, are known as the most coveted fruits globally due to their rich energy sources. Mango’s nutritional content, delicious flavour, exquisite taste, and vibrant colour distinguish it as the ultimate & King of the Fruit; (Shailendra Rajan *et al.*, 2021). Mango is not only rich in nutritional value and flavour but also contains significant amounts of vitamin A, vitamin C, carbohydrates, and minerals that are readily absorbed by the body (Lebaka *et al.*, 2021). The pulp of the mango fruit finds usage in creating beverages, mango nectar, concentrated fruit syrup, or enhancing the taste of ice cream, sorbets, and a variety of culinary products (Owino *et al.*, 2021).

Various diseases can impact mango cultivation throughout every phase of growth and development, spanning from nursery plants to fruits during storage and transportation (Muid *et al.*, 2003). Mangoes are affected by numerous fungal afflictions, including anthracnose, rot diseases (root, stem, penicillium, black, mutor, petaloid, Macrophoma, etc.), and powdery mildew, which impact mangoes, resulting in significant agricultural losses. (Ploetz.*et al.*, 2001).

Mango growing belts in India

Mangos are grown in nearly every Indian state. The following lists the growing belts by different states:

Table 1: Major Mango growing belts in India (Yadav 2015)

State	Growing belts
Andhra Pradesh	East and West Godavari, Krishna, Srikakulam, Chittoor Adilabad, Khamman, Vijaynagar, and Vishakhapatnam
Chhattisgarh	Bastar, Raipur, and Jabalpur
Gujarat	Surat, Junagarh, Bhavnagar, Valsad, Mehsana, and Khera
Haryana	Kurushetra and Karnal
Jammu & Kashmir	Udhampur, Kathwa, and Jammu
Jharkhand	Godda, Sahibganj, Dumka, Hazaribagh, Sindega, Ranchi, and Gumla
Karnataka	Bangalore, Tumkur, Kagu, Kolar
Kerala	Trissur, Malappuram, Kannur, and Palakkad
Madhya Pradesh	Jabalpur, Katn Balagha, Rajgari, Bastar, Ramnandgaon, Satna, Durg, and Bilaspur
Maharashtra	Raigarh, Sindhudurg, and Ratnagiri

Orissa	Gunpur, Malkanpur, Rayagada, Sonepur, Bolangir, Gajapati, Koraput, Dhenkanal, Ganjam, and Puri
Punjab	Ropar, Hoshiarpur, and Gurdaspur
Tamil Nadu	Madurai, Dharmapuri, Vellore, Tiruvallur, Theni
Uttaranchal	Almora, Nainital, Bageshwar, Udham Singh Nagar, Dehradun, and Haridwar
Uttar Pradesh	Varanasi, Saharanpur, Faizabad, Lucknow, and Bulandshahar
West Bengal	Nadia, Malda, and Murshidabad

Major Mango-growing varieties

Approximately 1,500 different mango types, including 1,000 commercial cultivars, are grown in India. Every major mango variety has a distinct flavour and taste.

According to the prevalence of various kinds in various states, they can be categorized as follows:

Table 2: Major Mango growing varieties in different states of India. (*MANGO*, n.d.), (ABC Fruits. 2022)

State	Varieties grown
Andhra Pradesh	Himayuddin, Suvernakhe, Neelum, Totapuri, Cherukurasam, Allumpur-Baneshan, Banganapalli, Bangalora, and Neelum
Bihar	Zardalu, Langra, Chausa, Dashehari, Fazli, Bathua, Bombai, Himsagar, Kishen Bhog, Sukul, GulabKhas
Goa	Mankurad and Fernandin
Gujarat	Kesar, Alphonso, Vanraj, Jamadar, Totapuri, Neelum, Dashehari, Langra, Rajapuri, and Vanraj
Haryana	Dashehari, Fazli, Chausa, Sarauli, and Langra
Himachal Pradesh	Langra, Chausa, and Dashehari
Jharkhand	Jardalu, Himsagar, Chausa, Gulabkhas, Amrapali, Mallika, Bombai, Langra
Karnataka	Alphonso, Pairi, Baganapalli, Neelum, Mulgoa, Bangalora, Totapuri
Kerala	Mundappa, Pairi, and Olour
Madhya Pradesh	Alphonso, Amrapali, Mallika, Bombay Green, Langra, Sunderja, Dashehari, Fazli, and Neelum
Maharashtra	Alphonso, Mankurad, Mulgoa, Pairi, Kesar, Gulabi, Vanraj, and Rajapuri
Orissa	Mallika, Amrapalli, Suvarnakhe, Neelum, Baneshan, and Langra
Punjab	Langra, Chausa, Malda, and Dashehari
Rajasthan	Langra, Chausa, Dashehari, and Bombay Green
Tamil Nadu	Banganapalli, Bangalora, Alphonso, Mulgoa, Neelum, Rumani, and

	Totapuri
Uttar Pradesh	Bombay Green, Chausa, Fazli, Safeda, Dashehari, Langra, and Amrapalli
West Bengal	Amrapali, Mallika, Fazli, Gulabkhas, Langra, Himsagar, Bombai, and Kishen Bhog
Chennai	Rumani
Uttarakhand	Bombay Green, Amrapali, Mallika, Chausa, Dashehari, and Ramkela
Chhattisgarh	Kesar, Langra, and Alphonso
Jammu & Kashmir	Malta, Fazli, Dasehari, and Amrapali

Hybrid species of Mango Plant

Genetic manipulation is used to create several hybrid plants, such as Amrapalli, Mallika Ratna, and many more, with the goal of increasing mango production and obtaining desirable characteristics. The major mango types names and different institutions names are listed below:

Table 3: Major Hybrid species of Mango Plant (Rathour *et al.* 2023), (Dinesh *et al.*, 2016)

Hybrids Name	Hybrid Species	Name Of Institution / University
Amrapalli	Dashehari x Neelum	IARI, New Delhi
Mallika	Neelum x Dashehari	IARI, New Delhi
Arka Aruna	Banganapalli x Alphonso	Indian Institute of Horticultural Research , Bangalore
Arka Puneet	Alphonso x Banganapalli	Indian Institute of Horticultural Research , Bangalore
Arka Neelkiran	Alphonso x Neelum	Indian Institute of Horticultural Research , Bangalore
Arka Anmol	Alphonso x Janardhan Pasand	Indian Institute of Horticultural Research , Bangalore
Ratna	Neelum x Alphonso	Fruit Research Station, Vengurla
Sindhu	Ratna x Alphonso	Fruit Research Station, Vengurla
Au Rumani	Rumani x Mulgoa	Fruit Research Station, Kodur
Manjeera	Rumani x Neelum	Tamil Nadu Agricultural University, Colmbatore
PKM 1	Chinnasuvernarekha x Neelum	Horticultural Research Station, Periyakulam
Al Fazli	Alphonso x Fazli	Fruit Research Station, Sabour
PKM 2	Neelum x Alphonso	Horticultural Research Station, Periyakulam
Neeleshwari	Neelum x Dashehari	Agricultural Experimental Station, NAU, Paria, Gujarat

Neeleshan	Neelam x Baneshan	Fruit Research Station, Sangareddy, Andhra Pradesh
Neeludin	Neelum x Himayuddin	Fruit Research Station, Sangareddy, Andhra Pradesh
Sonpari	Alphonso x Baneshan	Agricultural Experimental Station, Navasari , Agricultural University, Paria, Gujarat
Neelgoa	Neelum x Mulgoa	Fruit Research Station, Sangareddy, Andhra Pradesh
Arunika	Amrapali x Vanraj	Central Institute For Subtropical Horticulture, Lucknow
Sabri	Gulabkhas x Bombai	Bihar Agriculture University, Sabour, and Bhagalpur
Sunder Langra	Sardar Pasand x Langra	Bihar Agriculture University, Sabour, and Bhagalpur
Jawahar	Gulabkhas x Mahmood Bhar	Bihar Agriculture University, Sabour
Al Fazli	Alphonso x Fazli	Fruit Research Station, Sabour
Mahmood Bahar	Bombay Green x Kalapadi	Bihar Agriculture University, Bhagalpur
Neelphonso	Neelam x Alphonso	Agricultural Experimental Station, Navasari, Agricultural University, Paria, Gujarat
KMH 1	Cherukurasam x Khader	Fruit Research Station, Kodur

Causes of poor mango productivity

The losses of mango productivity depends on various factors such as Senility of mango trees, Poorly managed high-density orchards, Climatic change, Soil related constraints, nutritional constraints, and abiotic stresses (Ganeshamurthy *et al.* 2019), (Niaz AHMED *et al.* 2021)

Numerous diseases and pests affecting the various varieties of mangos are one of the main causes, which eventually cause catastrophic losses to the mango industry worldwide. In which a common disease is anthracnose (J.I. Hormaza *et al.* 2023).

Worldwide distribution of disease

Globally, India was a leading producer of mango, accounting for nearly 45% of the market with a production of around 25,115 tons. China closely followed, also having a 45% hold. Other nations collectively contributed to 5% or fewer of worldwide market share as of 2021. Within India, Uttar Pradesh is a significant contributor, producing around 4,782 tons of mango, which constitutes 24% of the national market hold. However, the production rate experiencing a decline due to various diseases, including Anthracnose, which is widespread in regions like Italy, Ethiopia, South Africa, and others

(APEDA *et al.*2007). As per the study by Dofuor AK *et al.* in 2023, there has been a considerable lost in output due to this disease - 30% in Ghana's districts, 39% in India, 50.28% in Gondunglegi, Indonesia, 25-30% in Bangladesh, 62.8% in Thailand, and 30-60% annual losses in China. In India, Himachal Pradesh has experienced significant yield losses of about 50.28%. Those factors have led to a decrease in mango production (Dofuor AK *et al.* 2023), (Yadav *et al.*, 2016).

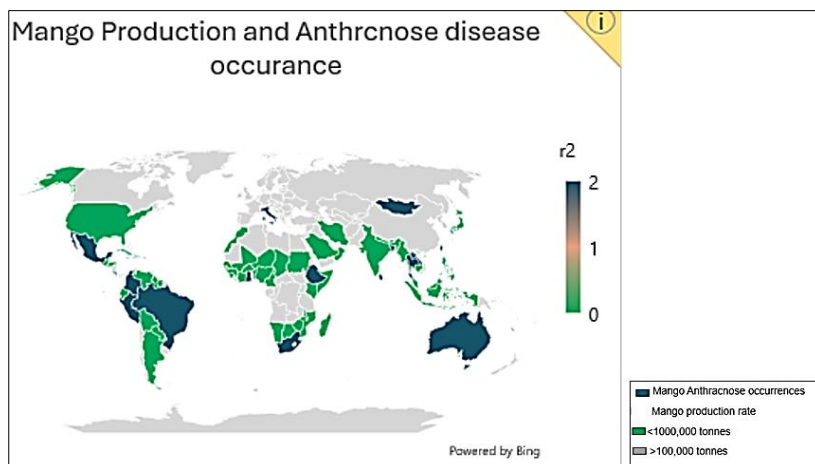


Fig 1: Mango Production and Anthracnose disease occurrence Taxonomy of *C. gloeosporioides* (Gautam *et al.*2014)

- Kingdom: Fungi
- Division: Ascomycota
- Class: Sordariomycetes
- Order: Phyllachorales
- Family: Phyllachoraceae
- Genus: Colletotrichum
- Species: *gloeosporioides*
- Scientific Name: *C. gloeosporioides* (Penz.) Penz. & Sacc.
- Teleomorph: Glomerella cingulata (Stoneman) Spauld. & H. Schrenk.

Epidemiology of anthracnose diseases

The disease cycle is divided into two phases (Uddin *et al.*, 2018).

Bio-trophic phases (infection starting phases) (Taïbi *et al.*, 2020).

- 1) **Dissemination:** The journey begins with the conidia, or spores, of the *Colletotrichum gloeosporioides* pathogen. Like tiny travellers, they are passively dispersed by the splash of rain or the flow of irrigation water.
- 2) **Inoculation:** These pathogenic agents settle on their new potential infection sites like leaves, panicles, and branch terminals.
- 3) **Infection and pathogen development:** After the spores have germinated, they begin a covert operation by penetrating the outer layer and surface of the immature fruits and developing tissues. The presence of infection is only noticeable once the spores break through the outer layer of ripened fruits and remain dormant until the climacteric fruits start to ripen.

Necrotrophic phases (infected cell killing phase)

- i) **Symptom and disease development:** The quiet invasion is promptly noticeable as black and sunken lesions rapidly spread across the affected areas of plants and fruits.
- ii) **Pathogen reproduction:** During rainy or humid weather, within the symptomatic tissue, the pathogen undergoes significant development. It produces sticky masses of conidia in acervuli, which are the fruiting bodies that signify the pathogen's growth. As the fungus continues to multiply, various diseases may potentially emerge throughout the season.
- iii) **Pathogen survival:** The fungus *Colletotrichum gloeosporioides* demonstrates remarkable durability, persisting through different seasons on impacted and leafless branch tips as well as fully grown leaves. During unfavourable conditions, it retreats into plant debris such as lifeless branches, stripped leaves, and neighbouring tree varieties, remaining inactive until the next opportunity for infection arises.
- iv) **Transmission:** In the end, spores form on the surface of infected tissue and then spread through insects, air currents, or water splashes, starting a new cycle of infection (Al Adawi. *et al.* 2006).

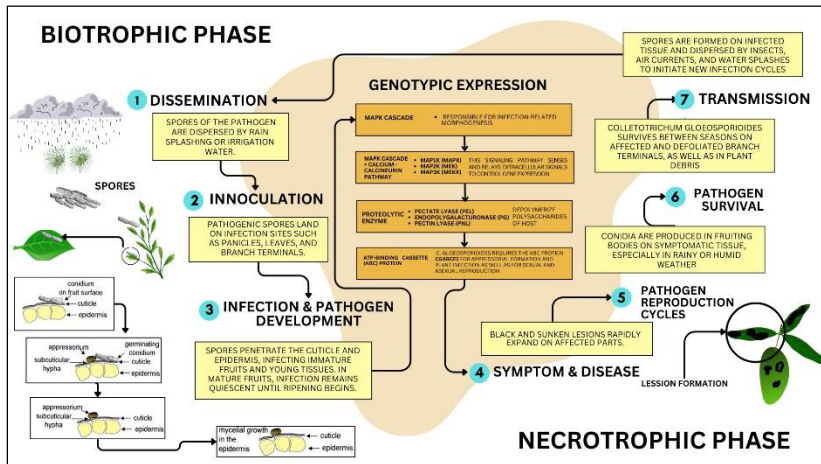


Fig 2: Two phases of Anthracnose disease cycle of Mango: Bio-trophic phases (Infection starting phases) and Necrotrophic phases (infected cell killing phase)

Mode of infection and transmission of *C.gloesporioides*

Infection

The infection mode of *C.gloesporioides* is hemi-biotrophic, involving sequential phases of biotrophy followed by necrotrophy. The process proceeds through the following stages.

- Initial interaction:** Initially, pathogens establish interaction with the host by developing a melanized appressorium, followed by penetration of the host cuticle.
- Biotrophic phase:** Following penetration, infection vesicles and primary hyphae develop. These structures, similar to haustoria observed in powdery mildews and rust fungi, operate without causing harm to the host. This phase of infection is recognized as the biotrophic stage (Munch et al., 2008).
- Necrotrophic phase:** Spore generation, sclerotia development, host plant concealment, ground sheltering, and migration to another sensitive host plant in the event of unfavorable conditions are some of the techniques (Dofuor AK et al., 2023).

Transmission

The stages of spore transmission occur as follows.

- i) **Spore transmission:** As previously discussed, spores can be carried by water, wind, or animals to infect susceptible hosts, facilitating the spread of the disease across relatively short distances.
- ii) **Seasonal dormancy:** In autumn and winter, when the majority of plants shed leaves and enter a dormant state, plant pathogenic fungi also employ corresponding dormancy strategies until the arrival of spring.
- iii) **Survival strategies:** The methods include spore production, sclerotia development, hiding inside the host plant, looking for cover on the ground, and, in the event of unfavourable conditions, shifting to another active host plant. (Akpanah D *et al.*, 2023).

Etymology of disease

Spore production is favoured by high humidity, frequent rainfall, and temperatures ranging from 24-32 °C, with a pH level between 5.8-6.5. Throughout mango production cycles, these conditions facilitated the development of anthracnose in leaves and panicles. *Colletotrichum gloeosporioides* typically stays inactive during the dry season, especially when temperatures are extreme, humidity is low, and there is ample sunlight (Kulshrestha *et al.*, 2015).

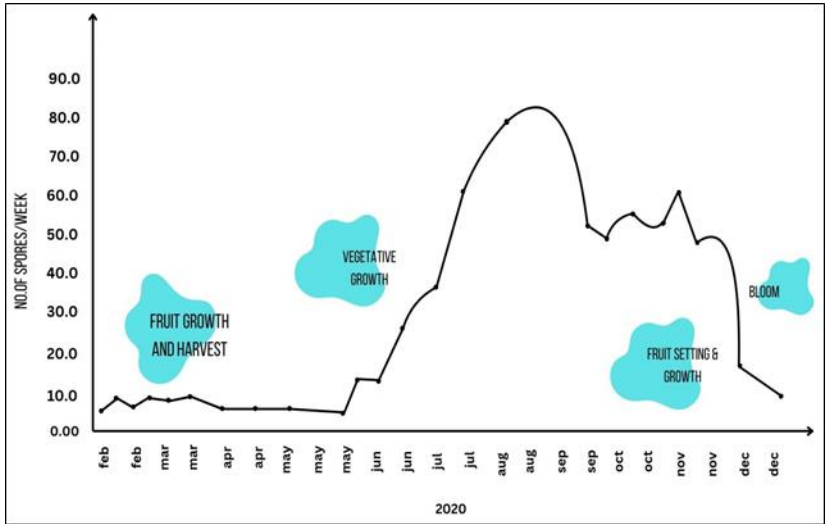


Fig 3: Graph showing different phases of *Colletotrichum gloeosporioides*

Symptomatology

The anthracnose fungus, a master of disguise, can produce what are termed latent infections. This implies that the fungus can stealthily infiltrate green fruit, entering a remaining dormant until the fruit ripens. As the fruit undergoes physiological changes linked to ripening, influenced by the anthracnose fungus awakens, leading to the development of lesions and subsequent spoilage of the fruit.

Leaves

Leaves bear the brunt of the infection, with spots commonly forming towards the margins. These spots, a palette of tan to dark brown hues, often bordered by a darker outline, can scatter across the leaf surface. Under the influence of damp conditions, these spots can enlarge to form extensive dead areas (A.K. *et al.*, 2008).

Fruits

The mature fruits, affected by anthracnose, exhibit depressed, conspicuous, dark brown to black decay spots, resulting in early fruit shedding. Rainfall during flowering and fruiting can cause extensive damage to flowers and young fruits due to anthracnose, resulting in significant losses of up to 35% of the harvested fruit. (Martínez *et al.*, 2009).

Twig

The twigs are not spared either. Severe, elongated, blackened lesions form on them, leading to dieback at the tips (A.K. *et al.*, 2008).

Stem

Stems too are susceptible, developing a typical black colour. The pathogen produces abundant spores that predominantly cover the most deteriorated areas of infection. (Nelson *et al.*, 2008)

Flower

The flowers, once vibrant, now bear black and irregular-shaped spots that grow and merge, leading to the demise of the flower panicles (A.K. *et al.*, 2008).



Fig 4: Symptoms of Anthracnose diseases on different parts of Mango tree

- a) **Leaf:** Palette of tan to dark brown hues.
- b) **Fruit:** Dark brown to black decay spots.
- c) **Twigs:** Severe, elongated, blackened lesions.
- d) **Stem:** Typical black and brownish colour.
- e) **Flower:** Black and irregular shaped spots.

Survivability percentage on different parts of mango plant

Based on a research study examining the prevalence and intensity of a disease affecting mango trees' leaves, it was found that the average occurrence of the disease on leaves, panicles, young fruits and other parts was 76%, 71%, 68% and 1% respectively. (Adunga G *et al.*, 2016)

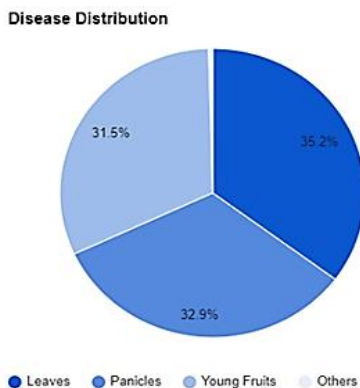


Fig 5: Showing Survivability percentage on different parts of mango plant

Host resistance

Mango anthracnose disease (MAD) is a highly damaging condition affecting mangoes, leading to substantial yield reductions in unattended plantations. Every commercially grown mango variety is vulnerable to MAD, and there is currently no universally resistant cultivar available. The disease can spread extensively before detection because it remains hidden during a prolonged latent period (Kwasi Asante *et al.*, 2023). Different mango cultivars exhibit different levels of susceptibility to anthracnose disease (Onyeani *et al.*, 2017). Some of them are as follows.

Table 1: Susceptibility of different cultivars of mango towards anthracnose

Mango Cultivar	Susceptibility to Anthracnose	Characteristics	References
Alphonso (Hapus)	Moderately susceptible	One of the most famous and prized Indian mango varieties.	Nallamani M <i>et al.</i> , 2023
Banganapalli	Moderately susceptible	Large, oblong fruit with a sweet and tangy flavor.	Boateng BO <i>et al.</i> , 2023
Chaunsa	Moderately susceptible	A popular North Indian variety known for its sweet taste.	Boateng BO <i>et al.</i> , 2023
Dasheri	Moderately susceptible	Small to medium-sized mangoes, sweet and fragrant.	Boateng BO <i>et al.</i> , 2023
Langra	Moderately susceptible	Medium-sized, sweet, and aromatic mangoes.	Boateng BO <i>et al.</i> , 2023
Kesar	Moderately susceptible	Known for its saffron-like aroma and sweet flavor.	Boateng BO <i>et al.</i> , 2023
Totapuri	Highly susceptible	Distinctly shaped, widely used for processing purposes.	R.G. Bhagwat <i>et al.</i> , 2015
Paheri	Highly susceptible	A late-season variety with a distinctive flavor.	Boateng BO <i>et al.</i> , 2023

Conclusion

Mango cultivation is an integral part of agricultural practices worldwide, offering significant nutritional and economic benefits. However, diseases, particularly anthracnose, pose a major threat to mango productivity. The analysis highlights the diverse growing belts and varieties in India, as well as hybrid species developed to improve yield. Despite these advancements, the persistent issue of disease necessitates ongoing research and implementation of effective management strategies. Addressing factors such as the senility of mango trees, poor orchard management, climatic changes, and soil constraints is crucial. Globally, collaborative efforts are required to mitigate the impact of diseases, ensuring the sustainability and growth of mango production. Continued innovation in disease management and cultivation techniques will be vital in preserving the legacy and future of mango cultivation.

Declaration

The authors assert that they have no conflicts of interest.

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References

1. Abera A, Lemessa F, Adunga G. Morphological characteristics of Colletotrichum species associated with mango (*Mangifera indica* L.) in Southwest Ethiopia. Food Sci Qual Manag. 2016;48:106-15.
2. Al Adawi AO, Deadman ML, Al Rawahi AK, Al Maqbali YM, Al Jahwari AA, Al Saadi BA, *et al.* Aetiology and causal agents of mango sudden decline disease in the Sultanate of Oman. Eur J Plant Pathol. 2006;116(4):247-54. <https://doi.org/10.1007/s10658-006-9056-x>
3. Alam A, Adhikary SK, Ahmed M. Morphological characterization of Colletotrichum gloeosporioides identified from anthracnose of *Mangifera indica* L. Asian J Plant Pathol. 2017;11(3):102-17. <https://doi.org/10.17311/ajppaj.2017.102.117>
4. Dinesh MR, Ravishankar KV, Sangma D. Mango breeding in India - Past and future. Indian Horticulture. 2016;11(1):1-12.
5. Dofuor AK, Quartey NK-A, Osabutey AF, Antwi-Agyakwa AK, Asante K, Boateng BO, *et al.* Mango anthracnose disease: The current situation and direction for future research. Front Microbiol. 2023;14:116-8203. <https://doi.org/10.3389/fmicb.2023.1168203>
6. Ganeshamurthy A. Enhancing mango productivity through sustainable resource management. Krishi; 2019. Available from: https://krishi.icar.gov.in/jspui/bitstream/123456789/23708/1/Enhancing%20mango%20productivity-Review_SPHJournal2018.pdf
7. Gautam AK. Colletotrichum gloeosporioides: Biology, pathogenicity, and management in India. J Plant Physiol Pathol. 2014;2(2):1-25. <https://doi.org/10.4172/2329-955x.1000125>
8. González-Fernández J, Hormaza J. Pest and diseases in mango (*Mangifera indica* L.). Mango.org. 2023. Available from: https://www.mango.org/wp-content/uploads/2020/08/Mango_Pests_and_Diseases_ENG.pdf

9. India production of mango. Apeda.gov.in. 2023. Available from: https://agriexchange.apeda.gov.in/India%20Production/India_Productio ns.aspx?cat=fruit&hscode=1050
10. Khan M, Ahmed N. Sustainable management of mango nutrition for better yield and quality. *Cercet Agron Moldova*. 2021;53(4):473-501. <https://doi.org/10.46909/cerce-2020-040>
11. Lebaka VR, Wee Y-J, Ye W, Korivi M. Nutritional composition and bioactive compounds in three different parts of mango fruit. *Int. J Environ Res Public Health*. 2021;18(2):741. <https://doi.org/10.3390/ijerph18020741>
12. Mango [Internet]. Nhb.gov.in. Available from: https://nhb.gov.in/report_files/mango/mango.htm
13. Mango varieties in India, production in India, economic importance. ABC Fruits; 2022. Available from: <https://abcfruits.com/mango-varieties-production-seasons-and-economic-importance/>
14. Misra A. Anthracnose of mango. In: *Crop Disease Identification and Management*; 2008. p. 116-7.
15. Münch S, Lingner U, Floss DS, Ludwig N, Sauer N, Deising HB. The hemibiotrophic lifestyle of *Colletotrichum* species. *J Plant Physiol*. 2008;165(1):41-51. <https://doi.org/10.1016/j.jplph.2007.06.008>
16. Nallamani M, Bahurudeen HM, Sobanbabu G, Suguna S. Management of mango anthracnose: An overview. *Int. J Agric Plant Sci*. 2023;2664-7664.
17. Nelson SC. Mango anthracnose (*Colletotrichum gloeosporioides*); 2008. Scholarspace.manoa.hawaii.edu. Available from: <http://hdl.handle.net/10125/12410>
18. Osorio J. Identification of *Colletotrichum* species causing anthracnose on Tahiti lime, tree tomato, and mango; 2014.
19. Owino WO, Ambuko JL. Mango fruit processing: Options for small-scale processors in developing countries. *Agriculture*. 2021;11(11):1105. <https://doi.org/10.3390/agriculture11111105>
20. Ploetz RC. Diseases of tropical fruit crops. CABI Publishing eBooks; 2003. <https://doi.org/10.1079/9780851993904.0000>
21. Ploetz RC. The major diseases of mango: Strategies and potential for sustainable management. *Acta Hort*. 2004;645(10):137-50. <https://doi.org/10.17660/actahortic.2004.645.10>

22. Bhagwat RG, Mehta BP, Patil VA, Sharma H. Screening of cultivars/varieties against mango anthracnose caused by *Colletotrichum gloeosporioides*. *Int J Environ Agric Res*. 2015;1:21-3.
23. Rajan S, Yadav A. Mango hybrids and improved cultivars. *Indian Hortic.*, 2021, 66(4).
24. Rathour TP. Mango cultivars, wild genotypes and their importance in India. *Just Agric.*, 2023, 3(9).
25. Sharma M, Kulshrestha S. *Colletotrichum gloeosporioides*: An anthracnose causing pathogen of fruits and vegetables. *Biosci Biotechnol Res Asia*. 2015;12(2):1233-46. <https://doi.org/10.13005/bbra/1776>
26. Taïbi A, Meile JC, Dieudonné H, Korsten L, Chillet M, Remize F. New bacterial agents to limit *Colletotrichum gloeosporioides* development on mango. *Adv. Microbiol.* 2020;10(12):691-712. <https://doi.org/10.4236/aim.2020.1012050>
27. Yadav A, Pandey DC. Geographical perspectives of mango production in India. *Imperial J Interdiscip Res*, 2016, 2.
28. Yadav D, Singh S. Mango: History origin and distribution. *J Pharmacogn Phytochem*. 2017;6(6):1257-62.

Chapter - 5

Anthracnose Disease in Mango: Its Impact, and Effective Management Strategies

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Chapter - 5

Anthracnose Disease in Mango: Its Impact, and Effective Management Strategies

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Abstract

India, home to around 1,000 varieties of mango cultivated for over 4,000 years, is a significant global player in mango production, contributing 38% of the world's mango yield. Plant pathogens, especially *Colletotrichum gloeosporioides* causing Anthracnose disease, pose severe threats to mango cultivation, leading to significant financial losses. Effective management of mango diseases involves both pre-harvest and post-harvest treatments. Chemotherapeutic approaches using fungicides like mancozeb and carbendazim are common, while eco-friendly methods, such as essential oils and plant extracts, show promise in reducing pathogen impact. Advances in molecular and morphological characterization of *C. gloeosporioides* aid in developing targeted treatments, and biodegradable polymers like chitosan offer innovative solutions for disease management. Various approaches, including chemical control and biocontrol, have been introduced to combat anthracnose. Ongoing research focuses on understanding host range, molecular characteristics, and susceptibility in different mango cultivars. This chapter offers current insights into the mechanisms of disease spread, the biological aspects of infection, and modern approaches to management. This comprehensive approach to understanding and combating mango diseases ensures sustainable agricultural practices and enhances global mango production.

Keywords: *Colletotrichum gloeosporioides*, management, characterization, anthracnose

Introduction

India is home to around 1,000 varieties of mango, cultivated for over 4,000 years. While only a few varieties are commercially grown, mango is produced in more than 90 countries worldwide. Asia, the native region of mango, leads in global production with 77%, followed by the Americas (13%)

and Africa (10%). India contributes significantly, with 44% of the area under mango cultivation and 38% of global production. The world productivity of mango is 7.74 M.T./ha, with India's productivity at 6.92 metric tonnes per hectare, compared to Brazil's 16 metric tonnes per hectare (Singh *et al.*, 2018).

Plant pathogens pose a significant threat to agricultural sustainability and cause severe economic losses. The rapid evolution of pathogens, environmental changes, and international trade contribute to the emergence of new plant diseases. Effective plant disease management requires timely and accurate pathogen identification. However, the diversity and spread of plant pathogens can hinder these efforts. Plant pathogens employ various strategies for host adaptability, transmission, and strain diversity. They interact with various species, including plants, endophytes, insects, pollinators, and other pathogens. The impact of diseases on different plants and the transmission and evolution of pathogens in agroecosystems and natural biosystems are often not well understood (Brent Sipes *et al.* 2024).



Fig 1: Common Diseases Affecting Mango Trees and Their Pathogens

One of the most common diseases in mangoes is Anthracnose diseases which is caused by *Colletotrichum gloeosporioides* (Penz.) which represents a significant pathogen genus that severely impacts mangoes, particularly during the processes of sorting, packaging, transportation, storage, and marketing, making it a major economic concern in the cultivation of mangoes. (R.G. Bhagwat *et al.*, 2015). *C. gloeosporioides* causes significant financial losses, amounting to between 30% to 60% in fruit yield, and potentially reaching 100% during periods of heavy rainfall or high humidity conditions

(Kamle M. *et al.*, 2016). Premature fruit drop is induced by this condition (U Damm *et al.*, 2012). In the initial stages of growth, the leaves start to display visible markings and darkening on their tips, leading to the premature shedding of flowers and a decline in fruit yield; as the fruits reach maturity, they develop sunken, dark, round patches (Phoulivong S, Cai L, Chen H *et al.*, 2010). Within biotic stress factors, anthracnose is instigated by *Colletotrichum gloeosporioides*.

Based on reports, fungal diseases cause spoilage in approximately 17.7% of mangoes during storage and marketing. (Sharma IM *et al.*, 1994). During rainy and humid conditions, *Colletotrichum gloeosporioides* can reach up to 75% in mango fruits. (Colón-Garay *et al.*, 2002). Anthracnose fungus frequently infects mango plants in their early stages, yet it remains dormant, exhibiting symptoms solely on mature fruits, post-harvest, or during storage and transportation. (Ayantu Tucho *et al.*, 2017)

Characterization of *Colletotrichum gloeosporioides*

Characterization on the molecular basis

In a study conducted by Kamle *et al.*, 45, molecular identification of *Colletotrichum gloeosporioides* isolates was achieved through Polymerase Chain Reaction (PCR) using species-specific primers, namely “CgF (TTGCTTCGGCGGGTAGGGTC) and CgR (ACGCAAAGGAGGCTCC GGA)”. In this study, all thirteen fungal isolates underwent molecular characterization via PCR utilizing universal ITS 1 and ITS 4 primers. The PCR products were subjected to 1% gel electrophoresis, yielding an amplicon size of approximately 560 bp. The amplicons underwent partial sequencing, and the resulting sequences were deposited in the NCBI GenBank database to acquire accession numbers. Based on these accession numbers, all thirteen isolates were identified as *Colletotrichum gloeosporioides*. Nevertheless, it's crucial to acknowledge that although PCR amplification of the ITS region is commonly employed in molecular techniques, it may not provide highly specific identification at the species level for fungi. Therefore, the ACTIN gene present in *Colletotrichum* sp. and specifically in *Colletotrichum gloeosporioides* was targeted and amplified by employing ACT gene primers (ACT512F, ACT783R) along with species-specific primers (CgF and CgR), the PCR reaction produced amplicons with sizes around 280bp and 380bp, further confirming the identity of the isolates (Lima WG *et al.* 2017), (Sharma G. *et al.* 2013).

Morphological characterization

Morphological studies involved a detailed assessment of different characteristics in all *C. gloeosporioides* isolates. The method outlined by

Menezes Assis specifically for *Colletotrichum gloeosporioides* was followed, utilizing the microculture technique to examine traits such as conidia shape and size, appressorium formation, and the presence of setae (Ganesan *et al.*, 38). The gene ACT was amplified with specific primers, resulting in PCR amplicons that confirmed the identity of the isolates. Incubation was conducted at a constant temperature with specific photoperiod conditions. Conidial measurements were documented using a micrometre eyepiece, highlighting morphological diversity based on observed shapes and sizes. Colonies of the fungus displayed varying colours when cultured on a PDA medium and produced distinct conidia shapes. Al-Gloeosporioides morphology affecting mango was thoroughly examined in a study exploring substantial variation among isolates obtained from different parts of mango plants across different geographical regions. (Abera *et al.*, 2016), (Tejashwini NK *et al.* 2023)

Physiological characterization

The presence of *C. gloeosporioides* in mango has revealed how the fungus adapts to different environmental and nutritional circumstances. This understanding is pivotal in devising effective strategies to combat anthracnose in mango cultivation (Doyle V. P *et al.*, 2013)

Anthracnose diseases management

Now a days for the management of anthracnose diseases in mango, there are various approaches some of which are listed below:

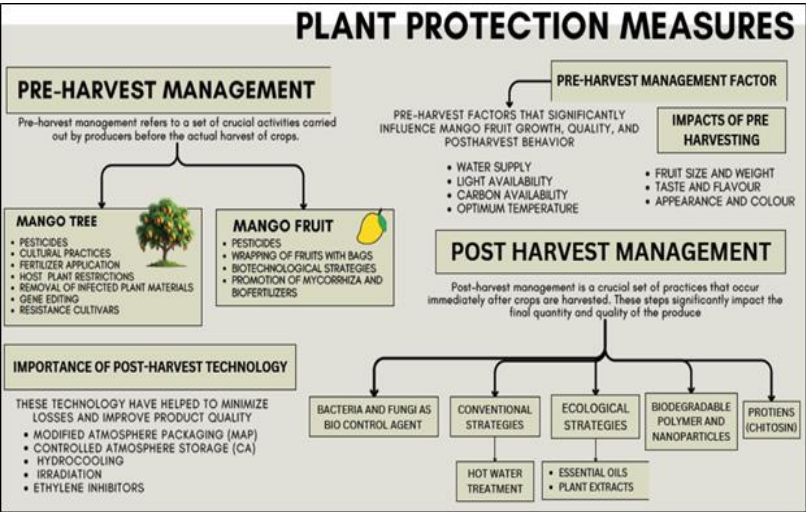


Fig 2: Plant Protection Measures for Mango Trees

Pre-harvesting treatments

Chemotherapeutic management of anthracnose in mango

Various methods of synthetic fungicides have been accepted to prevent mangoes from anthracnose disease (Shukla *et al.*, 79). At the time of flowering the latest technology of spraying fungicides minimizes the incidence of disease severity. There are various ways of applying chemotherapeutic agents such as dips, sprays, fumigants, treated wraps & box liners or in waxes coating. At a concentration of 0.1%, the non-systemic fungicide mancozeb suppressed the mycelial growth of *C. gloeosporioides* by 64.0%, whereas systemic fungicides like difenoconazole and propiconazole inhibited growth by over 90.7%. (Prashanth *et al.*, 88). Carbendazim, a systemic fungicide, demonstrated the highest effectiveness, while mancozeb, a non-systemic fungicide, exhibited the most potent inhibition against the pathogen *C. gloeosporioides*, which induces mango anthracnose disease. (Kolase *et al.*, 2014), (Lakshmi *et al.*, 2013)

Table 1: Effective Pesticides/Fungicides show inhibition against *C. gloeosporioides*

S. No.	Pesticides /Fungicides	Number of sprays	Spray Timing	Reference
1.	Mancozeb + copper	13 days	Panicle emergence onwards	Adunga <i>et al.</i> , 2016
2.	Insecticide + Mancozeb	Every 10 days	onwards from the flower bud	Nallamani M <i>et al.</i> , 2023
3.	Copper + Prochloraz	The variable experiences notable reduction during dry years.	Panicle emergence Onwards	José A. Mora-Aguilera <i>et al.</i> , 2021
4.	copper + Mancozeb/ chlorothalonil + insecticide	6 days	Sprays five time from induction to fruit Set	Sharma <i>et al.</i> , 2019
5.	Chlorothalonil 75% WP	14 days	Early stages of fruit development.	Dr. Sudha S <i>et al.</i> , 2021
6.	Mancozeb 63%WP	45 days	Immediately upon appearance of disease symptoms.	Sharma <i>et al.</i> , 2019
7.	Copper oxy chloride 50% WP	15-day interval	When the first symptoms of the disease appear or initial phases of fruit development	Sharma <i>et al.</i> , 2019
8.	Difenconazole 25% EC	15-day intervals	Immediately upon appearance of disease symptoms.	Bhavya <i>et al.</i> , 2018
9.	Hexaconzole 5% EC	15-day intervals	Start spraying when the first signs of disease appear.	Nallamani M <i>et al.</i> , 2023

11.	Tetraconazole 3.8% EC	15-day intervals	Start spraying when the first signs of anthracnose appear	Ranjitha N <i>et al.</i> , 2019
12.	Propiconazole 25% EC	15-day intervals	Start spraying when the first signs of anthracnose appear	Sharma <i>et al.</i> , 2019
13.	Thiophanate methyl 75% WP	12-15day Intervals	Start spraying when the first signs of anthracnose appear	Sharma <i>et al.</i> , 2019
14.	Myclobutanil 10% WP	12-15day Intervals	Apply the fungicide to protect young blossoms	Sharma <i>et al.</i> , 2019
15.	Captan 50% WP	12 to 15 days	Start spraying when the first signs of anthracnose appear	Dr. Sudha S <i>et al.</i> , 2021
16.	Azoxystrobin	14 days	Pre-harvest sprays significantly reduce the field inoculum (by 24.54%) on mango leaves.	Sharma <i>et al.</i> , 2019
17.	Zineb 68% +Hexaconazole 100%WP	15-day intervals	When symptoms become noticeable.	Devappa Venkat <i>et al.</i> , 2019
18.	Flusilazole + Carbendazim 37.5% WP	10–15-day intervals	When symptoms become noticeable.	Mevada and Kapadiya <i>et al.</i> , 2019
19.	Tebuconazole + Trifloxystrobin 75% WG	15-day intervals	When symptoms become noticeable	Sharma <i>et al.</i> , 2019

Post-harvesting treatment

Conventional strategies

Successful management of postharvest fruit diseases involves implementing effective postharvest practices, including maintaining orchard cleanliness, applying fungicides, and carefully controlling temperature during storage and transportation (Páez *et al.* 2022).

Hot water treatment (HWT)

This technique works well for some fruits where the water temperature doesn't cause the skin to burn or lose colour, change flavor, or soften the pulp. The temperature chosen for HWT offers great promise for reducing postharvest losses of different fruits because it is marginally higher than the target pathogen's thermal death point. According to several pieces of research, hot water treatment (HWT) is frequently used to control deterioration and preserve quality (Djioua *et al.*, 2009). Research has demonstrated that this method can maintain product quality while increasing profitability and market

value (Barrera Necha *et al.*, 2008). For example, fruit stored for nine days at 25 °C was exposed to HWT at 54 °C for four minutes, resulting in a 44% reduction in anthracnose. (Zhao *et al.*, 2013). However, a key challenge with short-term heat treatment is the need for high temperatures, which can potentially damage the product, necessitating precise temperature control and careful management.

Eco-friendly approach for management

Essential oils

Essential oils are generally regarded as safe natural food preservatives for consumption, as long as they do not alter the taste or quality of the food, and show no signs of harm, which should always be evaluated (Kudachikar *et al.*, 2017). Recently, there has been a significant rise in the adoption of essential oils in agriculture as a substitute for synthetic pesticides to safeguard crops. This shift is due to the various insecticidal, herbicidal, bactericidal, and fungicidal properties of essential oils, as well as their low toxicity to humans and environmental friendliness (Raveau *et al.*, 2020), (Bautista-Baños *et al.*, 2008). The utilization of aromatic plant essential oils to manage *Colletotrichum species* is a crucial focus of research (Kudachikar *et al.*, 2017). Unanswered questions surround the antifungal properties of these oils against *Colletotrichum* fungi, such as their mode of action, the specific effects of their key components, and their potential harm to other plant species. Therefore, it is essential to enhance research endeavours on essential oils that target these fungi, which pose significant threats to crop well-being (Kiranmayee *et al.*, 2020).

Table 2: Efficient Essential oil for inhibiting the growth of *C. gloeosporioides*

Essential Oil	Method of Extraction	Type of Evaluation	Minimum Inhibition Concentration	References
Oregano oil from Mexico (<i>Poliomintha longiflora</i>)	Steam distillation	<i>In vitro</i>	1.0 g/L	Andrade-Ochoa S <i>et al.</i> , 2019
Thyme (<i>Thymus vulgaris</i>)	Steam Distillation	<i>In vitro</i>	512-102µg/mL	Mevada <i>et al.</i> 2019 Chillet <i>et al.</i> 2019
Clove (<i>Syzygium aromaticum</i>) & Cinnamon (<i>Cinnamomum zeylanicum</i>)	N.M. commercial sample	<i>In vitro</i> & <i>In vivo</i>	66.7µL/L	Costa <i>et al.</i> , 2020
Oregano (<i>Origanum vulgaris</i>) and lemongrass	Hydro distillation	<i>In vitro</i>	20 -100µg/mL	Andrade-Ochoa S. <i>et al.</i> , 2018
<i>Corymbia citriodora</i> <i>Cymbopon citrus</i> , <i>Cymbopogon flexuosus</i> , <i>Curcuma longa</i>	Steam Distillation	<i>In vitro</i>	10–20 µL	Schwan <i>et al.</i> , 2019
Mint (<i>Mentha piperita</i> L.)	Hydro distillation	<i>In vitro</i>	0.12 -8.0 µL/mL	De Oliveira <i>et al.</i> , 2020
Rosewood oil (<i>Aniba duckei</i> Kostermans)	Hydro distillation	<i>In vitro</i>	2000µL/mL	Barros Gomes <i>et al.</i> , 2019
Almond (<i>Acrocomia aculeata</i>), Clove (<i>Caryophyllus aromaticus</i> L.) and neem (<i>Azadiractha indica</i> A. Juss)	Steam distillation	<i>In vitro</i>	-	Costa <i>et al.</i> , 2020
Swingle oil derived from <i>Murraya microphylla</i> (Merr.)	Hydrodistillation employing column chromatography using silica gel	<i>In vitro</i>	63.3 microliters of crude oil (equivalent to 3.24 microliters for Terpinen-4-ol).	Xia <i>et al.</i> , 118
Basil (<i>Ocimum basilicum</i>)	Supercritical CO ₂ Extraction	Evaluation Type	62.5 micrograms per millilitre to 250 micrograms per millilitre	Karunanayake <i>et al.</i> , 2014

Plant extracts

In the fight against *C. gloeosporioides*, plant extracts have the potential to replace synthetic fungicides with ecologically benign treatments due to their antibacterial capabilities. Numerous investigations have demonstrated the antifungal properties of plant extracts against *C. gloeosporioides* in both lab and real-world contexts, with the aim of controlling postharvest anthracnose in fruits and vegetables. *In vitro* tests showed that 15% concentrations of plant extracts from *Aegle marmelos*, *Ipomoea carnea*, and *Pongamia pinnata* inhibited the growth of *C. gloeosporioides*' mycelia by 74.85%, 63.23%, and 54.89%, respectively (Satapathy *et al.*,2021).

In a different investigation, (Kwodaga *et al.*, 2019) discovered that a 100% water-based extract from the fruit of the *Capsicum annum* dramatically decreased the mycelial growth of *C. gloeosporioides* by 76.1% and the germination of spores by 91.7%. However, España *et al.* found that ethanolic extracts inhibited *C. gloeosporioides* more than essential oils from *Eucalyptus* species. The majority of studies have looked at how several solvents work together in crude extracts (Oliveira *et al.*, 2017). The effects of different extraction methods and organic solvents, such as ethanol and methanol, on the effectiveness of plant extracts, have been the subject of several studies Researchers are investigating the application of edible coatings coated with plant extracts to enhance fruit preservation and fight fungal diseases, which holds potential for novel postharvest interventions. Plant extracts have demonstrated the ability to combat anthracnose in fruits, which presents new opportunities for the development of creative control methods for *C. gloeosporioides* in the food and agriculture sectors (Alves *et al.*, 2015).

Table 3: Effective Phyto-extract inhibiting the growth of *C. gloeosporioides*

Phyto-extracts	Mode of action	References
Eucalyptus	Fungal growth is impeded by oil	Chauhan <i>et al.</i> , 1990 España <i>et al.</i> , 2017
castor oil	Decrease electrolyte Absorption	Kwodaga <i>et al.</i> , 2019
Lantana leaves	Suppress Fungal Growth	Anamika <i>et al.</i> , 2015
<i>Azadirachta indica</i>	Therapeutic intervention	Raveau <i>et al.</i> , 2020, Ahmed <i>et al.</i> 2018
Garlic bulbs	Fungal growth Inhibition	Gahlot <i>et al.</i> , 2021
<i>Prosopis juliflora</i>	Inhibit fungal growth	Deressa <i>et al.</i> , 2015
Eucalptus	Inhibitory action	Hur <i>et al.</i> , 2000
Capsicum annum fruit	Inhibit the growth of fungal threads and prevent the seeds from sprouting	Kwodaga <i>et al.</i> , 2019
<i>Plantago sinaica</i>	Inhibit fungal growth	Kumar <i>et al.</i> , 2017

Extract from Sarmentosum Leaves	Halted the Mycelium Growth	Bussaman <i>et al.</i> , 2012
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Biodegradable polymers

Chitosan

Chitosan is a polymer that is produced when chitin, a naturally occurring polysaccharide, is deacetylated. It is frequently found in insects, crustaceans, and the mycelium of certain fungi. Compared to chitin, chitosan and its synthetic derivatives are more soluble in mild acid solutions, which simplifies the processing procedures. Due to their numerous advantageous qualities, they have garnered a lot of interest for a variety of uses in industries like food, medicine, agriculture, cosmetics, and water treatment. Additional materials, such as lipids and waxes, can be added to increase these properties. Clays can also be added to improve the films' flexibility and ability to repel water. Nevertheless, the mechanical and flavor aspects can be affected by this modification. Several of these issues have been resolved by using essential oils (EOs). Clays might be used to increase the films' flexibility and water resistance, but this modification might affect their mechanical and sensory qualities. Some of these problems have been lessened by the addition of essential oils (EOs) (Bautista-Baños *et al.*, 2017).

Edible coatings using chitosan

Thin layers of food-safe materials, usually biopolymers, are used to create edible films and coatings, which are applied to food surfaces to improve stability, aid in preservation, and appeal to consumers. According to Naser Tamimi *et al.* in 2021, the coatings give the food surface barrier, antibacterial, and antioxidant qualities. Natural polymers, such as proteins and polysaccharides and their nanocomposites are used to create coverings that are edible and biodegradable. This kind of preservation helps food's sensory properties last for a long time. Moreover, chitosan has been combined with essential oils to enhance qualities like insect repellent, water resistance, antioxidant activity, antibacterial and antifungal qualities, and several other advantageous biological qualities with a wide range of applications.

Other Polymers & nanoparticles

Testing has been done on a number of eco-friendly polymers besides chitosan to see whether or not they can manage anthracnose. Fruit quality can be preserved by using polymers that are beneficial in preventing fungal deterioration, such as gum Arabic, methylcellulose, aloe vera gel (rich in polysaccharides), and starch (Kumar *et al.*, 2020). Aloe vera gel solutions at

concentrations of 15%, 25%, and 50% have been shown in a study by Bill *et al* in 2014 to affect a variety of fungal infections, including *C. gloeosporioides*, in both natural and controlled conditions. In controlled conditions, the research demonstrated a 72.96% inhibition of the fungus. Furthermore, in real-world settings, aloe vera gel showed a dose-dependent response, showing substantial suppression of *C. gloeosporioides* after 72 hours.

Furthermore, the combination of aloe vera and thyme oil coatings proved to be fungicidal against *C. gloeosporioides*, as noted by Bill *et al* in 2014. Evaluations on arrowroot starch coatings infused with *Piper aduncum* essential oil highlighted their antifungal properties, especially at concentrations of 0.75% and 1%, as illustrated by Nogueira *et al* in 2021 and Sousa *et al.* in 2014 Studied the efficacy of on-site coverings using hydroxypropyl methylcellulose (HM) and beeswax (BW) on mangoes., leading to a considerable reduction in anthracnose incidence and delayed ripening. Similarly, Vieira *et al.*in 2020 reported significant outcomes from HM combined with 0.25% silver nanoparticles in managing anthracnose and extending the fruit's shelf life. Additionally, the antifungal activity of microcapsules containing gum arabic and thyme oil was observed to hinder fungal growth over time, as detailed by Cai *et al.*, 2010.

Table 4: Effective Nanoparticles for inhibiting the growth of *C. gloeosporioides*

Nanoparticles Mode of action		References
Composite of Chitosan with Silver Nanoparticles	Disturbance of plasma membrane and DNA	Chowdappa <i>et al.</i> , 2014
carbendazim-conjugated silver nanoparticles (Cz-AgNPs)	Protein denaturation and damage to the cell wall	Shivamogga <i>et al.</i> , 2020
Zinc oxide nanoparticles (ZnO NPs)	The influence of membrane leakage on cell survival	Ruangtong <i>et al.</i> , 2022
Composite of Chitosan-silver nanoparticles	Interrupting fungal cells.	Ruffo <i>et al.</i> , 2019
Copper oxychloride combined with silver nanoparticles	Inhibition of spore formation.	Ruffo <i>et al.</i> , 2019
Nano-sized particles of aloe vera gel	Membrane leakage and cell viability	Khaliq <i>et al.</i> , 2019
Copper nanofibers	Flattening and rupturing the mycelia.	Vestergaard <i>et al.</i> , 2023
Glycerol Nanoparticles	Leakage from membranes impacting cell viability.	

Maltol capped silver	Disrupting the cytoplasm of fungal cells.	Raghavendra <i>et al.</i> , 2020
Chitosan Nanoparticles	Interrupting fungal cells.	Bautista-Baños <i>et al.</i> , 2017

Proteins

The use of protein Agri-industrial waste to produce bioactive peptide compounds, which have been demonstrated to have antimicrobial and antioxidant activities. The reported findings indicate that films made from cottonseed protein hydrolysates exhibited an inhibitory effect against *C. gloeosporioides*, with an inhibition zone measuring 27.02 mm. Additionally, cottonseed byproduct liquid hydrolysate demonstrated growth inhibition of 70.78% (de Almeida *et al.*, 2020). Vieira *et al.* in 2020 conducted a study. Garcia examined the impact of cornzein nanoparticles on two *Colletotrichum* species in laboratory settings. They observed a reduction in fungal growth by 34.1% and 50% at concentrations of 1000 ppm and 500 ppm, respectively, which subsequently led to decreased fungal decay. A protein extract from the roots and stems of *Cereus jamacaru*, a Brazilian plant, demonstrated a 43% inhibition effect against *C. gloeosporioides*. Additionally, protein extracts from the roots and stems of *Cereus jamacaru*, a Brazilian plant, demonstrated a 43% inhibition effect against *C. gloeosporioides*. The observed inhibition was linked to noticeable alterations in fungal structure, such as increased membrane permeability, initiation of reactive oxygen species (ROS), and subsequent cell demise. Vieira *et al.* conducted a study showing that the protein present in *Clitoria ternatea* seeds, a tropical legume recognized for its strong resistance to pathogens and pests, suppressed the growth of *C. gloeosporioides* due to its high levels of a protein called 'finotin'. This protein also demonstrated effectiveness against various other types of filamentous fungi. Furthermore, the antimicrobial efficacy of proteins can be amplified through the use of nanoparticles (NPs), which show promising potential. The antimicrobial activity of proteins can also be enhanced by using nanoparticles (NPs), which are promising solutions. (Peralta-Ruiz *et al.*, 2023) tested gelatin coatings with silver NPs for the control of *C. gloeosporioides* *In vitro* and in situ on mangoes. The researchers noticed that fungal growth was completely inhibited on the Petri dishes; however, they did not observe significant inhibition on mangoes when compared to the control group (Nogueira *et al.*, 2021).

Utilizing fungi and bacteria as biocontrol agents to address postharvest anthracnose

Biological methods for disease management have become increasingly favored, leading to heightened research activity in this domain. These methods have also been sourced from various other origins, including the phyllosphere,

roots, and soil. (Alvindia *et al.*, 2015). Overall, numerous studies have indicated greater antifungal efficacy when biocontrol agents were administered before infection rather than following it. (Song *et al.*, 2016). Filamentous fungi, particularly *Trichoderma* species, have been utilized for *Colletotrichum* spp. management. Following investigations into their effectiveness for postharvest anthracnose control, formulations of antagonistic microorganisms have been suggested for commercial application. The efficacy of powdered *Meyerozyma caribbica* formulations has been shown in addressing postharvest issues in mangoes stored at a temperature of 25 °C (Aguirre-Güitrón *et al.*, 2019). Minimizing fruit weight loss and preserving the viability of the antagonistic microorganism throughout storage. (Peeran *et al.* 2014), (Barros-Castillo *et al.*, 2019).

Table 5: Effective Microbial treatment for inhibiting the growth of *C. gloeosporioides*

Antagonist	Mechanism of Biocontrol	Lesion Length Inhibition	References
Bacteria			
<i>Bacillus subtilis</i>	Produce protease, cellulase, and siderophore, which inhibit it's the growth	60 %	Sarmah <i>et al.</i> , 2023
	Cause the lysis the hyphae of the pathogen, thereby inhibiting its growth	58 %	Sarmah <i>et al.</i> , 2023
<i>Paenibacillus polymyxa</i>	Competing for nutrients and space, siderophore, lytic enzymes, VOCs	85%	Grady <i>et al.</i> , 2016
<i>Bacillus cereus</i>	Controlling pathogen growth through antibiosis and competition	37.5%	Tian, C <i>et al.</i> , 2022
<i>Pseudomonas fluorescens.</i>	Cause the lysis (breaking down) of the hyphae of the pathogen, thereby inhibiting its growth.	84.05%	Barman <i>et al.</i> , 2017
<i>Bacillus amyloliquefaciens</i>	induce the production of lytic enzymes in plants, which can mediate defense	60-100%	Liang <i>et al.</i> ,2022 Alijani <i>et al.</i> , 2022
<i>Bacillus thuringiensis</i>	Produces the inhibitory metabolites mainly include lipopeptides, proteins, and volatile compounds.	-	Nujthet <i>et al.</i> , 2023
Fungi			
<i>Cryptococcus laurentii</i>	Competition for nutrients and space	75.88%	Bautista <i>et al.</i> , 2017
<i>Trichoderma longibrachiatum</i>	Compete with the pathogen for space and nutrients, thereby limiting the growth of the pathogen.	27%	Kredics <i>et al.</i> , 2014

<i>Debaryomyces nepalensis</i>	resistance, lytic	4.7–93.4%	Taïbi <i>et al.</i> ,2020
<i>Trichoderma harzianum</i>	Secondary metabolites that can inhibit the activity of numerous plant diseases.	81.67%	Nujthet <i>et al.</i> , 2023 Kredics <i>et al.</i> , 2024
<i>Trichoderma viride</i>	Induced Systemic Resistance	76.45%	Nujthet <i>et al.</i> , 2024

Current research on anthracnose disease management

Anthracnose, a fungal affliction impacting a diverse array of plant life, results in significant financial detriments within agricultural and horticultural domains. Ongoing research endeavours are directed towards devising novel and enhanced methodologies for the disease's management. Prominent research foci include:

i) Biocontrol

- a) **Microbial antagonists:** The discernment and deployment of autochthonous microorganisms such as fungi, bacteria, and yeasts capable of impeding anthracnose pathogens (Negi *et al.*, 2023).
- b) **Phytoextracts and Volatile oils:** The scrutiny of phytoextracts and volatile oils for their fungicidal efficacy against anthracnose (Mevada *et al.*, 2019).
- c) **RNA Interference (RNAi):** The refinement of RNAi approaches to selectively incapacitate genes within the anthracnose fungus, attenuating its virulence or proliferation (Rajam *et al.*,2021)

ii) Host resistance

- a) **Cultivar selection for resistance:** The pinpointing of plant genetic markers conferring anthracnose immunity and their integration into the cultivation of resistant strains (Katoch *et al.*, 48), (Selvan T *et al.*, 2023).
- b) **Genomic editing:** The application of CRISPR-Cas9 and analogous genomic editing tools to instigate precise genetic alterations, bolstering plant resilience (Zhang C *et al.*, 2023).
- c) **Induced Systemic Resistance (ISR):** The provocation of a plant's endogenous defense stratagems to counteract anthracnose invasion. (Dale Walters *et al.*,2005)
- d) The utilization of CRISPR-Cas9 gene-editing technology stands out as a pivotal advancement, offering the potential to precisely and

efficiently enhance plant resistance to anthracnose, thereby promising a revolutionary impact on disease management strategies (Karmakar *et al.*, 2022).

iii) Advancements in fungicide development

- a) **Progressive exploration in fungicides:** Innovating the discovery and formulation of novel fungicides with precise targeting and reduced environmental impact in comparison to traditional fungicides (Yenjerappa *et al.* 2021).
- b) **Nanotechnology in pest control:** Investigating nanoparticles as efficient carriers for fungicides, enabling accurate application and decreased overall usage (Yousef *et al.* 2023).
- c) **Harnessing biopesticides:** Evaluating biopesticides, sourced from natural origins, as potential alternatives to synthetic fungicides. (Yonghuai Huang *et al.* 2020)

iv) Surveillance and Prediction

- a) **Advancing predictive models:** Developing computational models capable of predicting anthracnose outbreaks based on meteorological conditions and environmental variables (Isip *et al.*, 2021).
- b) **Utilizing remote sensing:** Applying satellite imagery and other remote sensing technologies to identify areas prone to anthracnose infection. (Yang, R *et al.*, 2023).
- c) **Development of biosensors:** Designing biosensors capable of on-site detection of anthracnose pathogens, facilitating swift intervention. (Palou *et al.*, 2024)

v) Comprehensive disease management

- a) **Diversified control approaches:** Implementing a comprehensive strategy combining cultural practices, biological control agents, and resistant cultivars to ensure sustainable and efficient anthracnose management. (Aljawasim *et al.*, 2023).
- b) **Precision agriculture techniques:** Embracing precision agriculture technologies to tailor disease management strategies according to specific field conditions, thereby optimizing resource utilization. (Chin *et al.*, 2023)

Conclusion

In conclusion, mango cultivation in India, a nation with over 1,000 varieties of this fruit, is significantly threatened by plant pathogens, particularly *Colletotrichum gloeosporioides*, the causative agent of anthracnose. This pathogen inflicts severe economic losses, necessitating comprehensive disease management strategies. Effective management includes both pre-harvest and post-harvest interventions. Pre-harvest approaches utilize synthetic fungicides like mancozeb and carbendazim, while post-harvest treatments involve conventional methods such as hot water treatment and eco-friendly alternatives like essential oils and plant extracts. Innovations in biodegradable polymers and nanoparticles also show promise in combating anthracnose. These methods highlight the importance of integrated disease management strategies to sustain mango production and mitigate economic losses.

Declaration

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References

1. Abera A, Lemessa F, Adunga G. Morphological characteristics of *Colletotrichum* species associated with mango (*Mangifera indica* L.) in Southwest Ethiopia. *Food Sci. Qual. Manag.* 2016;48:106-15.
2. Aguirre-Güitrón L, Calderón-Santoyo M, Bautista-Rosales PU, Ragazzo-Sánchez JA. Application of powder formulation of *Meyerozyma caribbica* for postharvest control of *Colletotrichum gloeosporioides* in mango (*Mangifera indica* L.). *Lwt.* 2019;113:108-271.
3. Alijani Z, Amini J, Ashengroph M, *et al.* Biocontrol of strawberry anthracnose disease caused by *Colletotrichum nymphaeae* using *Bacillus atrophaeus* strain DM6120 with multiple mechanisms. *Trop Plant Pathol.* 2022;47:245-59.
4. Aljawasim BD, Samtani JB, Rahman M. New insights in the detection and management of anthracnose diseases in strawberries. *Plants.* 2023;12:3704.

5. Alves KF, Laranjeira D, Câmara MP, Câmara CA, Michereff SJ. Efficacy of plant extracts for anthracnose control in bell pepper fruits under controlled conditions. *Hortic Bras.* 2015;33:332-8.
6. Alvindia DG, Acda MA. The antagonistic effect and mechanism of *Bacillus amyloliquefaciens* DGA14 against anthracnose of mango cv 'Carabao'. *Biocontrol Sci Technol.* 2015;25(5):560-72.
7. Tripathy AP, Satapathy RR, Rout M. Management of *Colletotrichum gloeosporioides* causing papaya anthracnose with best resulting plant extract and bio-agents. *The Pharma Innov J.* 2021;10(10):629-30.
8. Andrade-Ochoa S, Sánchez-Aldana D, Chacón-Vargas KF, *et al.* Oviposition deterrent, larvicidal and pupicidal activity of seven essential oils and their major components against *Culex quinquefasciatus* Say (Diptera: Culicidae). *Insects.* 2018;9:25.
9. Barman K, Asrey R, Singh D, Patel VB, Sharma S. Effect of *Pseudomonas fluorescens* formulations on decay and quality of mango (*Mangifera indica*) fruits during storage. *Indian J Agric. Sci.* 2017;87:1214-8.
10. Barrera-Necha LL, Bautista-Baños S, Flores Moctezuma HE, Estudillo AR. Efficacy of essential oils on the conidial germination, the growth of *Colletotrichum gloeosporioides* (Penz.) and control of postharvest diseases in papaya (*Carica papaya* L.). *Plant Pathol J.* 2008;7(2):174-8.
11. Barros Gomes PR, Oliveira Cunha MJ, De Sousa D, *et al.* Chemical study and antifungal activity of the essential oil of the branches of *Aniba duckei* Kostermans. *J Essent Oil Bear Plants.* 2019;22:1554-61.
12. Bautista-Baños S, Ventura-Aguilar RI, Correa Pacheco Z, Corona-Rangel ML. Chitosan: a versatile antimicrobial polysaccharide for fruit and vegetables in postharvest. A review. *Rev Chapingo Ser Hortic.* 2017;23(2):103-22.
13. Bautista-Rosales PU, Calderon-Santoyo M, Servín-Villegas R, Ochoa-Álvarez NA, Vázquez-Juárez R, Ragazzo-Sánchez JA. Biocontrol action mechanisms of *Cryptococcus laurentii* on *Colletotrichum gloeosporioides* of mango. *Crop Prot.* 2014;65:194-201.
14. Bill M, Sivakumar D, Korsten L, Thompson AK. The efficacy of combined application of edible coatings and thyme oil in inducing resistance components in avocado (*Persea americana* Mill) against anthracnose during post-harvest storage. *Crop Prot.* 2014;64:158-67.

15. Bussaman P, Namsena P, Rattanasena P, Chandrapatya A. Effect of crude leaf extracts on *Colletotrichum gloeosporioides*. J Entomol.; 2012. p. 1-6.
16. Chauhan HL, Joshi HU. Evaluation of phyto-extracts for control of mango fruit anthracnose. Indian Soc Tob Sci. 1990;533(105):455-9.
17. Chillet M, Minier J, Hoarau M, et al. Potential use of thymol to control anthracnose development in mango. Eur J Plant Pathol. 2019;155:943-52.
18. Chin R, Catal C, Kassahun A. Plant disease detection using drones in precision agriculture. Precis Agric. 2023;24:1663-82.
19. Kumar S. Chitosan-based nanocomposite films and coatings: Emerging antimicrobial food packaging alternatives. Trends Food Sci. Technol. 2020;97:196-209.
20. Chowdappa P, Gowda S, Chethana CS, Madhura S. Antifungal activity of chitosan-silver nanoparticle composite against *Colletotrichum gloeosporioides* associated with mango anthracnose. Afr. J Microbiol. Res. 2014;8(17):1803-12.
21. Colon-Garay J, Rivera-Vargas LI, McGovern R, Del PRR. Hypovirulent isolates of *C. gloeosporioides* induce resistance to anthracnose in detached mango fruits and seedlings. J Agric. Univ. P R. 2002;86(1&2):55-64.
22. Costa G, Buccini D, Arruda A, Favaro S, Moreno S. Phytochemical profile, anti-inflammatory, antimutagenic, and antioxidant properties of *Acrocomia aculeata* (Jacq.) Lodd. pulp oil. Food Sci. Technol. 2020;40:10.1590/fst.25319.
23. Walters D, Walsh D, Newton A, Lyon G. Induced resistance for plant disease control: Maximizing the efficacy of resistance elicitors. Am Phytopathol Soc. 2005;95:13-68.
24. De Oliveira Soares MG, Alves E, De Freitas AS, De Almeida CA, Da Silva Costa S. Control of anthracnose in banana with cassava starch film associated or not with essential oils. Rev Bras Cienc Agrar.; 2020, 15.
25. Deressa T, Lemessa F, Wakjira M. Antifungal activity of some invasive alien plant leaf extracts against mango (*Mangifera indica*) anthracnose caused by *Colletotrichum gloeosporioides*. Int. J Pest Manag. 2015;61(2):99-105.
26. Djoua T, Charles F, Lopez-Lauri F, et al. Improving the storage of minimally processed mangoes (*Mangifera indica* L.) by hot water treatments. Postharvest Biol Technol. 2009;52:221-6.

27. Do Prado Mattos A, Povh F, Rissato B, Schwan V, Schwan-Estrada K. In vitro antifungal activity of plant extracts, hydrolates, and essential oils of some medicinal plants and control of cucumber anthracnose. *Eur J Med Plants*; 2019. p. 1-9.
28. Doyle VP, Oudemans PV, Rehner SA, Litt A. Habitat and host indicate lineage identity in *Colletotrichum gloeosporioides* from wild and agricultural landscapes in North America. *PLoS One*; 2013. p. 8.
29. España MD, Arboleda JW, Ribeiro JA, Abdelnur PV, Guzman JD. Eucalyptus leaf by product inhibits the anthracnose-causing fungus *Colletotrichum gloeosporioides*. *Ind Crops Prod*. 2017;108:793-7.
30. Gahlot D, Meena NL, Trivedi A, Kumar S. *In vitro* efficacy of fungicides and phyto extracts against *Colletotrichum gloeosporioides* causing anthracnose of mango. *J Innov Sci*. 2021;29(3):217-21.
31. Grady EN, MacDonald J, Liu L, *et al*. Current knowledge and perspectives of *Paenibacillus*: a review. *Microb Cell Fact*. 2016;15:203.
32. Hamim I, Sipes B, Wang Y. Editorial: Detection, characterization, and management of plant pathogens. *Front Plant Sci*. 2024;15:1354-042.
33. Huang H, Tian C, Huang Y, *et al*. Biological control of poplar anthracnose caused by *Colletotrichum gloeosporioides* (Penz.) Penz. & Sacc. *Egypt J Biol Pest Control*. 2020;30:104.
34. Hur JS, Ahn SY, Koh YJ, Lee CI. Antimicrobial properties of cold-tolerant Eucalyptus species against phytopathogenic fungi and food-borne bacterial pathogens. *Plant Pathol J*. 2000;16(5):286-9.
35. Hussien A, Ahmed Y, Al-Essawy AH, Youssef K. Evaluation of different salt-amended electrolysed water to control postharvest moulds of citrus. *Trop Plant Pathol*. 2018;61(2):99-105.
36. Isip M, Alberto R, Biagtan A. Forecasting anthracnose-twister disease using weather-based parameters: geographically weighted regression focus. *Spat Inf Res*. 2021;29:727-36.
37. Kamle M, Kumar P. *Colletotrichum gloeosporioides*: Pathogen of anthracnose disease in mango (*Mangifera indica* L.). Springer International Publishing; 2016. p. 207-19.
38. Karmakar A, Taufiq S, Baig MJ, *et al*. Increasing disease resistance in host plants through genome editing. *Proc. Indian Natl. Sci. Acad*. 2022;88:417-29.

39. Karunanayake KOLC, Sinniah GD, Adikaram NKB, Abayasekara CL. Cultivar differences in antifungal activity and the resistance to postharvest anthracnose and stem-end rot in mango (*Mangifera indica* L.). Australas Plant Pathol. 2014;43:151-9.
40. Khaliq G, Abbas HT, Ali I, *et al.* Aloe vera gel enriched with garlic essential oil effectively controls anthracnose disease and maintains postharvest quality of banana fruit during storage. Hortic Environ Biotechnol. 2019;60:659-69.
41. Kolase SV, Kamble TM, Musmade NA. Efficacy of different fungicides and botanicals against blossom blight of Mango caused by *Colletotrichum gloeosporioides*. J Plant Prot. 2014;7(2):444-7.
42. Kredics L, Büchner R, Balázs D, *et al.* Recent advances in the use of Trichoderma-containing multicomponent microbial inoculants for pathogen control and plant growth promotion. World J Microbiol Biotechnol. 2024;40:162.
43. Kumar A, Kudachikar V. Antifungal properties of essential oils against anthracnose disease: a critical appraisal. J Plant Dis Prot. 2017;125. doi:10.1007.
44. Kwodaga J, Sowley E, Benjamin B. Antifungal activity of plant extracts against *Colletotrichum gloeosporioides* (Penz.), the causative agent of yam anthracnose disease. Arch Phytopathol Plant Prot. 2019;52:1-16.
45. Lakshmi BKM, Anitha Kumari D, Kumar AK, Dilip Babu J, Narayan Reddy P. Mitigating postharvest losses caused by anthracnose disease in mango by using bio-agents, botanicals and ISR chemicals. Acta Hortic. 2013;1012:661-70.
46. Li X, Zhu X, Zhao N, Fu D, Li J, Chen W, *et al.* Identification and phylogenetic correlation among *Colletotrichum gloeosporioides* pathogen of anthracnose for mango. Biocatal Agric. Biotechnol. 2013;2:285-7.
47. Liang YS, Fu JY, Chao SH, Tzean Y, Hsiao CY, Yang YY, *et al.* Postharvest application of *Bacillus amyloliquefaciens* PMB04 fermentation broth reduces anthracnose occurrence in mango fruit. Agriculture. 2022;12:16-46.
48. Liu C, Cao Y, Wu E, Yang R, Xu H, Qiao Y. A discriminative model for early detection of anthracnose in strawberry plants based on hyperspectral imaging technology. Remote Sens. 2023;15:46-40.

49. Lopez-Ortiz C, Reddy UK, Zhang C, Natarajan P, Nimmakayala P, Benedito VA, *et al.* QTL and PACE analyses identify candidate genes for anthracnose resistance in tomato. *Front Plant Sci.* 2023;14:1200-999.
50. Sudha S, Narendrappa T, Sivakumar G. Management of post-harvest anthracnose disease in mango using promising biocontrol agents. *Pharma Innov J.* 2021;SP-10(3):210-214.
51. Mevada R, Kapadiya H. Effect of various fungicides, biocontrol agents, phytoextracts and essential oils on postharvest management of banana anthracnose (*Colletotrichum musae* Berk. & Curt.). 2019;387-91.
52. Mora-Aguilera JA, Ríos-López EG, Yáñez-Zúñiga M, *et al.* Sensitivity to MBC fungicides and prochloraz of *Colletotrichum gloeosporioides* species complex isolates from mango orchards in Mexico. *J Plant Dis Prot*; 2021. p. 481-91.
53. Nallamani M, Mohammed Bahurudeen H, Sobanbabu G, Suguna S. Management of mango anthracnose: an overview. *J Int. J Agric. Plant Sci.* 2023;2664-7664.
54. Tamimi N, Nafchi AM, Hashemi-Moghaddam H, Baghaie H. The effects of nano-zinc oxide morphology on functional and antibacterial properties of tapioca starch bionanocomposite. *Food Sci. Nutr.*; 2021. doi:10.1002.
55. Negi R, Sharma B, Kaur S, *et al.* Microbial antagonists: diversity, formulation and applications for management of pest-pathogens. *Egypt J Biol Pest Control.* 2023;33:105.
56. Nogueira GF, Leme BdO, Santos GRS, Silva JV, Nascimento PB, Soares CT, *et al.* Edible films and coatings formulated with arrowroot starch as a non-conventional starch source for plums packaging. *Polysaccharides.* 2021;2:373-86.
57. Nujthet Y, Jantasorn A, Dethoup T. Biological efficacy of marine-derived *Trichoderma* in controlling chili anthracnose and black spot disease in Chinese kale. *Eur. J Plant Pathol.* 2023;166:369-83.
58. Nujthet Y, Kaewkrajay C, Kijjoa A, *et al.* Biocontrol efficacy of antagonists *Trichoderma* and *Bacillus* against post-harvest diseases in mangos. *Eur. J Plant Pathol.* 2024;168:315-327.
59. Oliveira LF, Oliveira ML, Gomes FS, Santana RC. Estimating foliar nitrogen in Eucalyptus using vegetation indexes. *Sci. Agric.* 2017;74:142-147.

60. Onyeani CA. Susceptibility of mango varieties to anthracnose fruit rot in Southwest Nigeria. *Am J Agric. Biol. Sci.* 2017;12(2):95-99.
61. Owino WO, Ambuko JL. Mango fruit processing: options for small-scale processors in developing countries. *Agriculture*, 2021, 11(1105).
62. Páez R. Management of anthracnose (*Colletotrichum gloeosporioides* (Penz.) Penz. and Sacc.) of mango (*Mangifera indica* L). *Fitopatología Colombiana*. 2022;24(1):21-8.
63. Peeran MF. Pathogenesis of *Colletotrichum lindemuthianum*, the incitant of anthracnose disease in beans mediated by macerating enzymes. *The Bioscan*. 2014;9:295-300.
64. Kiranmayee V, Mishra JP, Prasad R, Sekhar JC, Reddy VP, Kumar S. Isolation and in vitro evaluation of essential oils against anthracnose of pomegranate caused by *Colletotrichum gloeosporioides* (Penz.). *J Pharmacogn Phytochem*. 2020;9(3):37-40.
65. Peralta-Ruiz Y, Rossi C, Grande-Tovar CD, Chaves-López C. Green management of postharvest anthracnose caused by *Colletotrichum gloeosporioides*. *J Fungi*; 2023, 9(623).
66. Phoulivong S, Cai L, Chen H, McKenzie E, Abdelsalam K, Chukeatirote E, *et al.* Mango: *Colletotrichum gloeosporioides* is not a common pathogen on tropical fruits. *Fungal Divers*. 2010;6(6):44-53.
67. Prashanth A, Sataraddi AR, Naik MKMB, Patil RS. Evaluation of fungicides, bioagents and botanicals against pomegranate anthracnose. *Int J Plant Prot*. 2008;36(2):283-7.
68. Bhagwat RG, Mehta BP, Patil VA, Sharma H. Screening of cultivars/varieties against mango anthracnose caused by *Colletotrichum gloeosporioides*. *Int J Environ Agric Res*. 2015;1:21-3.
69. Prasad RR, Anamika A. Effects of plant leaf extract against *Colletotrichum gloeosporioides* (Penz) Sac. Causing post-harvest disease of papaya. *J Agric Sci*. 2015;10:55-39.
70. Rajam MV, Chauhan S. Host-induced gene silencing (HIGS): an emerging strategy for the control of fungal plant diseases. *Concepts Strategies Plant Sci*. 2021;10:10-07.
71. Ranjitha N, Devappa V, Sangeetha CG. Evaluation of fungicides against *Colletotrichum gloeosporioides* (Penz.) Penz. and Sacc., the incitant of mango anthracnose. *J Pharmacogn Phytochem*. 2019;8(3):812-4.

72. Raveau R, Fontaine J, Sahraoui AL. Essential oils as potential alternative biocontrol products against plant pathogens and weeds. *Foods*. 2020;9(3):365.
73. Ruangtong J, Jaithon T, Huehne S. Mango anthracnose: a serious threat to the mango industry. *J Innov Sci.*; 2022. p. 1-10.
74. Roberto SR, Youssef K, Hashim AF, Ippolito A. Nanomaterials as alternative control means against postharvest diseases in fruit crops. *Nanomaterials*. 2019;9(12):17-52.
75. Sarmah PP, Nath HK, Nath T. Endophytic *Bacillus* spp. from native chilli cultivars and their effect against fruit rot pathogen of *Bhut Jolokia* (*Capsicum chinense* Jacq.). *Egypt J Biol Pest Control*. 2023;33:119.
76. Selvan T, Panmei L, Murasing KK, Guleria V, Ramesh KR, Bhardwaj DR, *et al*. Circular economy in agriculture: unleashing the potential of integrated organic farming for food security and sustainable development. *Front Sustain Food Syst*. 2023;7:1170-380.
77. Sharma G, Kumar N, Weir BS, Hyde KD, Shenoy BD. The ApMat marker can resolve *Colletotrichum* species: a case study with *Mangifera indica*. *Fungal Divers*. 2013;61:117-38.
78. Sharma IM, Kaul JL. Studies on post-harvest diseases of mango and chemical control of stem end rot and anthracnose. *Indian Phytopathol*. 1994;47:197-200.
79. Sharma A, Sharma IM, Sharma MM. Efficacy and economics of fungicides for management of mango anthracnose. *Indian Phytopathol*. 2019;72:361-6.
80. Nagaraju RS, Sriram RH, Achur R. Antifungal activity of carbendazim-conjugated silver nanoparticles against anthracnose disease caused by *Colletotrichum gloeosporioides* in mango. *J Plant Pathol*. 2020;102(1):39-46.
81. Singh S. Production performance of fresh mango in India: a growth and variability analysis. *Int. J Pure Appl Biosci*. 2018;6:935-41.
82. Song D, Zhang Y, Zhang L. Sensitivities of poplar anthracnose fungi isolates to carbendazim and three C-14 α -demethylation inhibitors. *Chin J Pestic Sci*. 2016;18:567-74.
83. Taïbi A, Meile JC, Dieudonné H, Korsten L, Chillet M, Remize F. New bacterial agents to limit *Colletotrichum gloeosporioides* development on mango. *Adv Microbiol*. 2020;10:691-712.

84. Tejashwini NK, Mesta RK, Hubballi M, Athani SI, Jawadagi RS, Biradar IB. Cultural and physiological studies on *Colletotrichum gloeosporioides* and *Fusarium oxysporum* causing twister diseases in onion. The Pharma Innov J. 2023;12(6):3803-8.
85. Kumar A, Kudachikar V. Antifungal properties of essential oils against anthracnose disease: a critical appraisal. J Plant Dis Prot. 2017;125:10.1007.
86. Velásquez C, Prieto F, Palou L, *et al.* New model for the automatic detection of anthracnose in mango fruits based on Vis/NIR hyperspectral imaging and discriminant analysis. Food Measure. 2024;18:560-70.
87. Vestergaard MC, Nishida Y, Tran LT, Sharma N, Zhang X, Nakamura M, *et al.* Antifungal activity and molecular mechanisms of copper nanoforms against *Colletotrichum gloeosporioides*. Nanomaterials, 2023, 13(2990).
88. Vieira WAS, Bezerra PA, Silva AC, Veloso JS, Câmara MPS, Doyle VP. Optimal markers for the identification of *Colletotrichum* species. Mol. Phylogenet. Evol. 2020;10:10-16.
89. Vieira WAS, Lima WG, Nascimento ES, Michereff SJ, Câmara MPS, Doyle VP. The impact of phenotypic and molecular data on the inference of *Colletotrichum* diversity associated with *Musa*. Mycologia. 2017;109:912-34.
90. Weir BS, Johnston PR, Damm U. The *Colletotrichum gloeosporioides* species complex. Stud Mycol. 2012;73:115-80.
91. Xia YL, Cao Y, Tan XH, Fang Y, Dong CZ, Wang ZK, *et al.* Chemical composition and fungicidal activity of *Murraya microphylla* essential oil against *Colletotrichum gloeosporioides*.
92. Yenjerappa ST, Malipatil R, Amaresh YS, Sreedevi SC, Narayan RP. Efficacy of different fungicides by in vitro against *Colletotrichum gloeosporioides*, the causal agent of mango anthracnose. Int. J Chem. Stud. 2021;9(1):3408-12.
93. Huang Y, Huang H, Tian C, *et al.* Biological control of poplar anthracnose caused by *Colletotrichum gloeosporioides* (Penz.) Penz. & Sacc. Egypt J Biol. Pest Control., 2020, 30(104).
94. Yousef HA, Fahmy HM, Arafa FN, *et al.* Nanotechnology in pest management: advantages, applications, and challenges. Int. J Trop Insect Sci. 2023;43:1387-99.

Chapter - 6

Marketing of Mushroom Products

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Chapter - 6

Marketing of Mushroom Products

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Abstract

The mushroom industry presents a significant opportunity for economic growth, job creation, and nutritional enhancement in India. The marketing landscape for mushrooms presents a unique opportunity to capitalize on shifting consumer preferences towards eco-friendly and ethical food consumption. By understanding the diverse uses of mushrooms beyond mere culinary delights, such as their medicinal properties and role in reducing farm waste, marketers can craft compelling narratives that resonate with health-conscious consumers and environmentally-conscious advocates alike. Moreover, leveraging the versatility of mushrooms in various recipes and cuisines can expand market penetration beyond traditional curry and pickle dishes, appealing to a broader audience seeking culinary adventures and plant-based protein alternatives. In parallel, targeted marketing efforts should prioritize consumer education on the nutritional benefits and medicinal qualities of mushrooms. Furthermore, enhancing product packaging and branding to communicate freshness, quality, and sustainability can enhance shelf appeal and differentiate mushroom products in a crowded marketplace. To optimize distribution channels, marketers should explore partnerships with supermarkets, specialty retailers, and online platforms to ensure broader accessibility and availability of mushroom products. Sampling initiatives in high-traffic areas and strategic partnerships with restaurants and food service establishments can also drive trial and adoption among consumers, further fueling market growth. Overall, a holistic marketing approach that combines consumer education, product innovation, strategic partnerships, and omni-channel distribution strategies can unlock the full potential of the mushroom market in India, driving demand, consumption, and economic prosperity across urban and rural communities.

Keywords: Mushroom marketing, consumer education, nutritional benefits, product innovation, Omni-channel distribution strategies

Introduction

Marketing is more than just selling a product. It involves understanding what customer wants, and to get the product to his hands at the right time in the right manner. When it comes to mushrooms, the mushrooms are more than just a food, they can be used for medicinal purposes as well. It can also be used to reduce farm waste and motivate the farmers to be more involved in the market. Promoting the sale of mushrooms can provide job opportunities and income for rural communities, while ensuring that city dwellers have access to high-quality produce.

Mushrooms: Global scenario

As noted by Singh *et al.* (2020), worldwide mushroom production in 2018-19 was 43 million tonnes. The respective percentage contribution by each species is given in the table 1. The global production is expected to reach 50 MT by 2025. The Asian countries are the primary producers of mushrooms contributing to 82.8% which is a lion’s majority. And in that also, Mainland China is the largest producer of mushrooms in the world.

When it comes to consumption, Mushroom market contributes to a whopping amount of USD 55.38 billion in 2022, and it is expected to reach USD 105.20 Billion by 2029 (Mushroom Market: Global Industry Analysis and Forecast (2023-2029) (maximizemarketresearch.com). This indicates a bright future for mushroom cultivation. In addition, shifting consumer preference towards eco-friendly and ethical consumption of food and increased demand of vegan alternatives has put forth mushrooms as a reliable source of protein.

The Wire has reported on November 20, 2020 that a policy intervention is urgent for mushroom. When compared to the western countries, consumption of mushrooms in India is significantly less. The lack of regulations related to mushrooms is a disadvantage to the producer side for starting a business.

India is frontier in production of every agricultural products. However, coming to the position in production of mushroom and truffles in India, India holds 18th position (Indiastat, 2016). From Fig, 2, it is clear how China is holding a monopoly in the market with. Even many European countries are in the leading in the market.

Table 1: Mushroom production species-wise

Mushroom	Percentage
<i>Lentinula edodes</i> (shiitake)	26%

<i>Auricularia</i> spp.	21%
<i>Pleurotus ostreatus</i> (Oyster)	16%
<i>Agaricus bisporus</i> (Button)	11%
<i>Flammulina velutipes</i>	7%
<i>Pleurotus eryngii</i> (King Oyster)	5%
<i>Volvariella volvacea</i> (Paddy Straw Mushroom)	1%
Others	13%

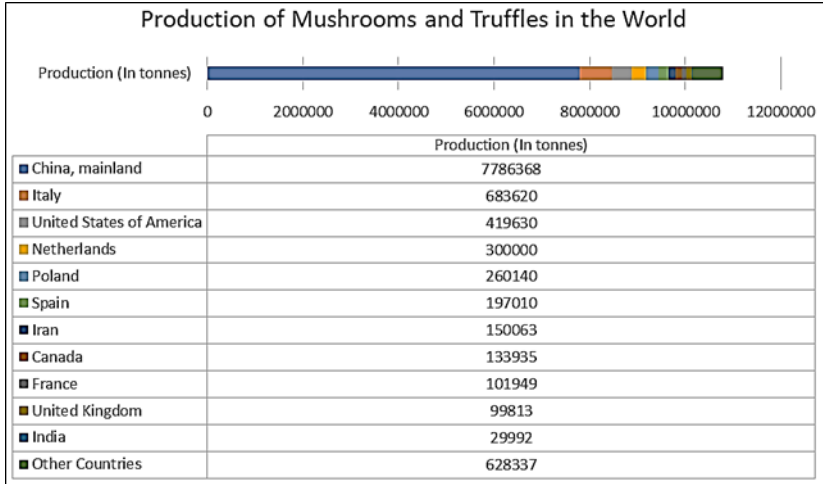


Fig 2: Production of Mushrooms and truffles in the World (Indiastat.com)

Mushrooms: Indian scenario

As per 2021-22 data, Bihar is leading in mushroom production in India, with 28,000 tonnes contributing more than 10% of the total production. The states in Northern and Northwest of India tops the chart in terms of production. However as mentioned earlier, even though the mushroom industry is blooming worldwide, India is not giving due importance it requires. India is having diverse climatic conditions, and the farmers of north-western states was able to exploit this and set up mushroom production units. This was beneficial for the farmers, as they could produce mushrooms with less and cheaper resources. The availability of good quality spawn and closeness to the market enabled for better sale of the commodities. However, this has resulted in excess arrival to the market during winter. The problem is adversary, especially for mushroom which is a highly perishable good.

As recorded by Sharma *et al.* (2017), In India, there are five mushroom species-white button mushroom (*Agaricus bisporus*), oyster (*Pleurotus* spp.), paddy straw (*Volvariella volvacea*), milky (*Calocybe indica*) and shiitake

(*Lentinula edodes*) are in commercial cultivation. Among these, the first three is contributing to 96% of production.

The mushroom import-export data by India reveals that the mushroom export business is still at a rudimentary stage.

Case study

Farooq *et al.* (2022) has conducted a case study on mushroom marketing in Hardoi district of Uttar Pradesh. The author has collected primary data through questionnaire. The area is having 88.6% of small and marginal farmers involved in mushroom cultivation. The study has concluded that due to lack of marketing system, farmers are unable to get adequate price. The farmers are sometimes forced to sell their produce by uneconomic minimum market price. The study stresses on the importance of farmer producer organizations and cooperative societies for production.

Chandra and Qureshi (2023) have conducted a study in the villages of Durg district, Chhattisgarh. The study has shown that at a lower investment cost, the family income could be improved by mushroom cultivation. This is because waste materials from the farm can be easily converted into nutrient-rich food source. Growing mushroom in indoor part time job that can be useful for converting unused space. It can also be used for employment as it is a labour-intensive, profitable business. The study has concluded that mushroom cultivation is a better job for landless farmers and small farmers to produce goods to support the family. Especially the survey has shown that women could generate considerable income after they have started growing mushrooms. The mushroom growing activity can be a potential activity to create economic and social impact by providing jobs and income to rural people, which ultimately leads to eradication of poverty. This can thereby act as a mode of generating economic growth in less developed areas and developing and underdeveloped nations.

Consumption of mushrooms in India

Shirur *et al.* (2014) has conducted a survey on mushroom consumption and purchasing behaviour in India. This has revealed that the consumption behaviour of the respondents was restricted to occasional use mushroom curry. Also, 4% has responded that they never consumed any mushroom products. 84% of the respondents has never tasted any recipe other than curry or pickle. It's not surprising that the vast majority of the respondents (79.85%) chosen White button mushroom as their first preference, which is also the largest producing mushroom in India. The study has noted that customers preferred quality of the mushrooms over price.

The author has also studied the reason for non-purchasing behaviour among the customers. The top 3 reasons are as follows; Non availability of mushrooms in the market, Mushrooms available in the market are not fresh, and Do not know how to make different mushroom recipes. Nearly one fourth of the respondents are unaware about the nutritional and medicinal qualities of mushrooms, despite having high education level.

Now from this study, it is clear as why the consumption of domestic market is low in India. The mushroom market in India is on vicious a loop of less demand-less production-more price. This has caused less innovations as well. As reported by Directorate of Mushroom Research, Solan, The consumption of mushrooms in China is 20-22 kg/person/year. In US and EU, the average consumption is 3kg/person/year. When it comes to India, the consumption is miserably low as 70% of the total 1.30 lakh tons is exported, and per capita consumption is only 70-80 g/person/year.

The less consumption in India is the reason why we could not produce enough to compete in the global market. The attitude of people towards their food habits needs to undergo a serious change for exploiting the income from mushrooms into our country.

Mushroom marketing: Why it is necessary

As addressed by honourable Prime Minister Shri Narendra Modi in his Independence Day speech, India is aiming to become a developed country by the year 2047. For achieving this, in addition to business development and driving investment to the cities, it is equally (or more) important to achieve development in the villages as well. The best way to achieve the same is to promote increased agricultural production and export from the villages. Now this may require use of every possible raw material and labour to maximize the production. This is where the mushroom production comes into picture. However, the coordination from the government, farmers, and industries are required to bring this dream into reality. Better facilities for transportation and storage of goods, and providing aids for production can be done from the government side. The private investment is needed for effective marketing and exports, and value addition. The farmers are directly involved in production and can bring innovation into this field.

Mushroom cultivation when compared with other agricultural production requires only less space and time. Plus providing artificial growing conditions for production can be done efficiently. This can reduce the dependence on climate and marketing at the right time ensuring maximum returns. The effective marketing strategy is the missing link for effective selling of the

product. The mushroom selling chain in India is limited to Producers-Retailers-Consumers or Producers-Consumers which is having its limitations. Also, majority of the farmers being seasonal producers, market gluts cause distress sale of the mushrooms. Although it is cost-effective, this limits the options available for farmers and year round production is not possible. So the first strategy should be educating the farmers about effective marketing strategies.

Changing consumption behaviour of the people

There is an old proverb-Need is the mother of all inventions. In case of opening the mushroom markets of India, first the demand should be there. Now this is the herculean task we need to overcome, as it requires changing the food consumption behaviour of the citizens. This however has to be encountered in multiple ways simultaneously.

The internet is a marketplace of ideas. And at this point of technology, internet has reached every households, which is knowingly or unknowingly is helping them to acquire new skills. Cooking channels are one of the most viewed category of videos in online social media platforms, and this can be an effective way of marketing. The Indian cuisine particularly in the mushroom arena is limited to only mushroom curry and pickles. This needs to be changed first. If the people are introduced with innovative ways of cooking delicious dishes with mushrooms, it is certain that they will try to do it themselves, and if effective, they will recommend it to their peers. This will introduce the people to new world of mushroom dishes, which can cause a slight change in the market.

The next part will be to motivate and educate the people to produce mushrooms in their households through training programmes. This can motivate the people to get involved in the process. Also unlike other crops, mushrooms can be produced in less area and less care, which can engage the urban households as well. This will increase their consumption of mushrooms, and thereby the demand for mushrooms in the market.

When it comes to villages, most of the people are already engaged in agricultural activities. The rubbles and agricultural wastes are best bedding material for mushrooms. The people should be motivated to use the same for mushroom production. This will reduce the cost for raw materials. Indirectly this will also cause reduction in pollution due to burning of stubbles. The production of mushrooms will be a side income for village people, and also will bring innovation into the field on production and creating new recipes.

The inclusion of mushrooms serves more than just increasing the income. As earlier mentioned, even the people with high level of education are unaware that mushrooms are having high nutritional content and can act as a source of protein. In a report by Khandelwal and Kavishwar (2018), protein deficiency in India is more than 80%. The mushrooms can considerably fill in this gap as a source of protein and other micronutrients.

Techniques to promote mushroom

Understanding the market landscape

Before diving into marketing techniques, let's grasp the broader context:

- **Global mushroom market:** The worldwide mushroom market is valued at approximately-\$50 billion. Major producers include China, the USA, the Netherlands, Italy, and Poland (mushroomsociety.in).
- **Indian scenario:** India produces around 1.45 lakh metric tons of mushrooms annually, contributing to about 0.4% of global production. However, per capita consumption in India remains low at approximately 70-80 grams per person per year by Wakchaure (2011).

Target Audience and Segmentation

Identify your potential customers within urban areas. Consider the following segments:

Target segmentation in the mushroom market involves dividing potential customers into distinct groups based on specific characteristics. Let's explore how this can be done:

- **Type-based segmentation:** Mushrooms, Shiitake Mushrooms, and Oyster Mushrooms are popular types in the market. Segmenting based on mushroom type allows businesses to tailor their marketing strategies to meet the preferences of each group.
- **Form-based segmentation:** Mushrooms are available in fresh and processed forms. Businesses can target consumers who prefer fresh mushrooms for cooking or those who opt for processed mushroom products like canned or dried mushrooms.
- **End-use segmentation:** Consider the purpose for which mushrooms are used:

Residential: Targeting individual consumers who buy mushrooms for home cooking.

Commercial: Focusing on restaurants, cafes, and other food service establishments.

Industrial: Addressing bulk buyers such as food processors or manufacturers.

- **Distribution channel segmentation:** Identify the channels through which mushrooms reach consumers:

Supermarkets/Hypermarkets: Targeting retail chains.

Specialty retailers: Focusing on specialty stores that sell organic or gourmet mushrooms.

Convenience stores: Reaching out to consumers seeking quick purchases.

Online stores: Addressing the growing e-commerce market.

- **Geographic segmentation:** Consider regional preferences and demand patterns. For instance, different regions may favor specific mushroom types or forms.
- **Demographic segmentation:** Analyze consumer demographics such as age, income, and lifestyle. For example, health-conscious individuals may prefer mushrooms due to their nutritional benefits.

Product Quality and Packaging (www.farmsnation.com)

- **Quality assurance:** Ensure consistent quality by adhering to best cultivation practices.
- **Attractive packaging:** Invest in eye-catching packaging that communicates freshness and quality.
- **Harvest timing:** Harvest mushrooms at the right stage of maturity. Overripe or underripe mushrooms may affect quality.
- **Hygiene:** Handle mushrooms with clean hands and use sanitized tools during harvesting and postharvest handling.
- **Cooling:** Rapidly cool freshly harvested mushrooms to slow down deterioration. Cold storage at 0-2°C (32-36°F) is ideal.
- **Grading:** Sort mushrooms based on size, color, and defects. Uniformity enhances market appeal.
- **Bruise prevention:** Mushrooms are delicate; avoid rough handling to prevent bruising.
- **Quality control:** Regularly inspect for signs of spoilage, mold, or off-odors.

Packaging techniques (www.farmsnation.com)

- **Plastic clamshells or bags:** Commonly used due to durability and moisture retention. However, consider eco-friendly alternatives.
- **Paper packaging:** Sustainable choice, but may offer less protection than plastic.
- **Mesh packaging:** Allows ventilation and prevents moisture buildup. Balance with protection from physical damage.
- **Wooden cases or bamboo baskets:** Used for bulk packaging and airfreight transport.
- **Mycelium packaging:** Innovative and sustainable. Mycelium-based materials are compostable and insulating.

Pricing Strategies

- **Competitive pricing:** Research prevailing market prices and set competitive rates.
- **Volume discounts:** Offer discounts for bulk orders to attract wholesalers and retailers. Some effective approaches are:

Market analysis

Before setting prices, perform a thorough market analysis

Research popular mushroom varieties in demand, such as Button, Shiitake, and Oyster mushrooms. Potential customers, including Supermarkets, Restaurants, Farmers markets and direct consumers interested in organic and locally-sourced foods. Analyze competitors by visiting local mushroom farms, reviewing their product range, pricing, and marketing strategies.

Understand pricing trends

Track historical and current prices of various mushroom types. This knowledge helps you set competitive yet profitable prices. Consider factors like location, type of mushrooms grown, and market demand.

Collaborate with restaurants and chefs

Work directly with chefs and restaurants to negotiate favourable pricing. Collaborate on menu development and incorporate your gourmet mushrooms into signature dishes. This can increase the perceived value and allow you to charge premium prices.

Distribution channels

- **Local markets:** Participate in weekly markets or mandis.
- **Online platforms:** Utilize apps like Farms nation to connect with urban buyers.
- **Wholesale distributors:** Collaborate with distributors who supply to supermarkets and restaurants.
- **Supermarkets and Hypermarkets:** These retail giants play a significant role in mushroom distribution. They offer a wide range of mushroom varieties to consumers. In developed regions like Europe and North America, supermarkets and hypermarkets have higher penetration.
- **Grocery stores:** Local grocery stores are essential distribution channels. They cater to individual consumers and households. Mushrooms can be prominently displayed in these stores to attract buyers.
- **Restaurants and Food service establishments:** Restaurants, cafes, and hotels are major buyers of mushrooms. They use them in various dishes, making them a crucial market segment. Establishing relationships with chefs and restaurant owners is essential.
- **Vendors and Street markets:** Street vendors and local markets provide direct access to consumers. Fresh mushrooms can be sold at stalls or mobile carts. These channels are especially relevant in densely populated areas.
- **Sampling and Awareness campaigns:** Distribute free mushroom samples in big malls, variety stores and at events.
- **Social media platforms:** Utilize platforms like Facebook, TikTok, Instagram, YouTube, and Twitter to showcase your mushroom products. Engage with potential customers and share recipes and health benefits.
- **Export opportunities:** Explore export possibilities to countries with high mushroom demand. Understand global trade dynamics and consumer preferences.
- **Track performance:** Monitor sales, customer feedback, and market trends.

Distribution and support

Mushroom distribution channels should also undergo a major change. As noted by Wakchaure (2011), mushroom marketing in India is not yet

organized. This limits the various avenues where the farmer can sell his produce. As any other commodity, mushroom also needs to be transported to farther distances. This has to be done through multitude of approaches.

Forming cold storage facilities is essential for storage for a longer term. This along with transport and processing facility will help reach the products to bigger cities. Also this means that logistics and distributors focusing exclusively on these kinds of developments are to be formed. Also the distribution should not only be restricted to retail markets, but also to vegetable stores, supermarkets or even directly to restaurants.

Public sector undertaking is necessary for initial thriving of this innovative idea. For this, minimum support price can be adopted for motivating the farmers. Also measures need to be undertaken to ensure the constant supply of market throughout the year. This will ensure that the price in the market will be having less seasonal variation and ensure year-long availability.

Conclusion

Judging the market trend, mushroom market is having a better future. However as indicated by statistics, the production and support to this industry in India is very low. The prime reason is that the consumption pattern of people doesn't primarily involve any mushroom based cuisines. This is the first thing that needs to be taken care of. For this, a long term vision is required from the government. The consumers should be introduced to different delicacies of mushrooms so that they are motivated to go to the market. And provided there is some demand in the market, the people should be motivated to conduct the cultivation in their households. The farmers on the other hand, should be exposed to method demonstration and trainings so that they can cost-effectively produce more and this followed by a constant demand in the market and adequate support from the government will ensure there is constant supply at a reasonable price. And given the domestic market is strong enough, the additional possibilities of export and processing can be explored.

References

1. Chandra O, Qureshi Y. Empowerment in agricultural proceeds through Mushroom Cultivation-A Case Study in the villages of Durg. District, Chhattisgarh, India; 2019.
2. Farooq A, Zechariah J, Kumar A, Tripathi P, Barker N. A Study on Marketing of Mushroom in Hardoi District of Uttar Pradesh. *Int. J Adv. Agric. Sci. Technol.* 2022;9(4):27-35.

3. Farmsnation.com. Scope of mushroom cultivation in India and where can you sell your mushrooms. Available from: <https://farmsnation.com/marketing/scope-of-mushroom-cultivation-in-india-and-where-can-you-sell-your-mushrooms/>.
4. Krishijagran.com. Beginner's guide for profitable business in mushroom farming. Available from: <https://krishijagran.com/agripedia/beginners-guide-for-profitable-business-in-mushroom-farming/>.
5. Mushroomsociety.in. Production and marketing of mushrooms. Available from: https://mushroomsociety.in/mushroomeLearning/pluginAppObj_2913/ProductionMarketing-BLA.pdf.
6. Khandelwal S, Kavishwar A. Indians are protein deficient, and it needs immediate attention. Forbes India; 2018.
7. Maximizemarketresearch.com. Mushroom Market: Global Industry Analysis and Forecast; 2023-2029. Available from: <https://maximizemarketresearch.com/market-report/global-mushroom-market/287/>.
8. Sharma VP, Annepu SK, Gautam Y, Singh M, Kamal S. Status of mushroom production in India. Mushroom Res. 2017;26(2):111-120.
9. Shirur M, Ahlawat OP, Manikandan K. Mushroom consumption and purchasing behaviour in India: A study among selected respondents. Mushroom Res. 2014;23(2):225-231.
10. Singh M, Kamal S, Sharma VP. Status and trends in world mushroom production-III-World Production of different mushroom species in 21st century. Mushroom Res, 2020, 29(2).
11. Wakchaure GC. Production and marketing of mushrooms: global and national scenario. In: Mushrooms-Cultivation, Marketing and Consumption. 2011. p. 15-22.
12. www.indiastat.com

Chapter - 7

The Silent Killer in Rice: Understanding Akiuchi Disease

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Chapter - 7

The Silent Killer in Rice: Understanding Akiuchi Disease

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Abstract

Akiuchi disease, a severe condition impacting rice crops, has extensive repercussions beyond reduced yields. The economic impact on farmers is considerable, affecting their livelihoods and food security. Environmentally, the disease leads to soil degradation and heightened greenhouse gas emissions. Various factors contribute to the development of Akiuchi, including poorly drained, heavy clay soils high in organic matter, extended periods of waterlogging or intermittent flooding, excessive nitrogen fertilization, and the inherent susceptibility of certain rice cultivars. Effectively managing Akiuchi disease necessitates a comprehensive approach. The use of organic amendments and soil conditioners, along with biofertilizers, can enhance soil health and inhibit anaerobic bacterial growth. Additionally, breeding for tolerance and leveraging genetic resources are long-term objectives aimed at developing resistant rice varieties. Ongoing research is crucial for the development of sustainable Akiuchi management strategies. Important research areas include elucidating the biochemical pathways involved in the disease, employing genomic approaches for breeding, integrating pest management methods, and assessing the impact of climate change on disease prevalence. Addressing these areas can mitigate the impact of Akiuchi disease on production and ensure the sustainability of rice farming.

Keywords: Akiuchi, rice, management, biofertilizer

Introduction

Akiuchi disease, commonly referred to as the "silent killer" or "buzone disease", poses a major threat to rice cultivation globally. This stealthy disease is characterized by its gradual onset and subtle symptoms, making early detection difficult. It primarily affects rice crops in poorly drained, heavy clay soils that are rich in organic matter. These conditions promote the proliferation of anaerobic bacteria, which decompose organic material and generate hydrogen sulfide (H_2S) as a byproduct. Highly soluble in water, H_2S is

absorbed by rice roots, disrupting cellular processes and causing stunted growth, yellowing leaves, and reduced tillering. As the disease progresses, more severe symptoms such as root discoloration, reduced root mass, and poor grain filling become apparent, significantly diminishing crop yields. The economic impact of Akiochi disease is substantial, imposing a heavy burden on farmers by reducing their incomes and jeopardizing food security.

Definition and Etiology

Akiochi is a physiological condition predominantly affecting rice plants, primarily induced by hydrogen sulfide (H_2S) toxicity. This detrimental gas is generated in waterlogged soils with anaerobic conditions due to the decomposition of organic matter by specific bacteria. The root system of rice plants is particularly sensitive to H_2S , leading to disruptions in nutrient uptake and a subsequent decline in overall plant health.

Symptoms and Diagnosis

Timely identification of Akiochi is essential for effective management. The condition presents through a range of distinct symptoms:

- **Discoloration:** Lower rice leaves exhibit a bluish-green or purplish tint as a result of H_2S accumulation.
- **Stunted growth:** Plants affected by the condition show reduced growth rates compared to healthy counterparts.
- **Root rot:** The root system experiences progressive decay, turning black and emitting a foul odor.
- **Reduced tillering:** Akiochi-infected plants have a markedly lower number of tillers, or secondary shoots.
- **Yield loss:** A significant decrease in grain yield is the most noticeable symptom of the disorder.

Precise diagnosis of Akiochi typically requires a combination of field observations and laboratory analyses. Soil samples are tested for hydrogen sulfide (H_2S) concentration, while plant tissue analysis is conducted to detect the presence of toxic elements.

The biochemical pathways involved in Akiochi disease

Indeed, gaining insight into the complex biochemical pathways involved in Akiochi disease is crucial for devising effective mitigation strategies. Below is an overview of the key processes:

1. Anaerobic respiration and H₂S production

- In waterlogged rice fields, anaerobic bacteria dominate the decomposition of organic matter in the soil.
- This decomposition process generates hydrogen sulfide (H₂S) gas as a byproduct.
- Enzymes, particularly sulfate-reducing enzymes, are crucial in the production of H₂S.

2. H₂S Uptake and root damage

- Due to its high solubility, hydrogen sulfide (H₂S) is easily absorbed by rice roots.
- Once inside the plant, H₂S interferes with cellular respiration by inhibiting cytochrome oxidase, an enzyme critical for the mitochondrial electron transport chain.
- This inhibition decreases ATP production, which is the primary energy source for cellular functions, thereby impeding various cellular processes.
- Additionally, H₂S can react with essential elements such as iron, which impairs the plant's nutrient uptake.

3. Impaired nutrient uptake and plant growth

- Hydrogen sulfide (H₂S) interferes with the proper functioning of root cell membranes, impairing the absorption of essential nutrients such as nitrogen, phosphorus, and potassium.
- This disruption negatively affects plant growth and development, resulting in stunted growth and diminished tillering, which is the formation of new shoots at the plant's base.

4. Disruption of photosynthesis

- The precise mechanisms by which hydrogen sulfide (H₂S) exposure impacts photosynthesis are not completely understood; however, there is evidence suggesting an indirect effect.
- This indirect effect could be attributed to decreased stomatal conductance, which affects the regulation of gas exchange through leaf pores, or to impaired activity of enzymes involved in photosynthesis.

5. Oxidative stress and defense responses

- Exposure to hydrogen sulfide (H₂S) leads to an increase in the production of reactive oxygen species (ROS), which are detrimental molecules.
- In response, the plant activates defense mechanisms to neutralize these ROS.
- However, this ROS scavenging process further depletes the plant's already strained cellular energy reserves.

The effects of Akiuchi disease

The repercussions of Akiuchi disease go beyond diminished crop yields. The financial strain on farmers is significant, impacting their livelihoods and food security. Furthermore, the environmental consequences are considerable, as the disease can lead to soil degradation and increased greenhouse gas emissions.

Factors contributing to Akiuchi

Multiple factors converge to create an environment favorable for the development of Akiuchi disease:

- **Soil conditions:** Soils with high clay content, rich in organic matter, and poorly drained provide optimal conditions for anaerobic bacteria.
- **Water management:** Extended periods of waterlogging or intermittent flooding intensify the problem.
- **Fertilization:** Overuse of nitrogen fertilizers can promote the proliferation of anaerobic bacteria.
- **Cultivar susceptibility:** Certain rice varieties show greater resistance to Akiuchi disease compared to others.

Management strategies

An integrated approach is essential for the effective management of Akiuchi disease. Critical strategies include:

1. Cultural practices

- **Enhanced drainage:** Improving soil drainage by constructing ditches or using raised beds can alleviate waterlogging.
- **Water management:** Regulating irrigation practices, including periodic drying, can help prevent anaerobic conditions.

- **Crop rotation:** Integrating non-rice crops into the rotation can break the disease cycle.
- **Residue management:** Effective management of rice residues can decrease the organic matter content in the soil.

2. Nutrient management

- **Balanced fertilization:** Applying suitable amounts of nitrogen and other nutrients can enhance plant growth while avoiding conditions that favor anaerobic bacteria.
- **Organic amendments:** Adding organic matter such as compost or manure can improve soil structure and drainage.

3. Chemical control

- **Soil amendments:** Using lime or gypsum can increase soil pH and decrease the availability of hydrogen sulfide.
- **Biofertilizers:** Introducing beneficial microorganisms into the soil can inhibit the growth of anaerobic bacteria.

4. Host plant resistance

- **Breeding for tolerance:** Developing rice varieties with natural resistance to Akiochi is a long-term objective.
- **Genotypic variation:** Identifying and utilizing genetic resources with tolerance traits can expedite breeding programs.

Research and future perspectives

Ongoing research is crucial for developing more effective and sustainable strategies for managing Akiochi disease. Key areas of focus include:

- **Understanding disease physiology:** Comprehensive research into the biochemical pathways associated with Akiochi can reveal potential control strategies.
- **Developing resistant varieties:** Utilizing genomic approaches and advanced breeding techniques can speed up the creation of rice cultivars resistant to Akiochi.
- **Integrated pest management (IPM):** Employing a combination of cultural, biological, and chemical control methods can improve overall disease management.
- **Climate change adaptation:** Examining the effects of climate change on Akiochi prevalence and developing suitable adaptation strategies is essential.

References

1. Moletti M, Giudici ML, Villa B. Rice Akiochi-brown spot disease in Italy: agronomic and chemical control. *Cahiers Options Méditerranéennes*, 1996, 15(3).
2. Padmanabhan SY. The present position and control of rice diseases in India. *Proc. Indian Acad. Sci. New Delhi: Springer India*. 1959;49(5):349-62.
3. Khush GS, Virmani SS. Breeding rice for disease resistance. *Prog. Plant Breed*. 1985;1:239-79.
4. Akiochi disease in rice: is weather a factor? [Internet]. Available from: <https://www.farmprogress.com/rice/akiochi-disease-in-rice-is-weather-a-factor->
5. Pests and diseases of rice [Internet]. Knowledge Bank IRRI. Available from: <https://www.knowledgebank.irri.org/step-by-step-production/growth/pests-and-diseases/diseases>

Chapter - 8

Root Disease Complex in Mulberry

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Chapter - 8

Root Disease Complex in Mulberry

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Abstract

The mulberry plant is susceptible to root disease complex, caused by the synergistic interaction of root-knot nematode and fungal pathogens. This complex significantly impacts sericulture, leading to more than 15 per cent crop loss and reduced leaf quality. Effective management strategies, including physical, cultural, biological and chemical methods, are crucial to mitigate infestations. This chapter highlights the disease symptoms, spread and management options to ensure sustainable mulberry production.

Keywords: Mulberry, root disease complex, root-knot nematode, fungal pathogens, sericulture

Introduction

Mulberry (*Morus alba* L.) is a highly adaptable perennial species that thrives in diverse climatic conditions, from temperate to tropical regions, and can grow well in various soil types. As the primary food source for silkworms, mulberry foliage plays a crucial role in determining the quality of raw silk. The silkworms rely solely on mulberry leaves for their nutrients, and the quantity and quality of leaves produced per unit area directly impact cocoon production and raw silk quality. Mulberry leaf is one of the most critical factors in sericulture, accounting for approximately 38-40 per cent of the total factors influencing a successful cocoon crop. The quality of mulberry leaves is significantly impacted by various biotic and abiotic stress factors. Biotic stresses, including diseases caused by pathogens and infestations by insect and non-insect pests, can substantially degrade the nutritive value of the leaves. Mulberry plants, with their perennial, fast-growing and lush green characteristics, are particularly susceptible to attracting these pests and diseases, making them more vulnerable to damage.

The rhizosphere soil is a hotspot of microbial activity, characterized by a significantly higher population of microorganisms. Notably, the rhizosphere soil of mulberry plants supports a markedly higher abundance of fungi and

nematodes. Among the nematodes, the root-knot nematode (RKN), *Meloidogyne incognita* (Kofoed and White) Chitwood is of high economic importance as it is known to cause 20-50 per cent loss in leaf yield apart from deteriorated quality and in India it was first reported on mulberry by Narayanan *et al.* (1966). RKNs are insidious parasites that target underground roots, making them challenging to detect. As a result, damage symptoms often go unnoticed until severe infestation has occurred. Infested plants exhibit stunted growth, marginal necrosis, and yellowing of leaves. The characteristic feature of RKN infestation is the formation of knots or galls on the roots, which impede water and nutrient uptake, leading to poor plant growth and development. Each RKN female lays 200-300 ellipsoidal eggs, covered in a gelatinous substance, which hatch and release larvae into the soil under favourable conditions. The life cycle is completed in 30-40 days, with 2-3 cycles occurring annually. Optimal growth and development conditions for RKNs include temperatures between 15-30°C and soil moisture levels of 40-60%. As generations repeat, the size and number of galls increase, damaging parenchyma tissue and creating cracks and holes that invite secondary root infections.

The mulberry root rot is another highly destructive and epidemic disease that poses a significant threat to mulberry cultivation and may induce around 12-14 per cent crop loss. This soil-borne disease, triggered by a solitary or combined infection of fungal pathogens, notably *Fusarium solani*, *Fusarium oxysporum*, *Lasiodiplodia theobromae* and *Macrophomina phaseolina*, can rapidly spread through irrigation water and contaminated farm implements, ultimately leading to widespread plant mortality. The disease typically manifests as isolated foci within mulberry gardens, subsequently expanding to adjacent areas. Its occurrence is reported throughout the year, irrespective of soil type, although its severity is exacerbated in soils with low moisture and organic matter content. Symptoms of the disease manifest initially above ground as sudden wilting and defoliation, progressing from the lower branches upwards. Below ground, symptoms include root decay, characterized by blackening due to fungal spores and mycelium, leading to complete root system collapse and plant mortality. Pruned affected plants either fail to sprout or produce small, pale yellow leaves with rough surfaces. Severely infected plants lose soil anchorage and can be easily uprooted. If immediate control measures are not employed, then there is a chance of increased crop loss.

The root disease complex, resulting from the synergistic interaction of multiple pathogens, has emerged as a significant threat to mulberry production, involving the concurrent infection of RKN (*M. incognita*) and root

rot fungal pathogens (*F. solani*, *F. oxysporum*, *L. theobromae* and *M. phaseolina*). When mulberry plants exhibit over 100 galls/plant due to *M. incognita* infection and mixed populations of *L. theobromae* and *F. solani*, the disease complex intensity is high, leading to 80-90 per cent plant mortality, particularly in sandy soils under irrigated conditions, as the nematode facilitates the entry of root rot pathogens, exacerbating disease severity and ultimately causing plant death. Generally, more than 15 per cent of crop loss would be recorded depending on the location and disease severity.

Above-ground symptoms

- a) Yellowing of leaves at the early stage of infection.
- b) Stunted growth and reduced leaf lamina.
- c) Affected plants may fail to sprout after pruning.
- d) Sprouted plants bear small, pale yellow, and wrinkled leaves.
- e) Wilting or browning of leaves, starting from the lower portion of branches and progressing upwards.
- f) Defoliation of leaves, leading to wilting of the whole plant.
- g) Patches of wilting plants appear in the garden when the disease is severe.

Below-ground symptoms

- a) Decayed roots turn black due to fungal mycelia/spores below the bark.
- b) Severe infestation leads to decay of the entire root system.
- c) Plants lose their grip in the soil and can be easily uprooted.
- d) Uprooted plants show more than 100 galls/plant due to nematode attack.
- e) Roots are decayed/rotted due to fungal attack.
- f) Conducting tissues become black/brown due to rotting of tissues as the disease progresses.

The root disease complex in mulberry spreads primarily through contaminated soil, infected farm implements and run-off irrigation water. Secondary sources of infection include the plantation of infected saplings, inter-cropping with susceptible crops and the growth of susceptible weeds in and around mulberry gardens. The disease thrives in conditions characterized by temperatures between 20-35 °C, soil moisture levels below 40% and a pH range of 4-8, ultimately leading to devastating consequences for mulberry production.

Management of root disease complex in mulberry

The root disease complex management can be broadly classified into four strategies: physical, cultural, biological and chemical methods.

1. Physical method

Planting disease-free saplings is the hassle-free approach, others include deep ploughing and hot water treatment. Summer deep ploughing/digging to 30-40 cm depth in the infected gardens can reduce the nematode population. The spores of root rot pathogen and eggs and larvae of RKN are sensitive to heat and exposure to higher temperatures through soil solarisation will reduce their population. The hot water treatment of saplings (48 C for 20 minutes) before plantation.

2. Cultural method

The disinfection of implements, use of trap crops and soil amendments can be considered under cultural method. The farm implements should be disinfected with 5% formalin solution or treated with boiling water before use to avoid cross-contamination and spreading of disease in healthy fields. The field application of neem/pongamia oil cake @ 2 MT/ha/year in four split doses during the fertilizer application is recommended. The planting of marigold (*Tagetes patula*) as intercrop at a 30 cm distance in between mulberry plant rows would reduce the RKN population.

3. Biological method

Biological approaches provide a targeted, safe and environmentally friendly solution, prioritizing non-target species, beneficial insects, animals and human safety while minimizing ecological impact. The utilisation of bacterial and fungal bioagents like *Pseudomonas fluorescens*, *Bacillus subtilis*, *Trichoderma harzianum*, *Trichoderma viride*, *Purpureocillium lilacinum*, *Pochonia clamydosporea*, etc. along with the FYM or seri-compost can reduce the incidence of root disease complex. Various bio-consortiums developed by CSRTI, Mysore viz., 'Bionema' (*Pochonia clamydosporea*: Neem oil cake: FYM :: 1:24:200) @ 200 g/plant, 'Raksha' (*Trichoderma harzianum*: FYM :: 1:50) @ 500 g/plant, 'Bio-mix' (*Trichoderma harzianum* and *Trichoderma viride*: Pongamia oil cake: FYM :: 1:10:50) @ 200 g/plant and 'Nemahari' (75% plant components and 25% chemicals) @ 40 kg/ha/application (40 kg Nemahari + 400 kg FYM) have also been found effective in managing RKN and root rot disease separately and also when occurred as disease complex.

4. Chemical method

The chemical control method for root disease complex in mulberry involves applying 1.5 g Furadan (Carbofuran 3 G) + 10 g Dithane M-45 (Mancozeb 75% WP) per plant around the root system, after removing soil to a depth of 15 cm, as soon as initial symptoms appear. In cases of severe disease, affected/dead plants should be uprooted and burnt, and the diseased pit treated with the same chemical combination before planting a new sapling, whose roots should be dipped in 0.1% Dithane M-45 solution for 30 minutes. This application should be repeated three times a year, at 4-month intervals. Additionally, surrounding plants in the diseased area should also be treated with the chemical combination as a preventive measure. It's essential to delay feeding leaves treated with nematicide and fungicide to silkworms for 40-45 days after application to ensure safety.

Conclusion

The root disease complex poses a significant threat to mulberry cultivation, compromising sericulture productivity and sustainability. Understanding the disease symptoms, spread, and management options is critical for developing effective control strategies. Integrated management approaches, combining physical, cultural, biological and chemical methods, offer a promising solution to mitigate infestations and ensure sustainable mulberry production. Adoption of these strategies will help minimize crop loss, maintain leaf quality and support the sericulture industry's growth and development.

References

1. Arunakumar GS, Revanna S, Kumar V, Yadav VK, Sivaprasad V. Studies on scanning electron microscopy and fungal association with root-knot nematode in major mulberry growing areas of Southern Karnataka. *Journal of Entomology and Zoology Studies*. 2018;6(4):511-518.
2. Bharath KB, Vinoda KS, Banuprakash KG, Kavitha TR. Evaluation of Bioagents for their Efficacy in Management of Root-knot Nematode Infesting Mulberry. *International Journal of Environmental and Agriculture Research*. 2024;10(7):1-7.
3. Dandin SB, Giridhar K. *Handbook of Sericulture Technologies*. Central Silk Board, Bangalore, India. 2014;160-172.
4. Nishita Naik V, Pratheeshkumar PM. Nemahari-A plant based product for management of root-knot disease in mulberry. *Technical Bulletin (01)*, Central Sericultural Research and Training Institute, Mysore, India; 2015.

5. Padma S, Bongale U, Sukumar J, Mallikarjunappa R, Pushpa R. Root-knot nematode *Meloidogyne incognita* (Kofoed and White) Chitwood in mulberry: A review. *Sericologia*. 2008;48(1):1-21.